机械与电子工程学院
College of Mechanical and Electronic Engineering

# 第7章

# 应力应变分析 强度理论

## Chapter 7 Analysis of Stress and Strain Failure Criteria

刘洪萍

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§ 7-2 平面应力状态分析-解析法
Analysis of plane stress-state

§ 7-3 平面应力状态分析-图解法
Analysis of plane stress-state

§ 7-4 三向应力状态分析
Analysis of three-dimen sional stress-state

§ 7-6 广义胡克定律 Generalized Hook’s law

§ 7-7 复杂应力状态的变形比能
Strain-energy density in general tress-state

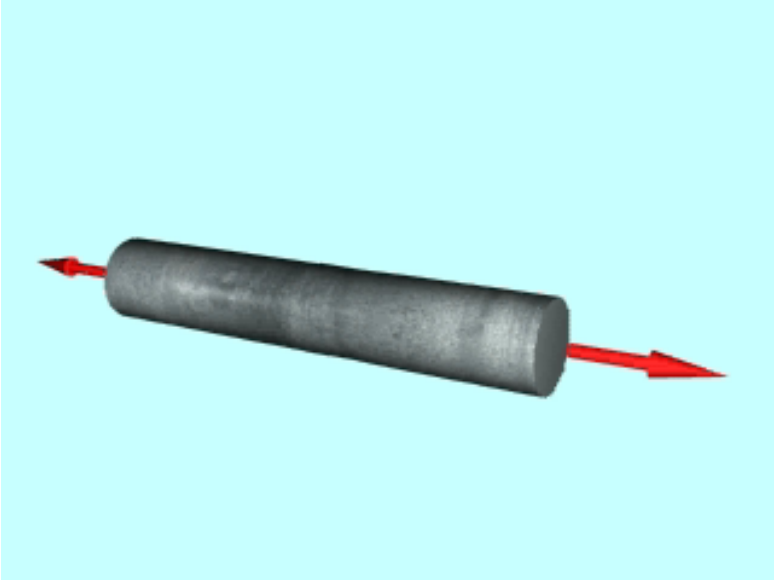
§ 7-8 强度理论 Failure criteria

# §7-1 应力状态概述 (Introduction of stress-state)

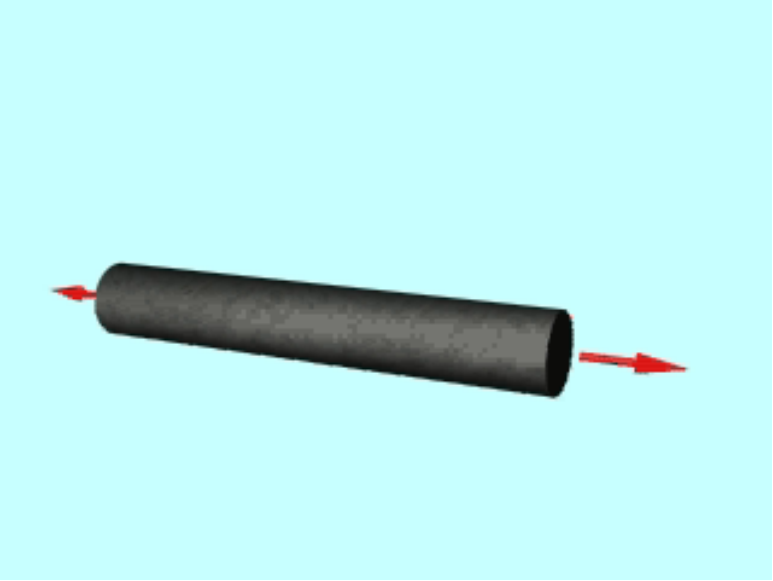
## 一、应力状态的概念 (Concepts of stresses-state)

1、回顾

低碳钢和铸铁的拉伸



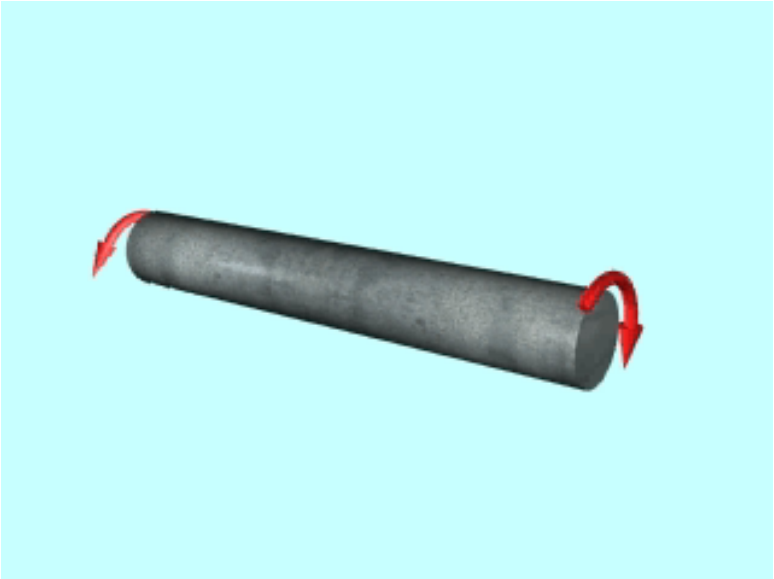
低碳钢



铸铁

塑性材料拉伸时为什么会出现滑移线？

# 低碳钢和铸铁的扭转

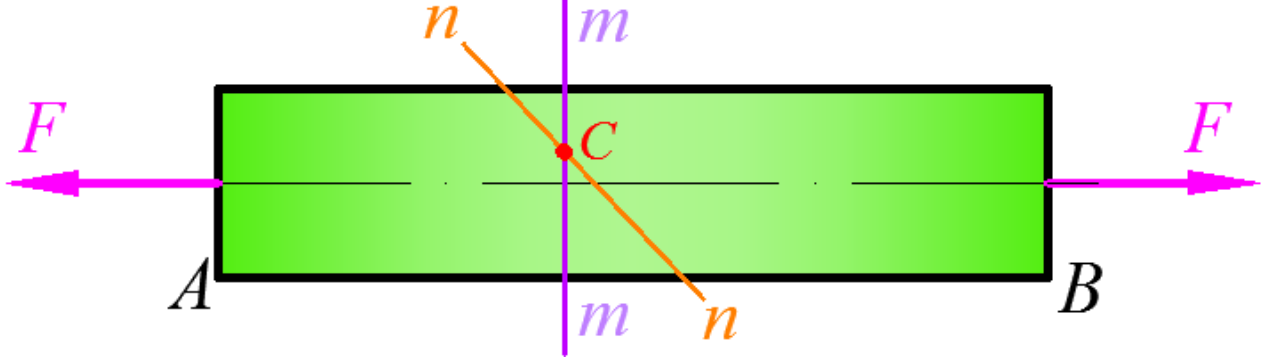


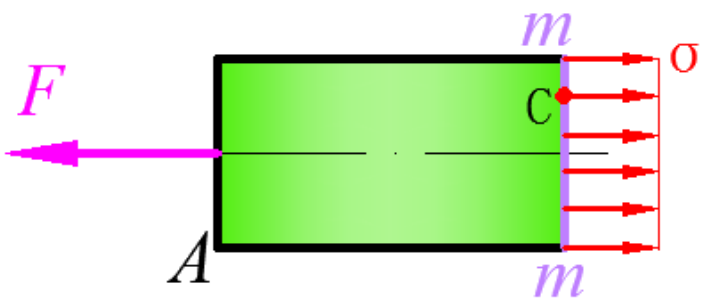
低碳钢

铸铁

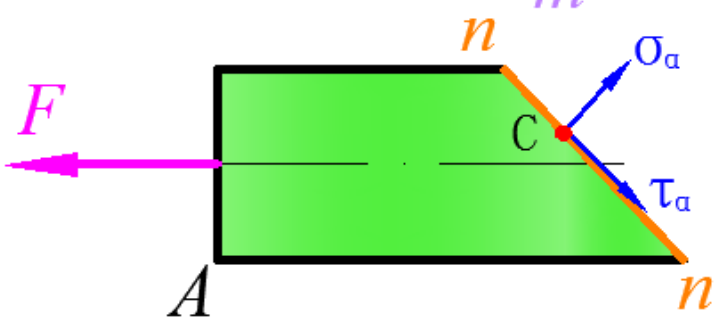
为什么脆性材料扭转时沿45°螺旋面断开？

杆件拉伸



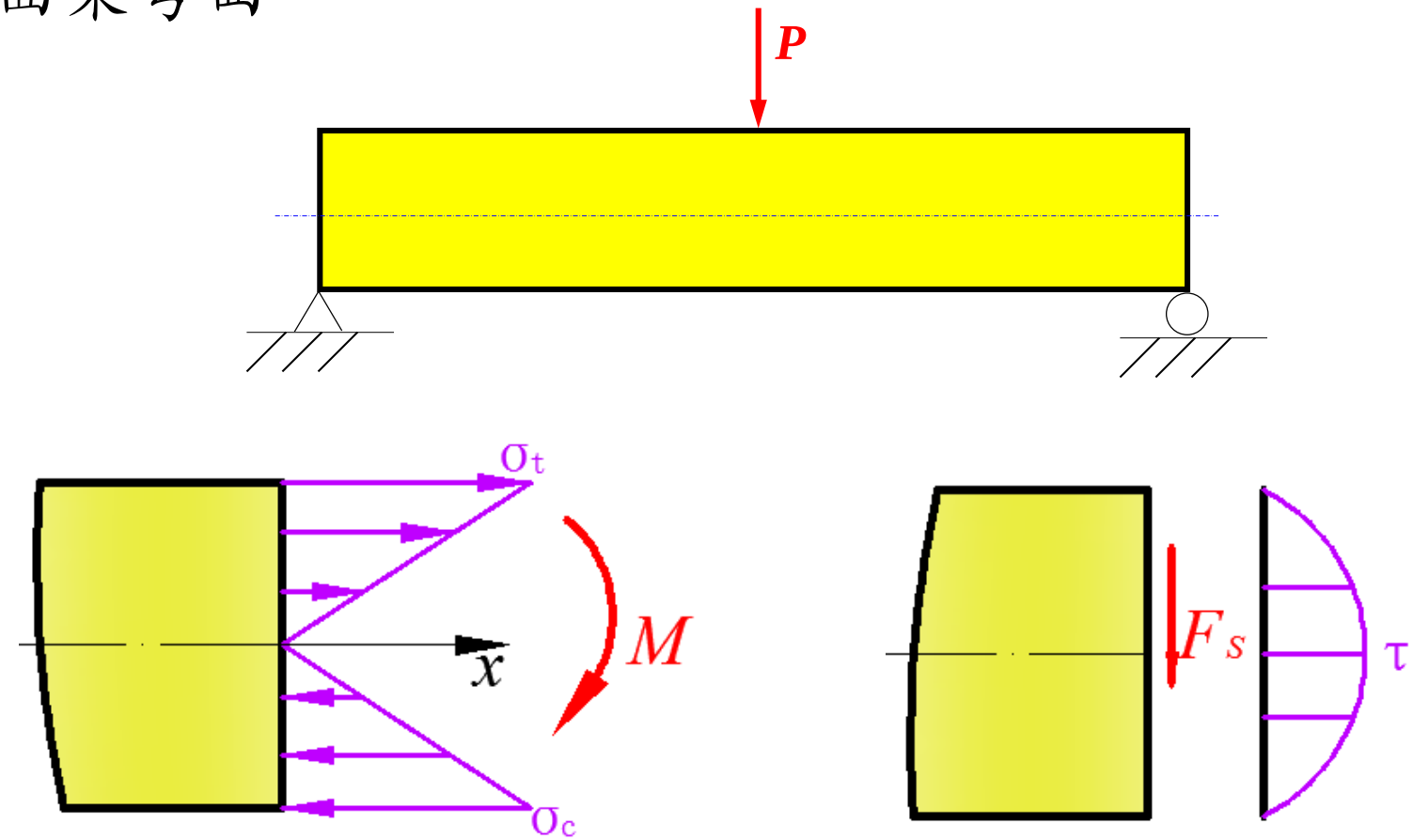

$$\sigma = \frac{F}{A}$$




$$\begin{cases} \sigma_\alpha = \sigma \cos^2 \alpha \\ \tau_\alpha = \dfrac{\sigma}{2} \sin 2\alpha \end{cases}$$

此例表明：即使同一点在不同方位截面上，它的应力也是各不相同的，此即**应力的面的概念**。

矩形截面梁弯曲




横截面上正应力分析和切应力分析的结果表明：同一面上不同点的应力各不相同，此即**应力的点的概念**。

2、重要结论（Important conclusions）

（1）拉中有剪，剪中有拉；

（2）不仅横截面上存在应力，斜截面上也存在应力；

（3）同一面上不同点的应力各不相同；

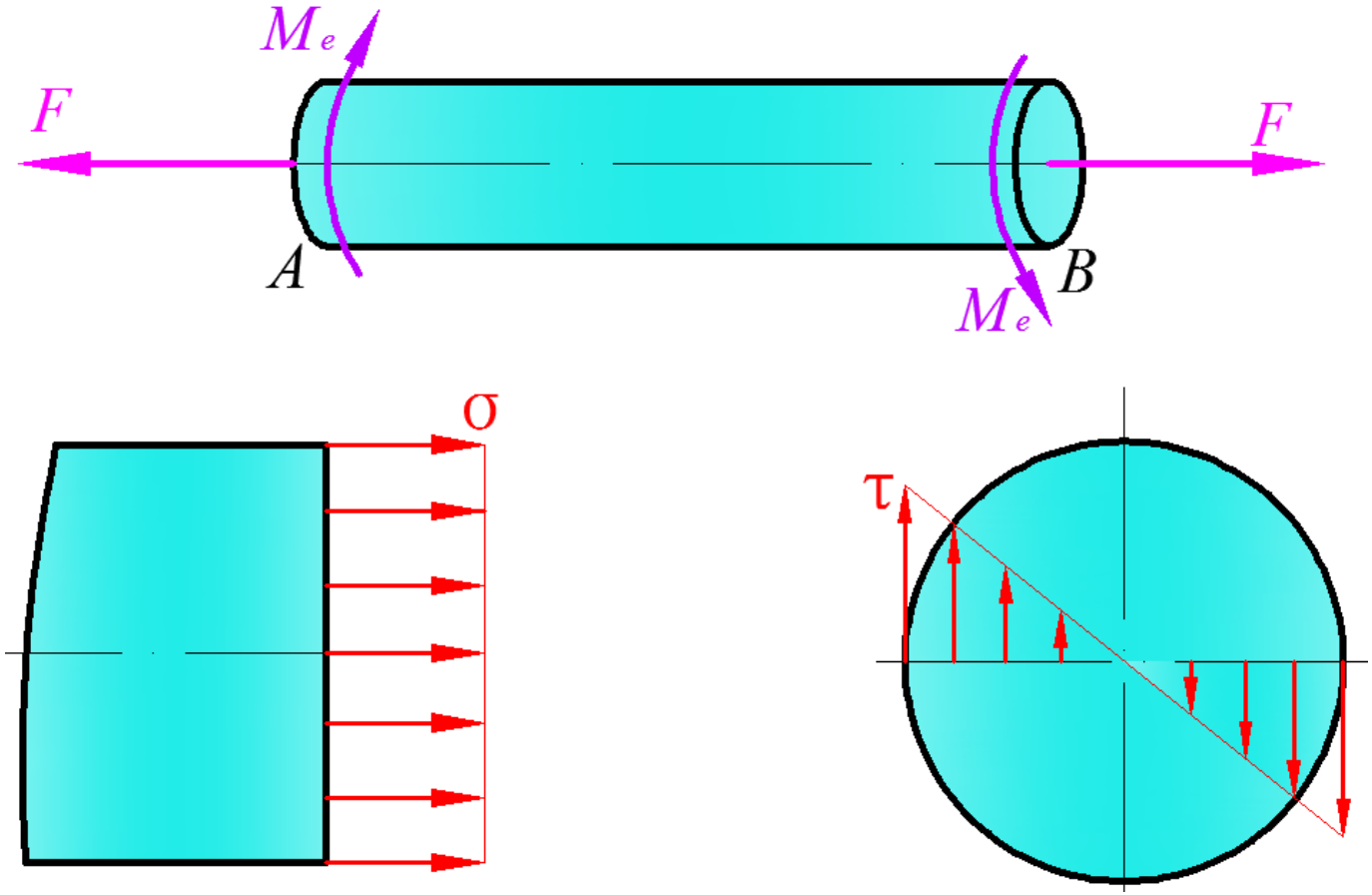
（4）同一点不同方向面上的应力也是各不相同。

**应 力**

哪一个**面**上？
哪一**点**？

哪一**点**？
哪个方向**面**？

# 圆轴拉伸扭转组合变形




强度条件是 $\sigma_{\max} \le [\sigma]$ $\tau_{\max} \le [\tau]$ ?

## 3、一点的应力状态（state of stresses of a given point）

过一点不同方向面上应力的情况，称之为这一点的**应力状态**，亦指该点的**应力全貌**。

## 4、研究目的

找出危险点，确定最大正应力和剪应力数值及所在截面的方位，以便研究构件破坏原因并进行失效分析，为建立**复杂变形**下的**强度条件**提供更广泛的**理论基础**。

## 二、应力状态的研究方法

（The method for investigating the state of stress）

1、**单元体**（Element body）

围绕构件内一点所截取的**微小正六面体**。

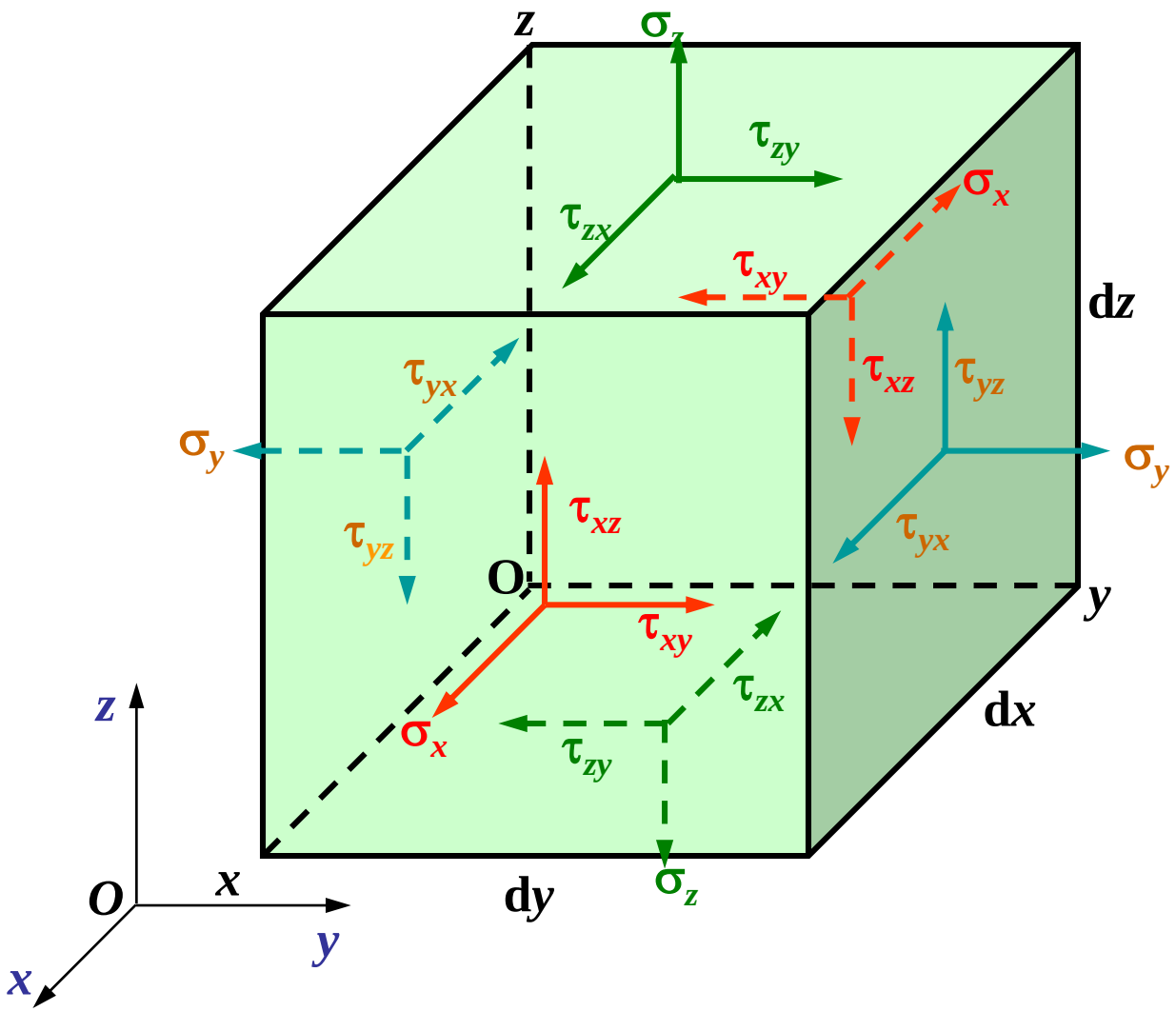
单元体上的应力分量

（1）应力分量的角标规定：第一角标表示应力**作用面**，第二角标表示应力**平行的轴**，两角标相同时，只用一个角标表示。

$$\tau_{yz} = \tau_{zy} \qquad \tau_{zx} = \tau_{xz} \qquad \tau_{xy} = \tau_{yx}$$

（2）面的方位用其**法线方向**表示。

（3）单元体各个面上的应力已知或可求。



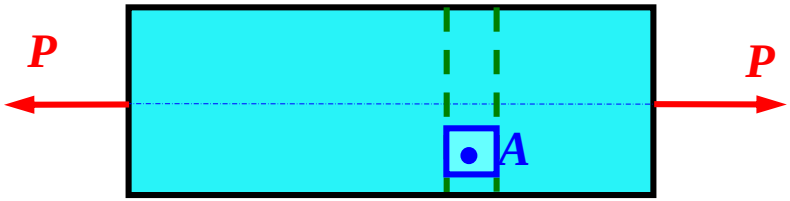

2、单元体特征（Element characteristic）

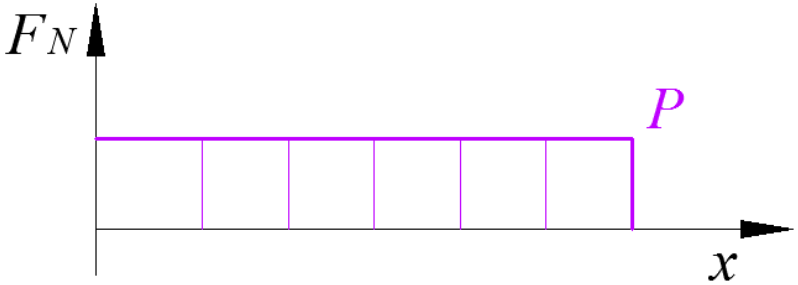
（1）单元体的尺寸无限小，**每个面上应力均匀分布**。

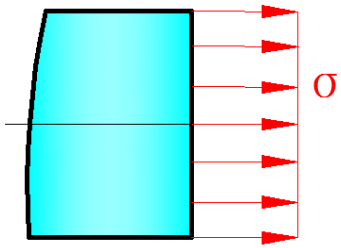
（2）任意**一对平行平面上的应力相等**。

（1）危险点的位置。

（2）用单元体表示危险点的应力状态。

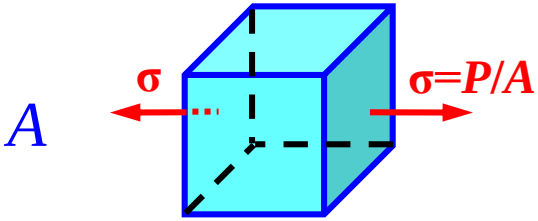


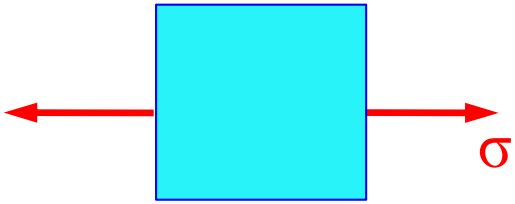




所有一点都为危险点

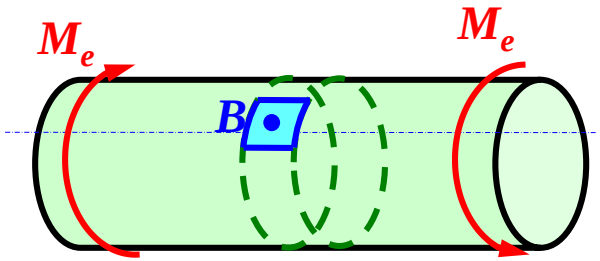
一对横截面，两对纵截面

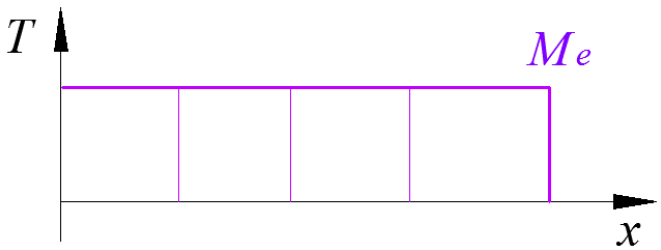


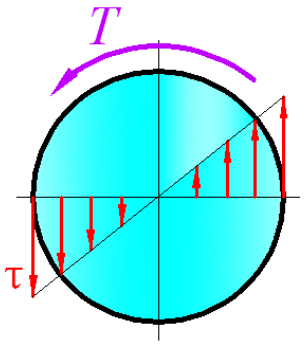



（1）危险点的位置。

（2）用单元体表示危险点的应力状态。

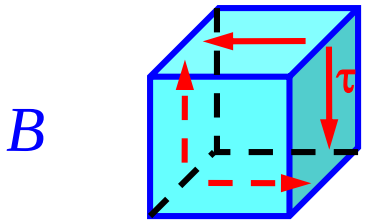


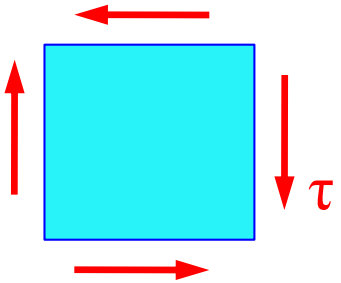




圆轴表面各点为危险点

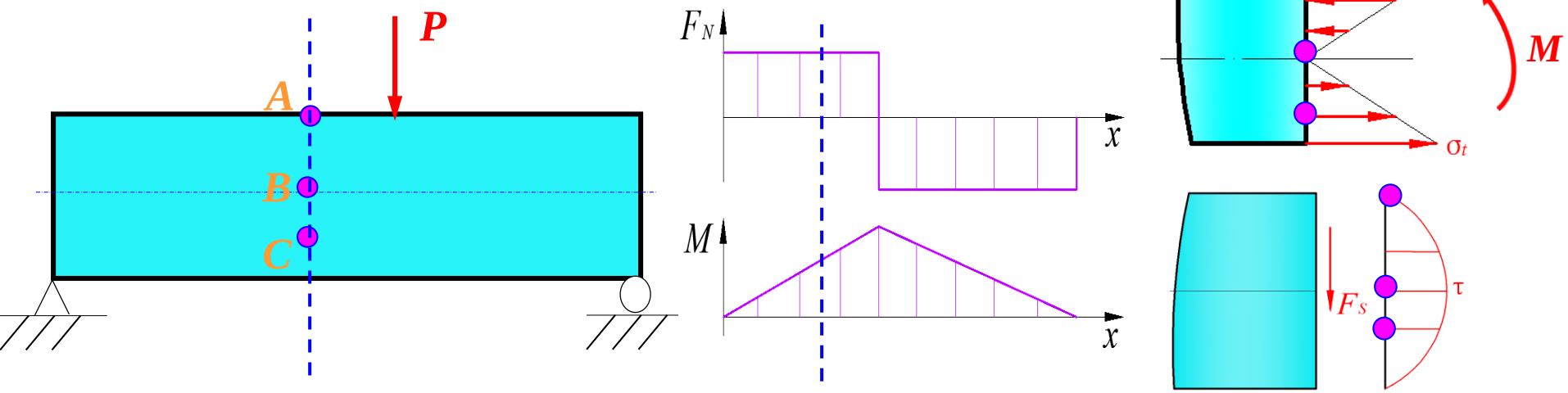
*b*) 横截面，周向面，直径面各一对



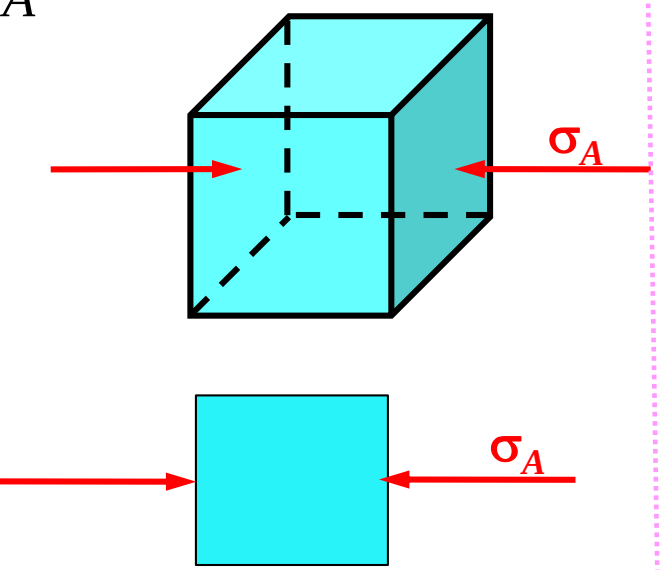

$$\tau = \frac{T}{W_t} = \frac{16T}{\pi d^3}$$



用单元体表示A、B、C三点的应力状态。

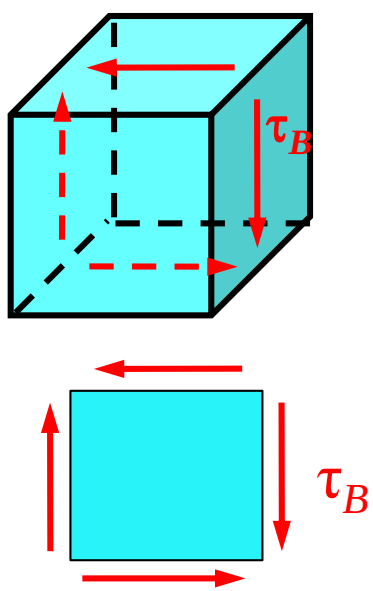



A



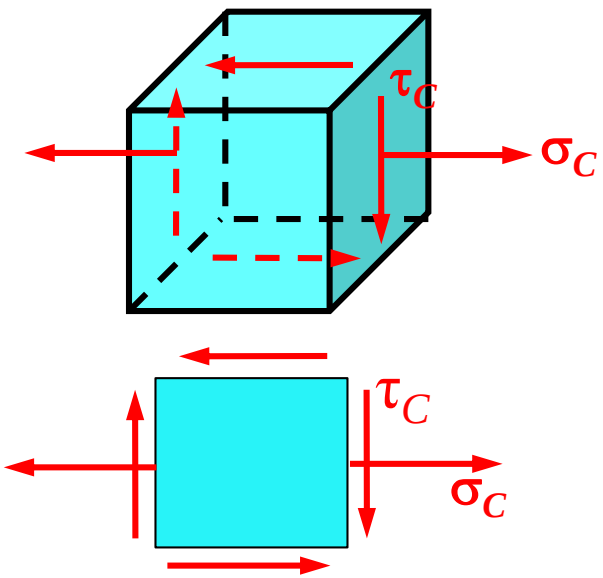

$$\sigma_A = \frac{M}{W_z}$$

B



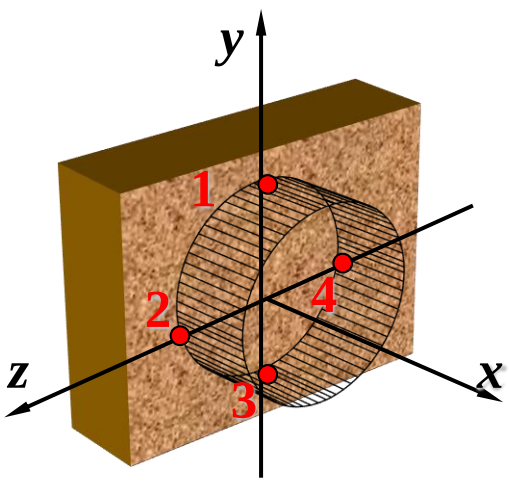

$$\tau_B = \frac{3}{2}\frac{F_S}{A}$$

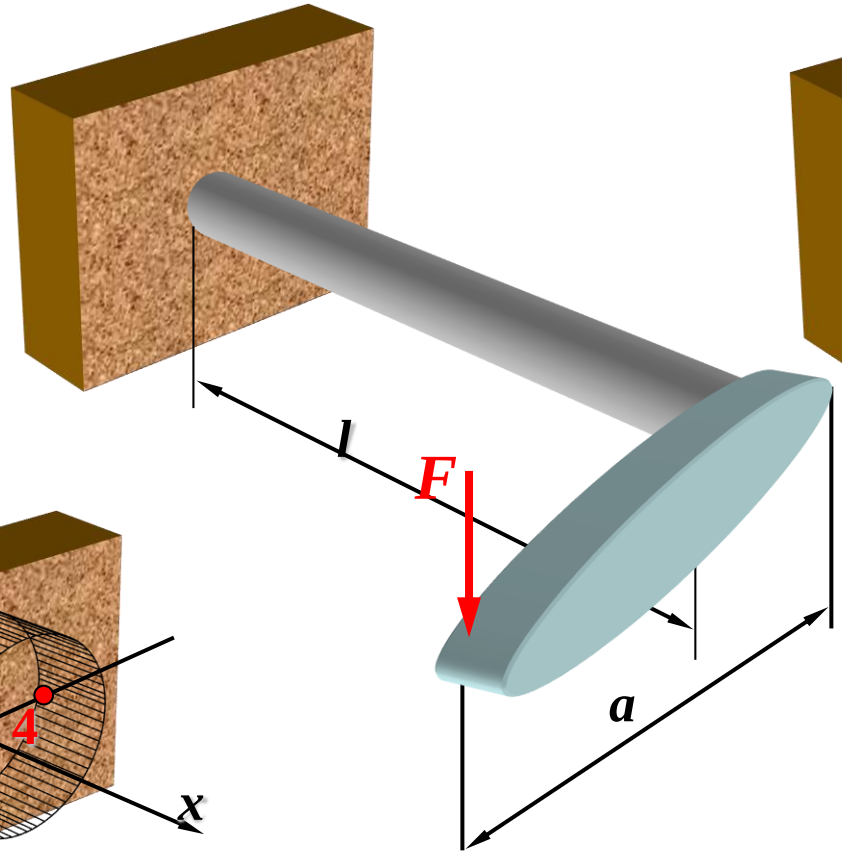
C

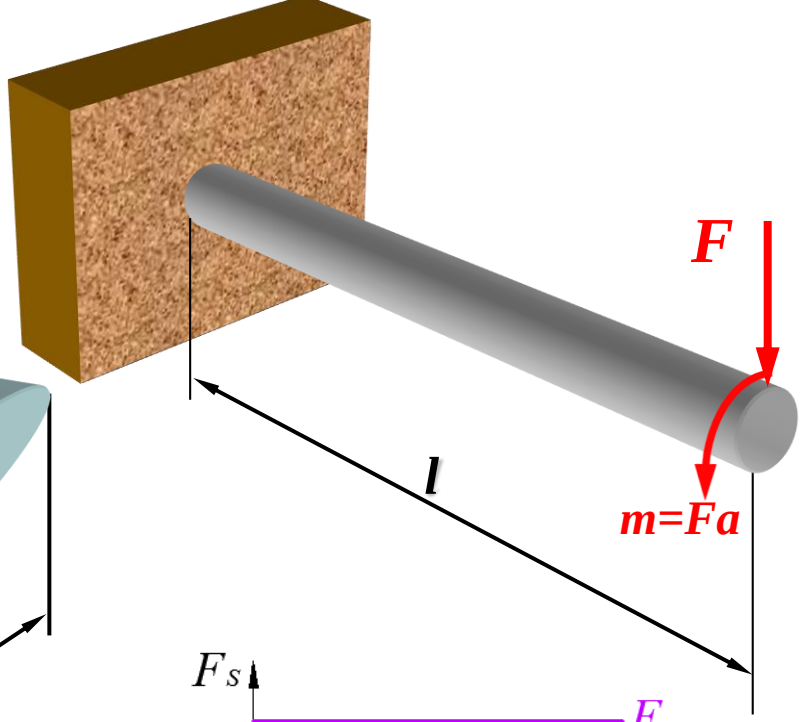


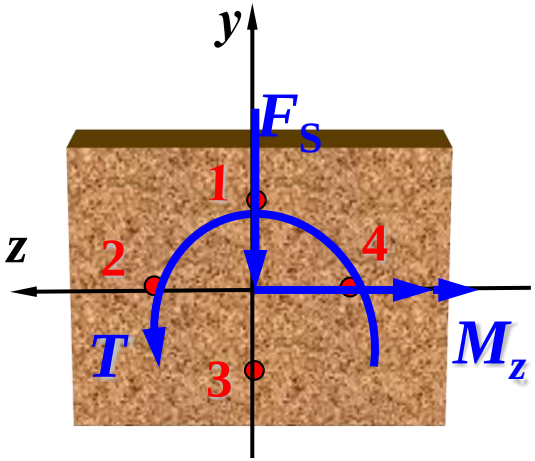

$$\sigma_C = \frac{My}{I_z} \qquad \tau_C = \frac{F_S S_z^*}{bI_z}$$

画出固定端处四点的单元体。



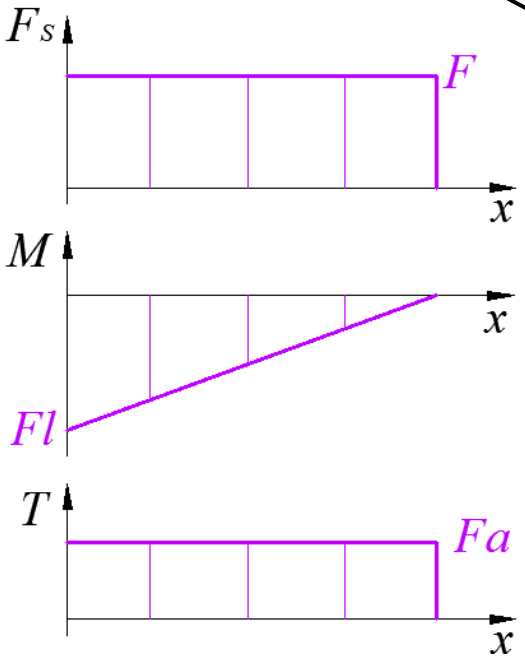



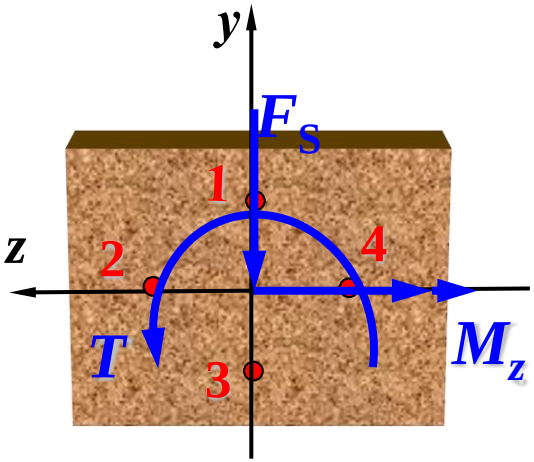



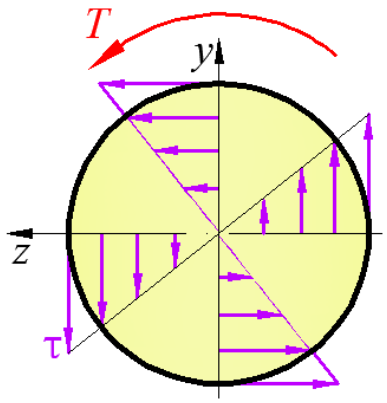

$T=Fa$

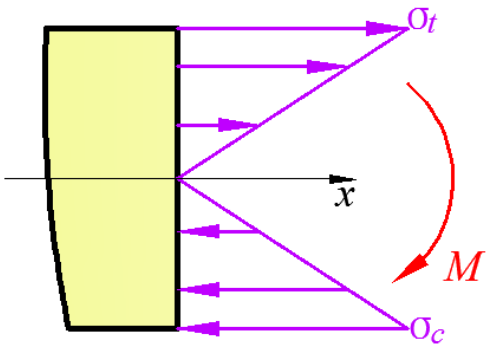
$M=Fl$

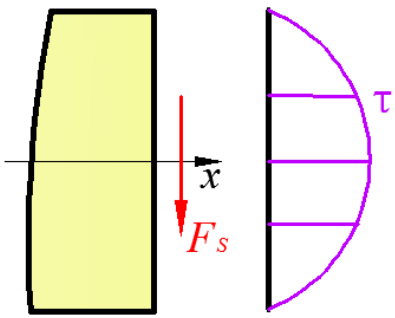
$F_s=F$



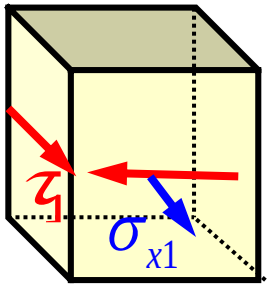



1

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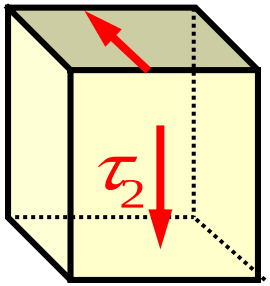


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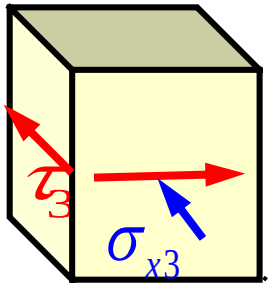


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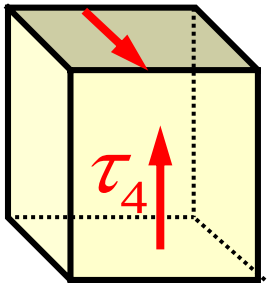



$$\sigma_{x1} = \frac{M_z}{W_z} = \frac{32Fl}{\pi d^3}$$

$$\tau_1 = \frac{T}{W_t} = \frac{16Fa}{\pi d^3}$$

$$\tau_2 = \frac{T}{W_t} + \frac{4F_S}{3A}$$

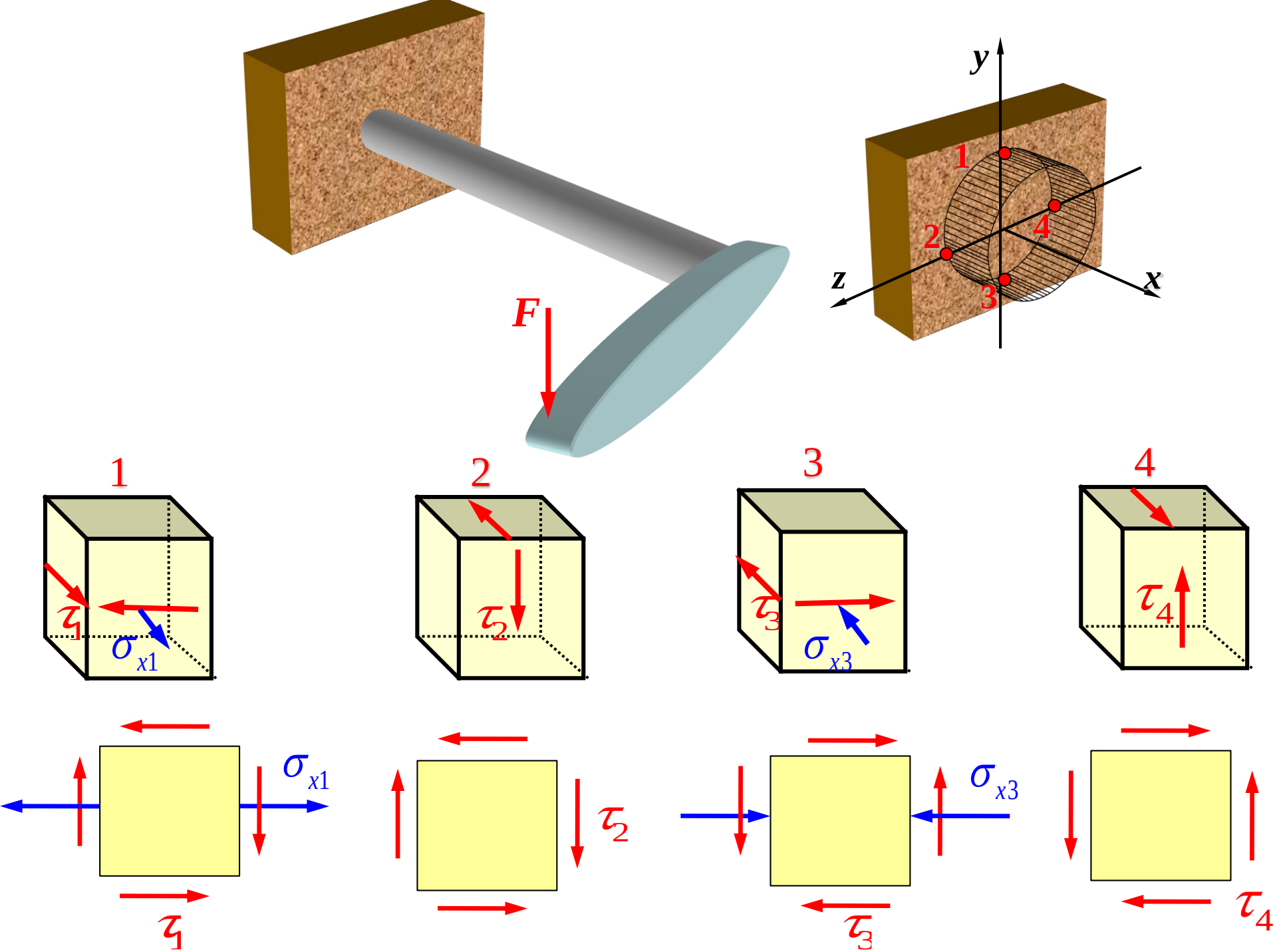
$$= \frac{16Fa}{\pi d^3} + \frac{16F}{3\pi d^2}$$

$$\sigma_{x3} = \frac{M_z}{W_z} = \frac{32Fl}{\pi d^3}$$

$$\tau_3 = \frac{T}{W_t} = \frac{16Fa}{\pi d^3}$$

$$\tau_4 = \frac{T}{W_t} - \frac{4F_S}{3A}$$

$$= \frac{16Fa}{\pi d^3} - \frac{16F}{3\pi d^2}$$


F
y
z
x
1
2
3
4
1
2
3
4
$\tau_1$
$\sigma_{x1}$
$\tau_2$
$\tau_3$
$\sigma_{x3}$
$\tau_4$
$\sigma_{x1}$
$\tau_1$
$\tau_2$
$\sigma_{x3}$
$\tau_3$
$\tau_4$

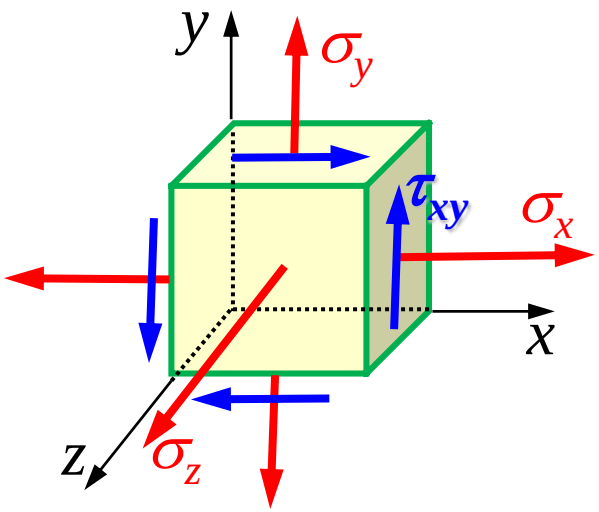
## 三、应力状态的分类

（The classification of stresses-state）

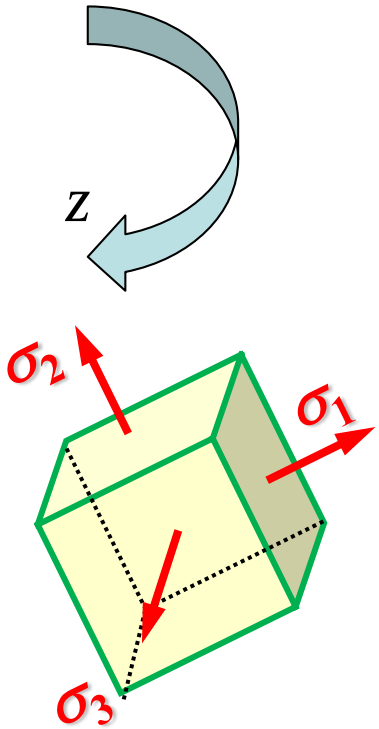
1、主平面（Principal plane）

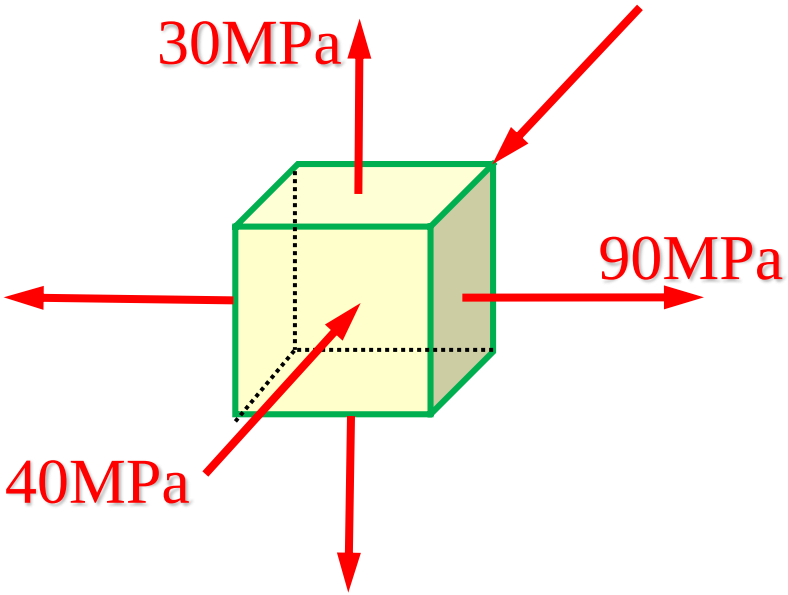
切应力为零的截面。

主应力：主平面上的正应力。




说明：一点处必定存在这样的一个单元体，三个相互垂直的面均为主平面，三个互相垂直的主应力分别记为$\sigma_1$，$\sigma_2$，$\sigma_3$且规定按代数值大小的顺序来排列，即 $\sigma_1 \geq \sigma_2 \geq \sigma_3$ 。






$$\sigma_1 = 90\text{MPa}, \quad \sigma_2 = 30\text{MPa}, \quad \sigma_3 = -40\text{MPa}$$

## 2、应力状态的分类（The classification of stresses-state）

（1）**单向**应力状态

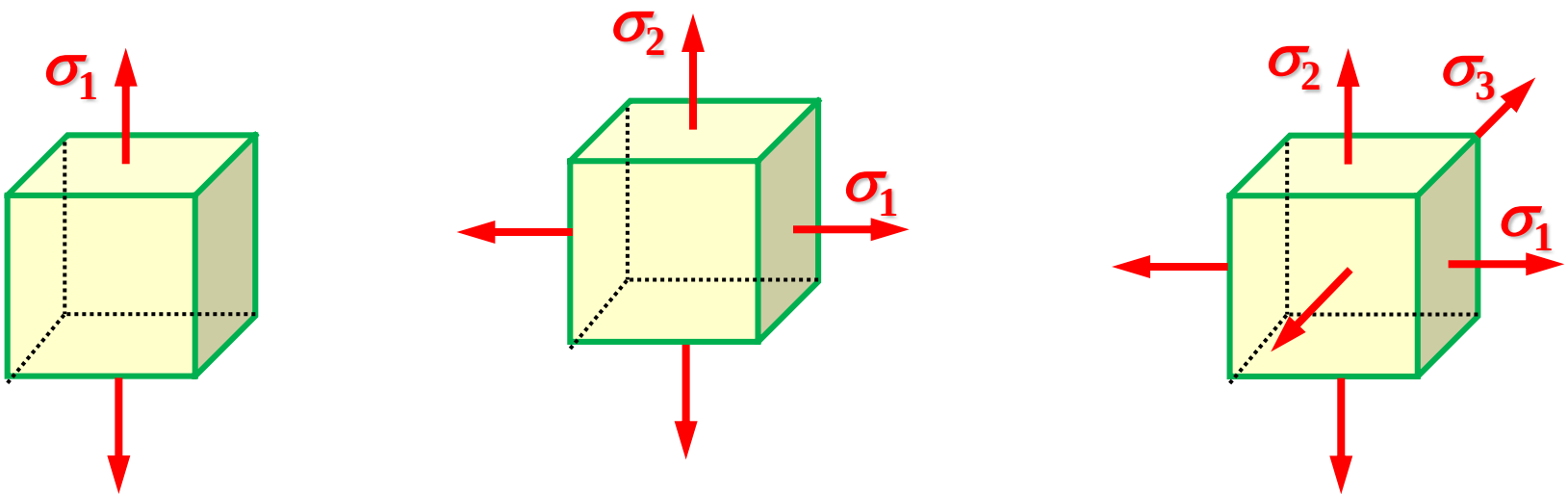
三个主应力 $\sigma_1$、$\sigma_2$、$\sigma_3$ 中只有**一个不等于零**。

（2）**二向**（平面）应力状态

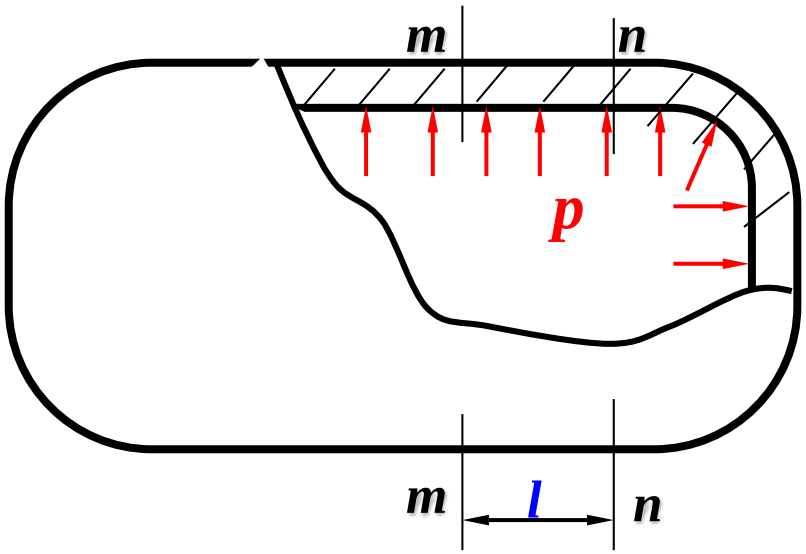
三个主应力$\sigma_1$、$\sigma_2$、$\sigma_3$ 中有**两个不等于零**。

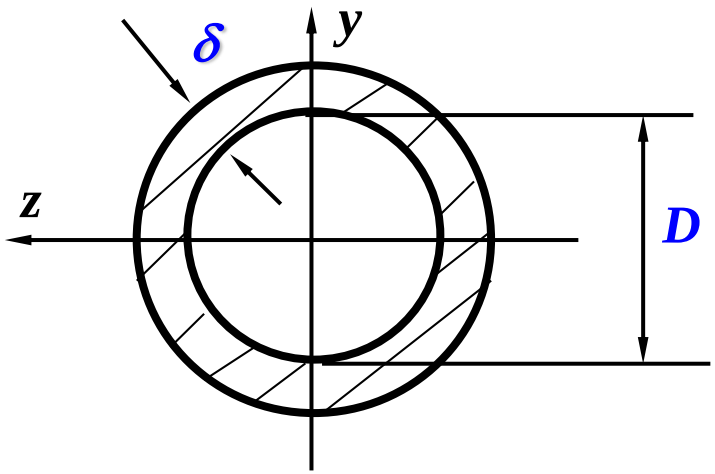
（3）**三向**（空间）应力状态

三个主应力$\sigma_1$、$\sigma_2$、$\sigma_3$ **均不等于零**。

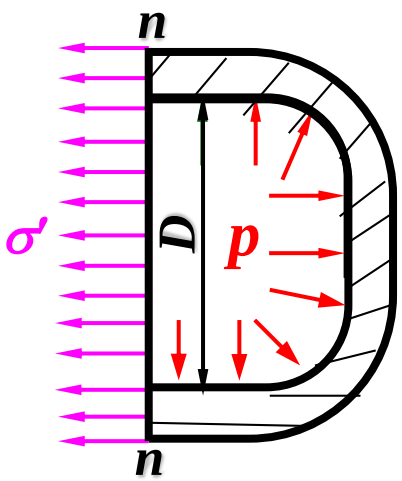


# 二向应力状态：薄壁圆筒受内压时的应力状态




（1）沿圆筒轴线作用于筒底的总压力为$F$



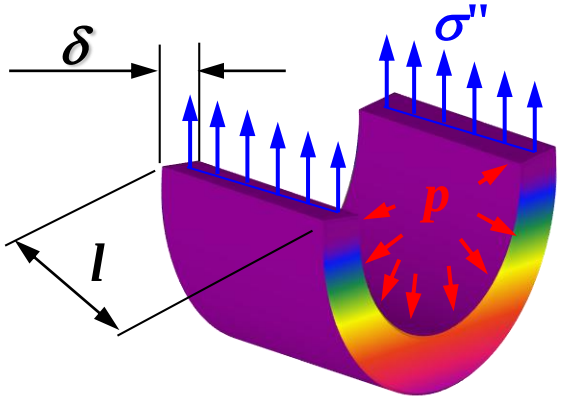

$$F = p \cdot \frac{\pi D^2}{4}$$

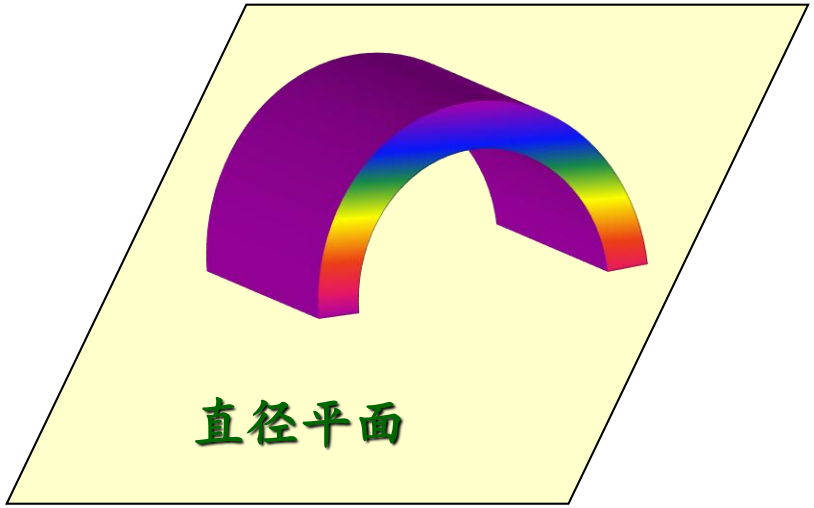
$$A = \pi D \delta$$

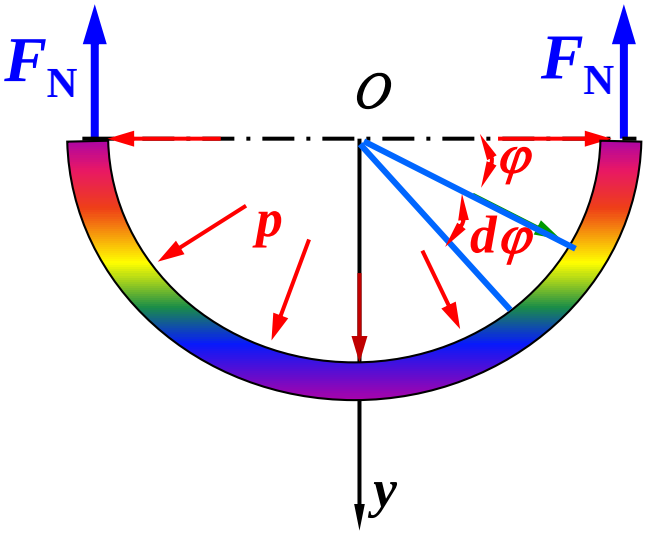
$$\sigma' = \frac{F}{A} = \frac{p \cdot \frac{\pi D^2}{4}}{\pi D \delta} = \frac{pD}{4\delta}$$

轴向应力

（2）假想用一直径平面将圆筒截分为二，并取下半环为研究对象




$$\sum F_y = 0$$

$$2F_N = 2\sigma''\delta l$$

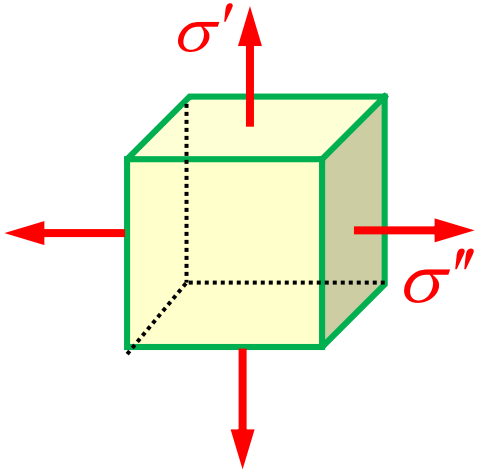
$$\int_0^{\pi} pl \cdot \frac{D}{2} \sin\phi \, d\phi = plD$$

$$2\sigma''\delta l - plD = 0$$

$$\sigma'' = \frac{pD}{2\delta}$$

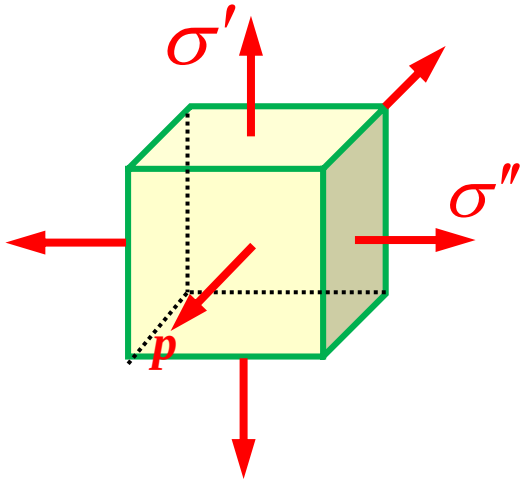
周向应力

薄壁圆筒受内压

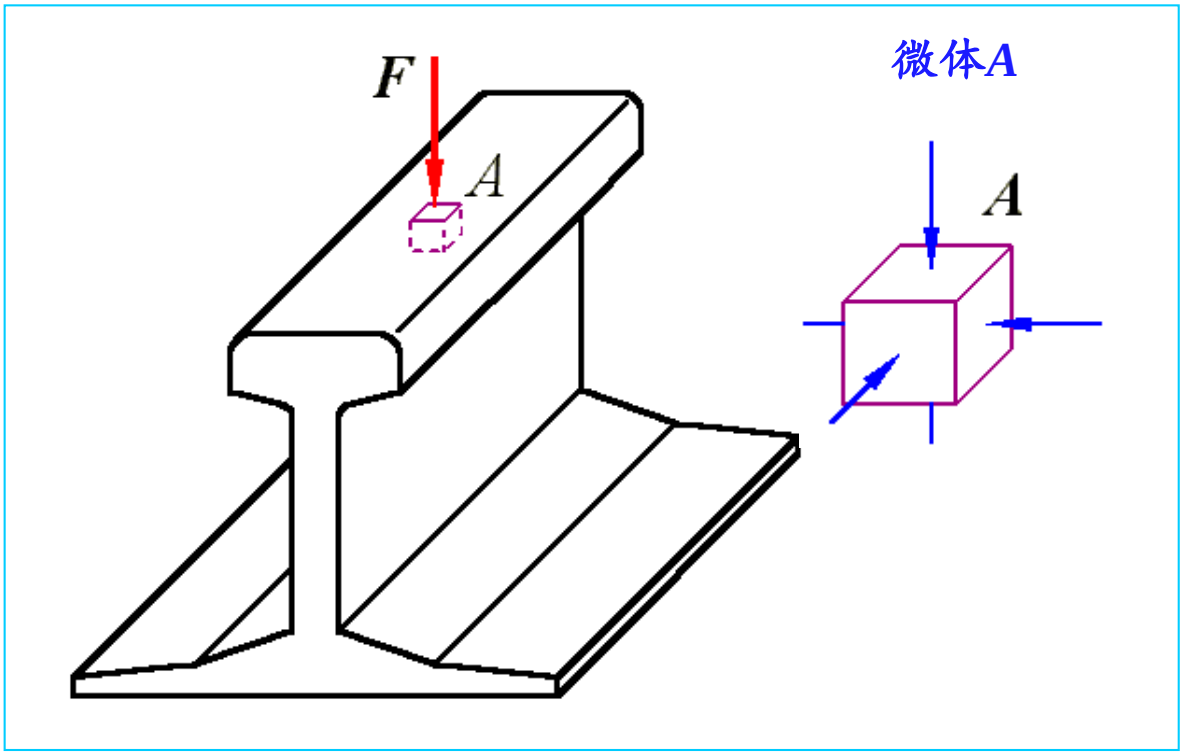



$$\sigma_1 = \sigma'$$
$$\sigma_2 = \sigma''$$
$$\sigma_3 = 0$$

厚壁壁圆筒受内压



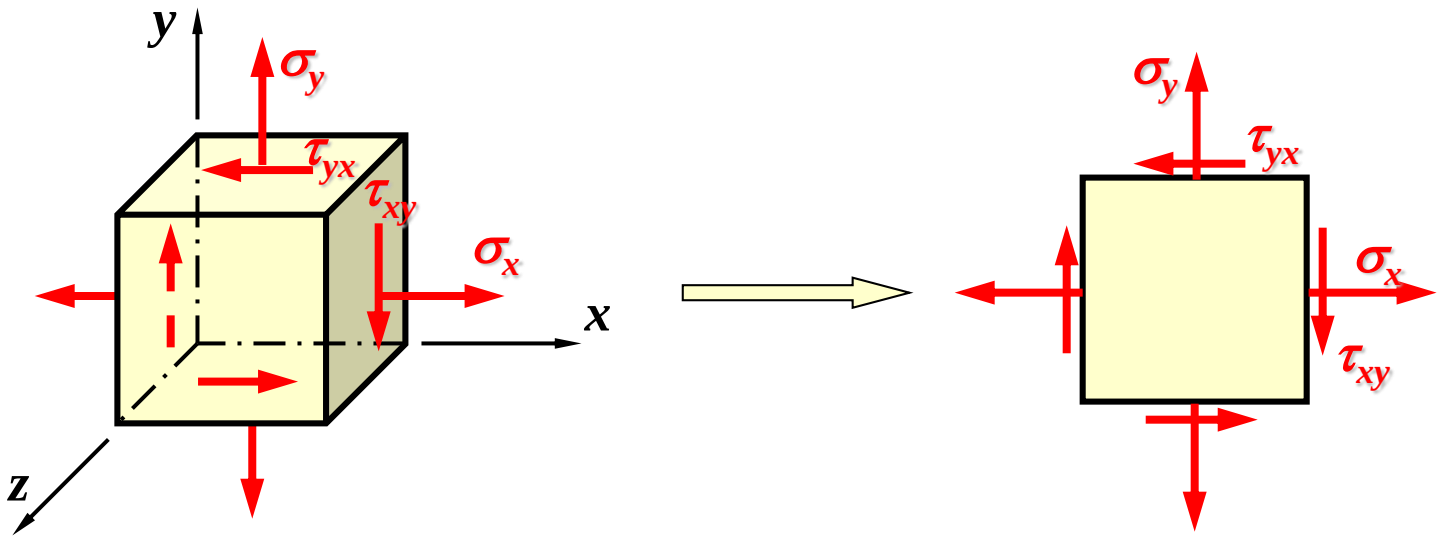

$$\sigma_1 = \sigma'$$
$$\sigma_2 = \sigma''$$
$$\sigma_3 = -p$$




钢轨与车轮的接触处

# §7-3 二向应力状态分析-解析法

（Analysis of plane stress-state）



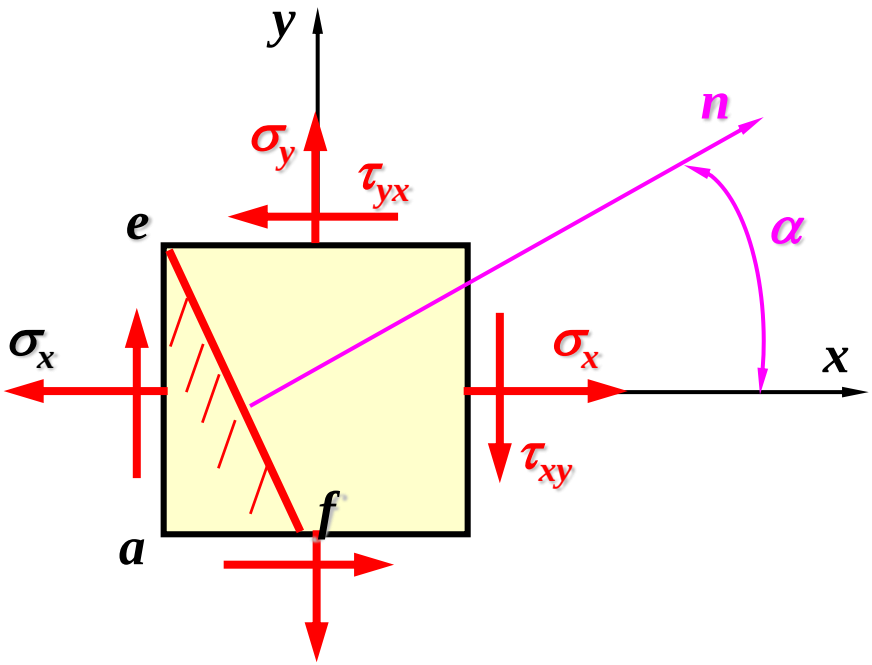

二向应力状态的普遍形式如图所示。

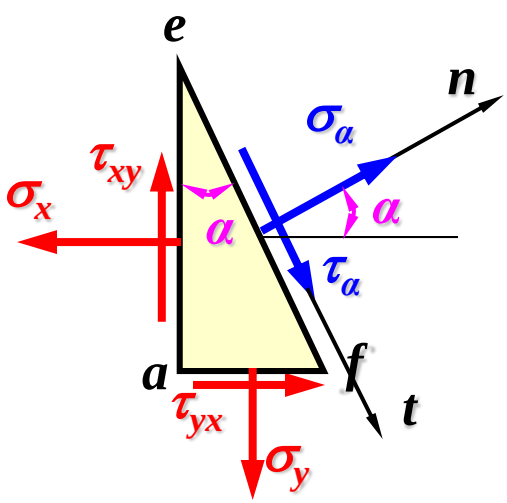
单元体上有$\sigma_x$，$\tau_{xy}$和$\sigma_y$，$\tau_{yx}$。

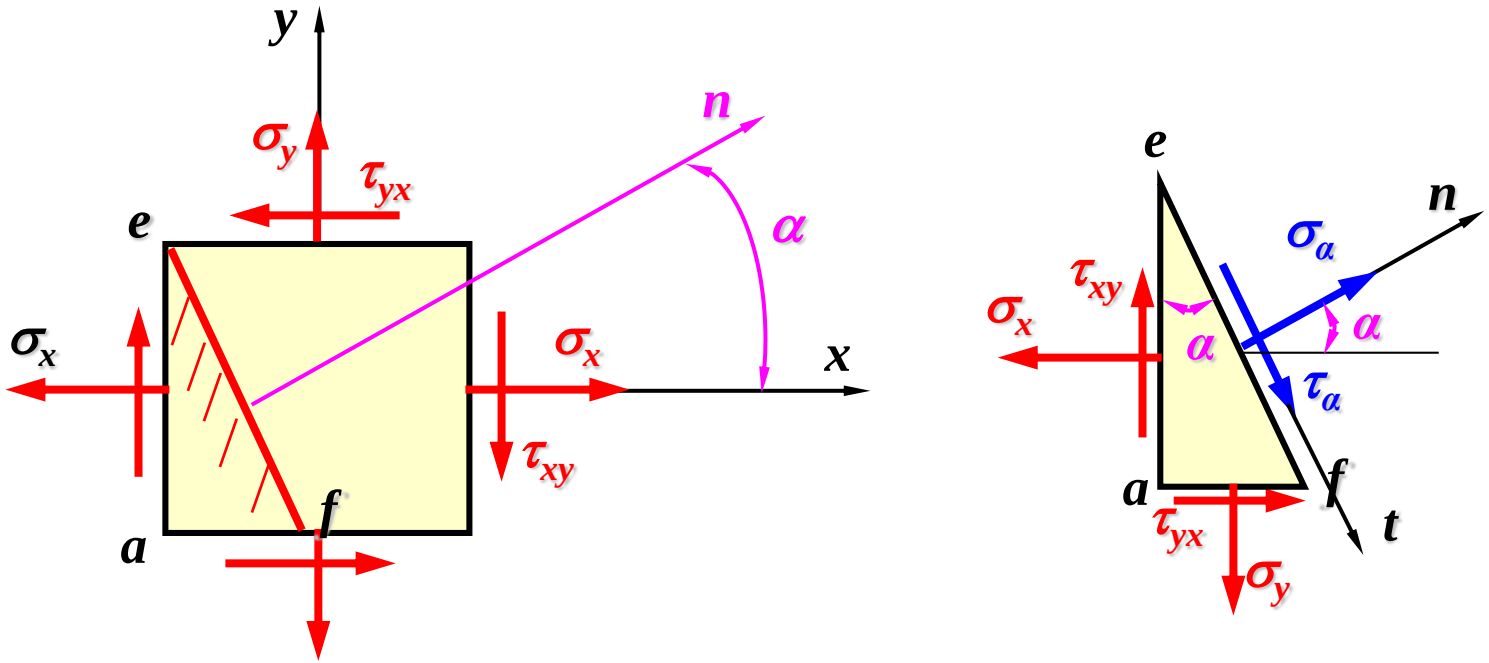
# 一、斜截面上的应力（Stresses on an oblique section）

## 1、截面法（Section method）

假想地沿斜截面*e-f*将单元体截开，留下左边部分的单体元 *eaf* 作为研究对象。






2、符号的确定（Sign convention）

（1）α：由*x*轴转到外法线*n*，**逆时针**转为正。

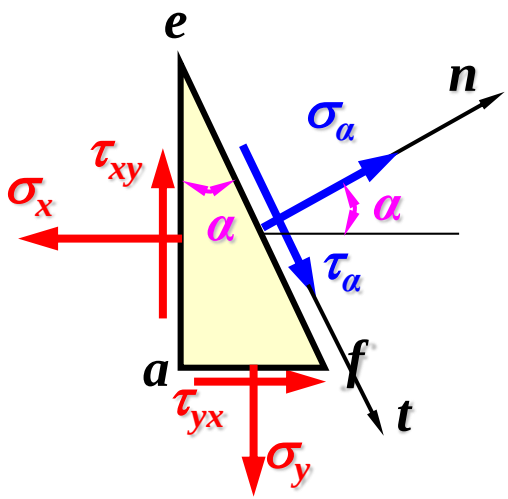
（2）σ：正应力仍规定**拉应力**为正。

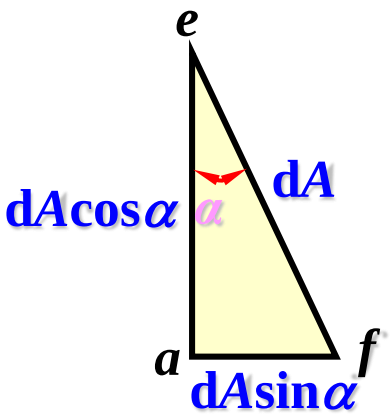
（3）τ：切应力对单元体内任一点取矩，**顺时针**转为正。

## 3、任意斜截面上的应力

是**力平衡**，而

不是**应力平衡**。




$$\sum F_n = 0 \quad \sigma_\alpha \mathrm{d}A + (\tau_{xy}\mathrm{d}A\cos\alpha)\sin\alpha - (\sigma_x \mathrm{d}A\cos\alpha)\cos\alpha + (\tau_{yx}\mathrm{d}A\sin\alpha)\cos\alpha - (\sigma_y \mathrm{d}A\sin\alpha)\sin\alpha = 0$$

$$\sum F_t = 0 \quad \tau_\alpha \mathrm{d}A - (\tau_{xy}\mathrm{d}A\cos\alpha)\cos\alpha - (\sigma_x \mathrm{d}A\cos\alpha)\sin\alpha + (\tau_{yx}\mathrm{d}A\sin\alpha)\sin\alpha + (\sigma_y \mathrm{d}A\sin\alpha)\cos\alpha = 0$$

$$\tau_{xy} = \tau_{yx}$$

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

不难看出$\sigma_\alpha + \sigma_{\alpha+90^\circ} = \sigma_x + \sigma_y$，$\tau_{\alpha+90^o} = -\tau_\alpha$ 。

即两相互垂直面上的正应力之和保持一个常数，切应力互等。

各项参数代入公式时应为代数值。

## 二、最大正应力及方位（Maximum normal stress and it's direction）

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

### 1、最大正应力的方位

令 $$\frac{\mathrm{d}\sigma_\alpha}{\mathrm{d}\alpha} = -2\left[\frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha\right] = 0$$

$$\boxed{\tan 2\alpha_0 = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y}} \longrightarrow \begin{cases} \alpha_0 \\ \alpha_0 + 90^\circ \end{cases}$$

$\alpha_0$ 和$\alpha_0$+90° 确定两个互相垂直的平面，一个是最大正应力所在的平面，另一个是最小正应力所在的平面。

$\alpha = \alpha_0$ 时，切应力为零，所以这是主平面。

## 2、最大正应力

将 $\alpha_0$和 $\alpha_0$+90° 代入公式

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\boxed{\begin{cases} \sigma_{\max} \\ \sigma_{\min} \end{cases} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}}$$

**$\sigma_{max}$、$\sigma_{min}$、0**是该点处的三个**主应力**。

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

$\sigma_{max}$、$\sigma_{min}$、0三个排序决定$\sigma_1$、$\sigma_2$、$\sigma_3$。

$\sigma_x \geq \sigma_y$时，$\left\{ |\alpha_0| \quad |\alpha_0 + 90^o| \right\}_{\min}$ ⟹ $\sigma_{max}$所在平面。

## 二、最大切应力及方位

(Maximum shearing stress and it's direction)

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

### 1、最大切应力的方位

令 $$\frac{\mathrm{d}\tau_\alpha}{\mathrm{d}\alpha} = 2\left[\frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha\right] = 0$$

$$\tan 2\alpha_1 = \frac{\sigma_x - \sigma_y}{2\tau_{xy}} \longrightarrow \begin{cases} \alpha_1 \\ \alpha_1 + 90^\circ \end{cases}$$

$\alpha_1$ 和 $\alpha_1+90^\circ$ 确定两个互相垂直的平面，一个是最大切应力所在的平面，另一个是最小切应力所在的平面。

## 2、最大切应力

将 $\alpha_1$和 $\alpha_1$+90° 代入公式

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

$$\begin{cases} \tau_{\max} \\ \tau_{\min} \end{cases} = \pm\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\boxed{\begin{cases} \tau_{\max} \\ \tau_{\min} \end{cases} = \pm\frac{1}{2}\left(\sigma_{\max} - \sigma_{\min}\right)}$$

$$\tan 2\alpha_1 = -\frac{1}{\tan 2\alpha_0}$$

$$\boxed{\alpha_1 = \alpha_0 + \frac{\pi}{4}}$$

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

$$\begin{cases} \sigma_{\max} \\ \sigma_{\min} \end{cases} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

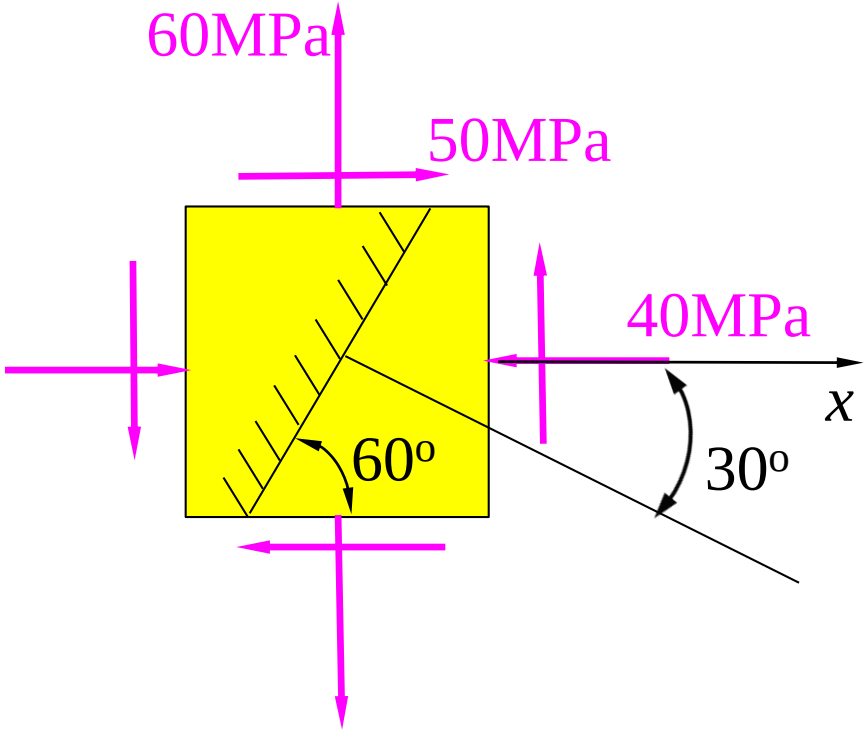
$$\tan 2\alpha_0 = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\alpha_1 = \alpha_0 + \frac{\pi}{4}$$

$$\begin{matrix} \tau_{\max} \\ \tau_{\min} \end{matrix} = \pm\frac{1}{2}(\sigma_{\max} - \sigma_{\min})$$

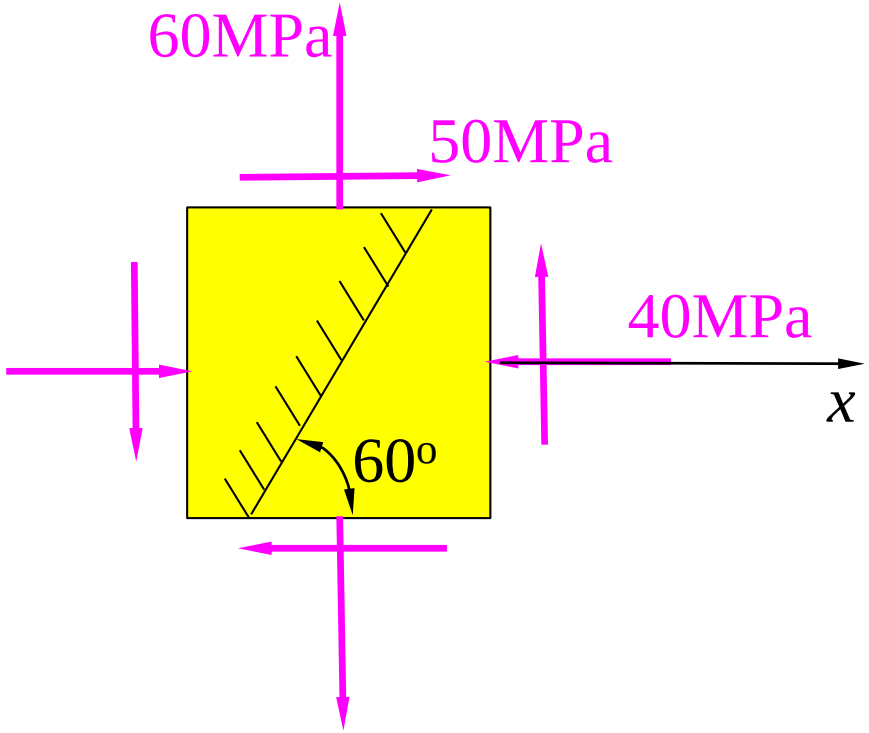
$\sigma_x \geq \sigma_y$ 时，$\left\{ |\alpha_0| \quad |\alpha_0 + 90^o| \right\}_{\min} \Longrightarrow$ $\sigma_{max}$所在平面。

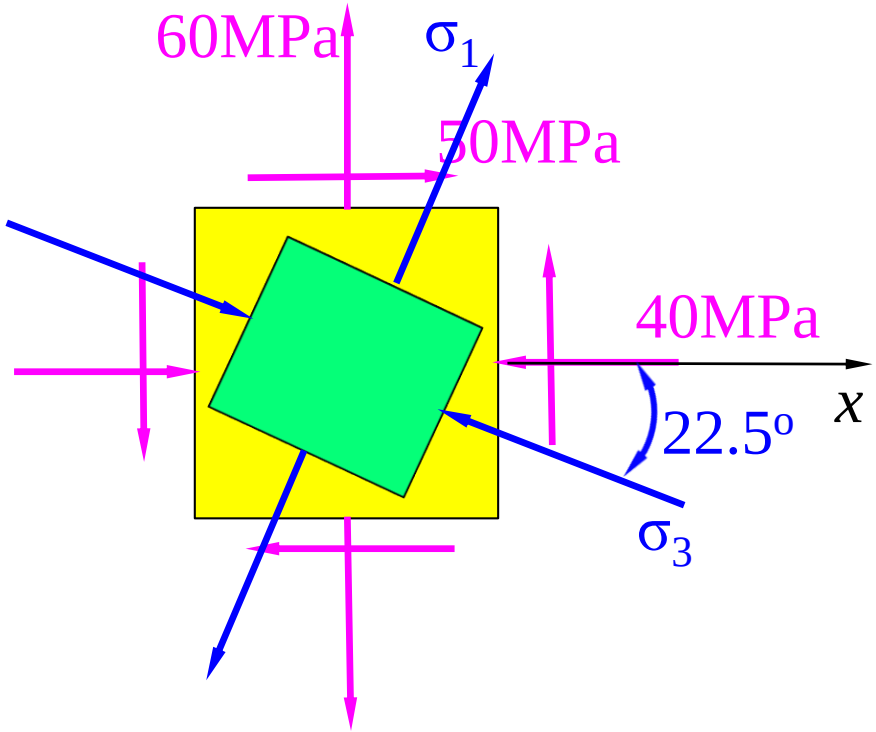
**例题7-1** 单元体的应力状态如图。求（1）斜截面上的应力；（2）主应力大小，主平面位置；在单元体上绘出主平面位置及主应力方向；（3）图示平面内最大切应力。




$\sigma_x =$ $\sigma_y =$

$\tau_{xy} =$ $\alpha =$




$$\sigma_{-30} = -58.3\text{MPa}, \quad \tau_{-30} = 18.3\text{MPa}$$

$$\sigma_1 = 80.7\text{MPa}, \quad \sigma_2 = 0, \quad \sigma_3 = -60.7\text{MPa}$$

$$\alpha_0 = -22.5^0, \quad \tau_{\max} = 70.7\text{MPa}$$

解: $\sigma_x = -40\text{MPa}$ $\sigma_y = 60\text{MPa}$

$\tau_{xy} = -50\text{MPa}$ $\alpha = -30^{\circ}$

（1）斜截面上的应力

$$\sigma_{-30^\circ} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$= \frac{-40+60}{2} + \frac{-40-60}{2}\cos(-60^\circ) - (-50)\sin(-60^\circ)$$

$$= -58.3\text{MPa}$$

$$\tau_{-30^\circ} = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha = \frac{-40-60}{2}\sin(-60^\circ) +$$

$$(-50)\cos(-60^\circ) = 18.3\text{MPa}$$

（2） 主应力大小和主平面的方位

$$\tan 2\alpha_0 = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y} = -\frac{2\times(-50)}{-40-60} = -1$$

$$2\alpha_0 = \begin{matrix} -45^\circ \\ 135^\circ \end{matrix} \qquad \alpha_0 = \begin{matrix} -22.5^\circ \\ 67.5^\circ \end{matrix}$$

因为$\sigma_x < \sigma_y$，所以$\alpha_0$= -22.5° 与$\sigma_{min}$所在平面对应。

$$\begin{cases} \sigma_{max} \\ \sigma_{min} \end{cases} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} = \begin{cases} 80.7\text{MPa} \\ -60.7\text{MPa} \end{cases}$$

$$\sigma_1 = 80.7\text{MP} \qquad \sigma_2 = 0 \qquad \sigma_3 = -60.7\text{MPa}$$

（3） 最大切应力 $\tau_{max} = \frac{1}{2}(\sigma_{max} - \sigma_{min}) = 70.7\text{MPa}$

# 解析法

优点：计算精确

缺点：公式太多

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

$$\begin{cases} \sigma_{\max} \\ \sigma_{\min} \end{cases} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\alpha_0 = -\frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\alpha_1 = \alpha_0 + \frac{\pi}{4}$$

$$\begin{matrix} \tau_{\max} \\ \tau_{\min} \end{matrix} = \pm\frac{1}{2}(\sigma_{\max} - \sigma_{\min})$$

$\sigma_x \geq \sigma_y$ 时，$\left\{ |\alpha_0| \quad |\alpha_0 + 90^o| \right\}_{\min} \Longrightarrow$ $\sigma_{max}$所在平面。

# §7-3 二向应力状态分析-图解法

(Analysis of plane stress-state with graphical means)

## 一、应力圆

$$\sigma_\alpha = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

$$\sigma_\alpha - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2}\cos 2\alpha - \tau_{xy}\sin 2\alpha$$

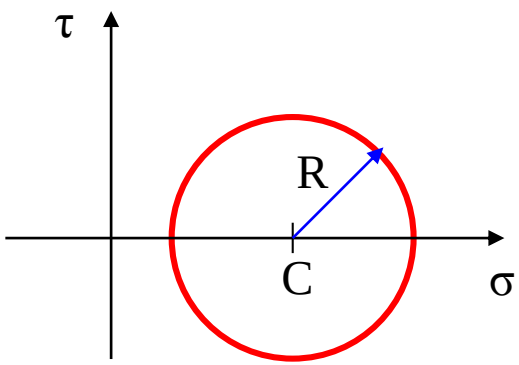
$$\tau_\alpha = \frac{\sigma_x - \sigma_y}{2}\sin 2\alpha + \tau_{xy}\cos 2\alpha$$

$$\left(\sigma_\alpha - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_\alpha^2 = \left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{xy}^2$$

$$\left(\sigma_\alpha - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_\alpha^2 = \left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{xy}^2$$

因为$\sigma_x$，$\sigma_y$，$\tau_{xy}$皆为已知量，所以上式是一个以$\sigma_\alpha$，$\tau_\alpha$为变量的**圆周方程**。当斜截面随方位角$\alpha$变化时，其上的应力$\sigma_\alpha$，$\tau_\alpha$在$\sigma$-$\tau$直角坐标系内的轨迹是一个圆。

$$\left(\sigma_\alpha - \frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_\alpha^2 = \left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{xy}^2$$

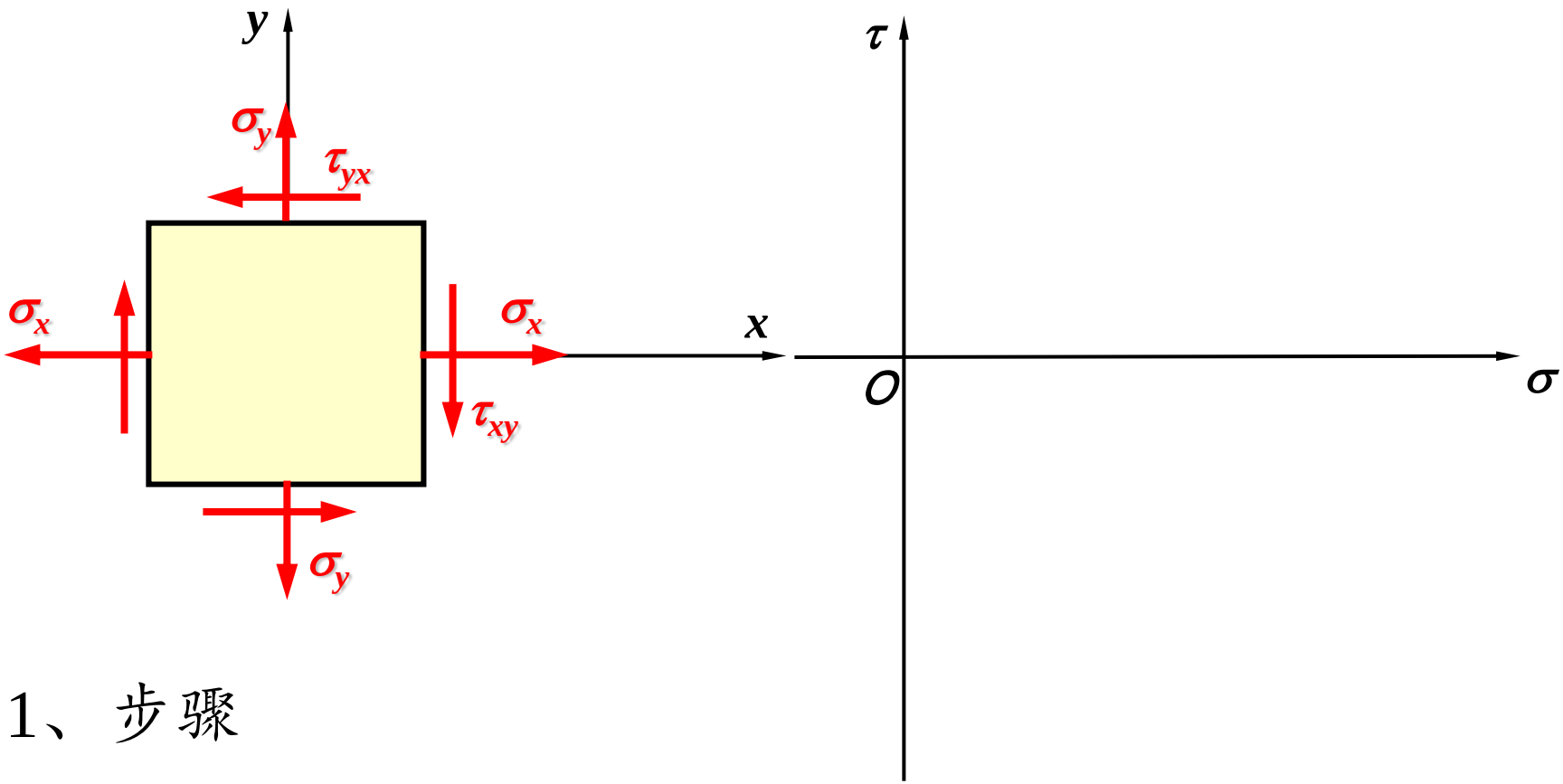



1、圆心的坐标 $C\left(\frac{\sigma_x + \sigma_y}{2}, 0\right)$

2、圆的半径 $R = \sqrt{\left(\frac{\sigma_x + \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

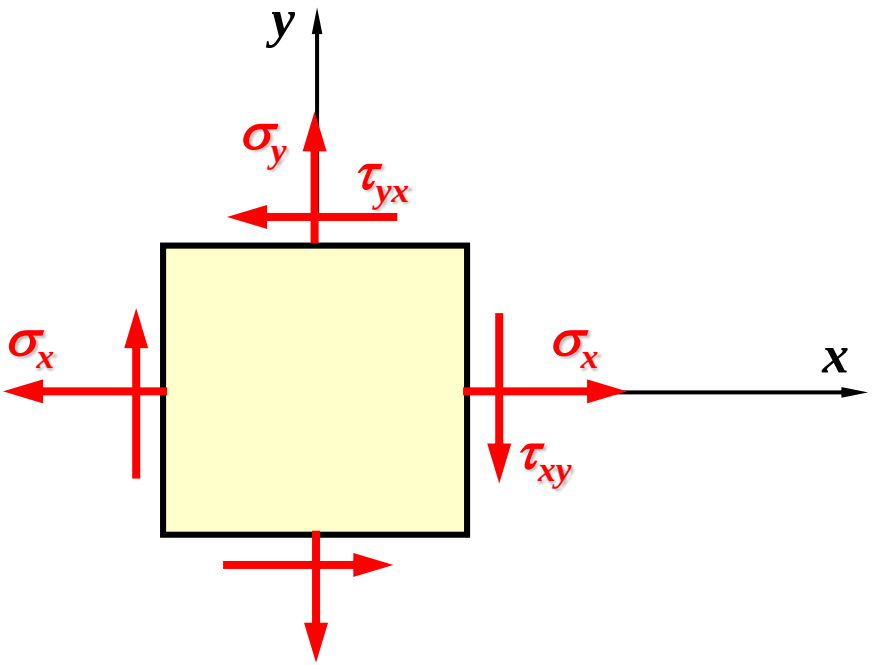
此圆习惯上称为**应力圆**。

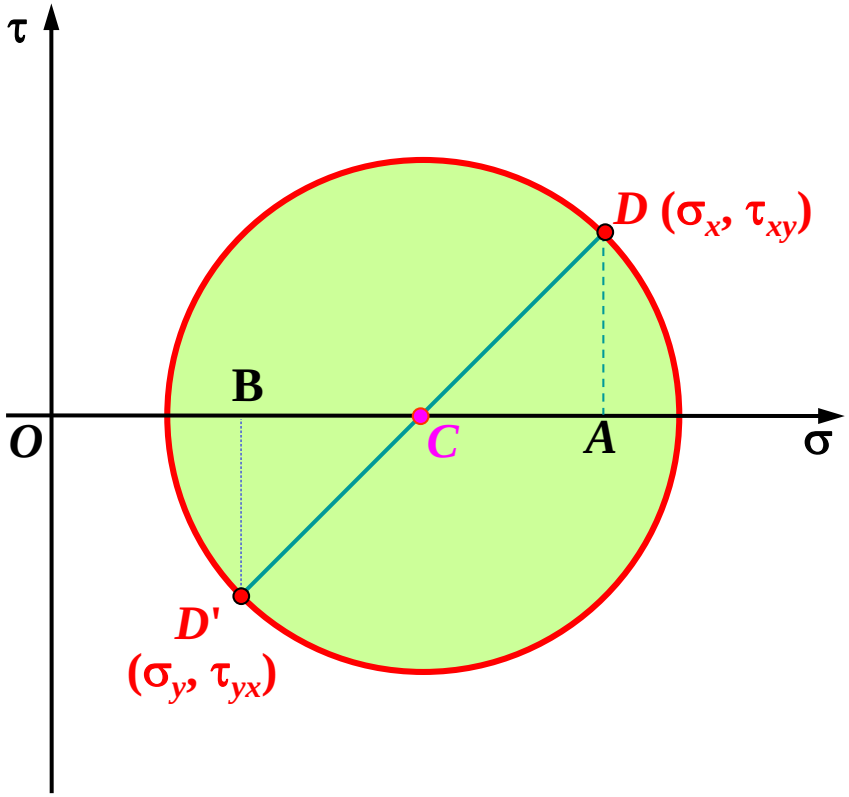
## 二、应力圆作法（The method for drawing a stress circle）




1、步骤

（1）建σ-τ坐标系，选定比例尺。




（2）在 σ - τ 坐标系中作出点 $D\left(\sigma_x,\tau_{xy}\right)$ 和点 $D'\left(\sigma_y,\tau_{yx}\right)$ 。

（3）连接 $D$、$D'$两点的直线与σ 轴相交于$C$ 点。

（4）以$C$为圆心， $CD$ 为半径作圆，该圆就是相应于该单元体的应力圆。

2、证明

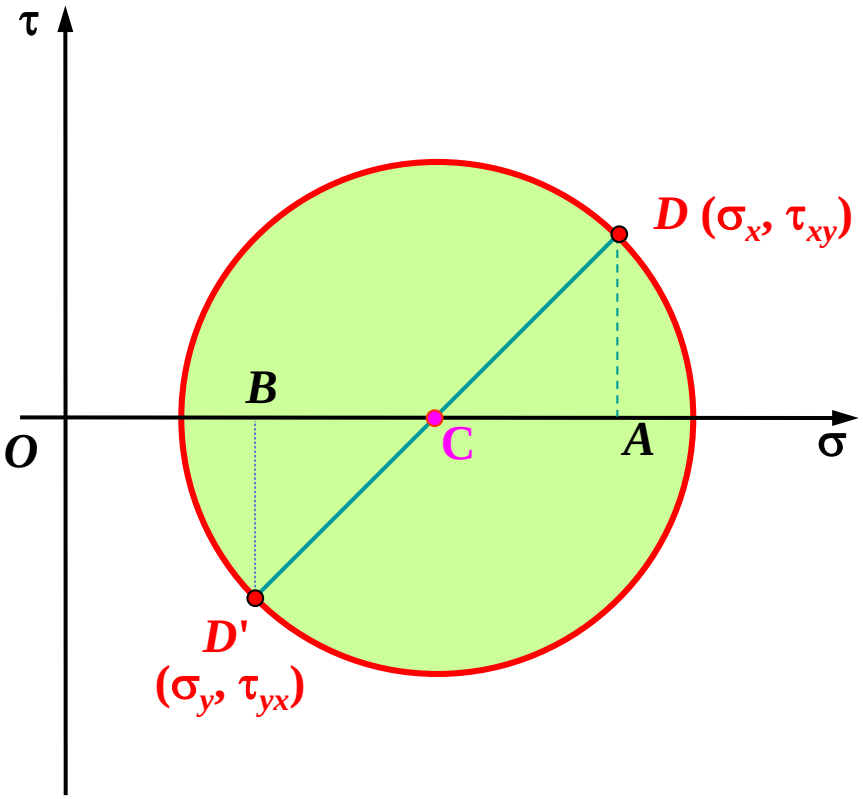
（1）该圆圆心C坐标原点为

$$C\left(\frac{\sigma_x+\sigma_y}{2},0\right)$$

（2）该圆半径为

$$D\left(\sigma_x,\tau_{xy}\right)\quad D'\left(\sigma_y,\tau_{yx}\right)$$

$$CD=\sqrt{\left(\frac{\sigma_x+\sigma_y}{2}-\sigma_x\right)^2+\tau_{xy}^2}=R$$



## 三、应力圆的应用（Application of stress-circle）

**点面对应**

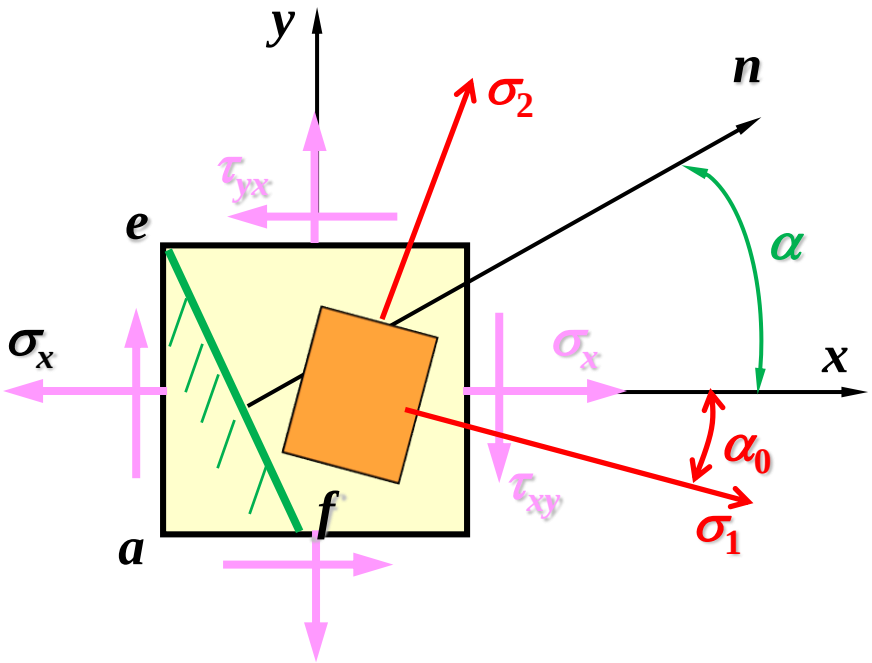
单元体某一**面上的应力**， 对应于应力圆上某**一点的坐标**。

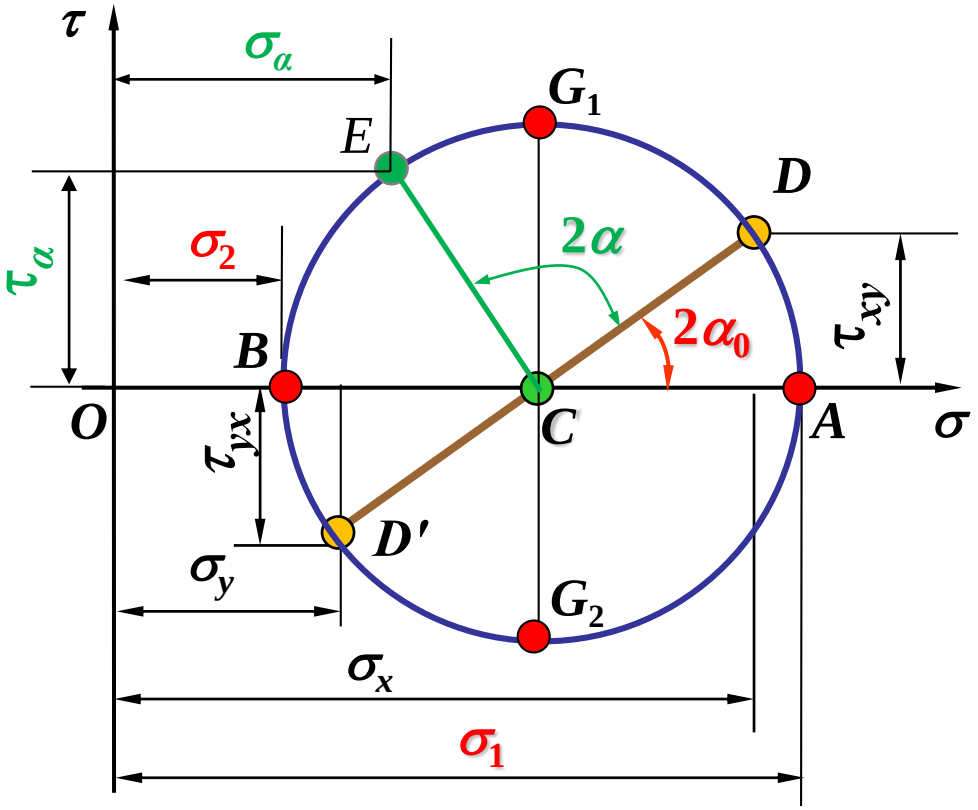
**转向相同**

两者的**转向一致**。

**转角加倍**

应力圆上任意两点所引半径的夹角等于单元体上对应两截面**夹角的两倍**。




点面对应

转向相同

转角加倍

(1)$x$、$y$为法线的面与点$D$、$D'$；

(2)斜截面$\alpha$与应力圆$2\alpha$；

(3)主应力与$A$、$B$，主平面与$2\alpha_0$；

(4)切应力与$G_1$、$G_2$。

## 四、解析法与图解法

1、解析法

优点：精确

缺点：公式多 --- 7个公式。

2、图解法

优点：简单

缺点：数值不精确。

意义：（1）通过明晰的几何关系帮助记忆公式。

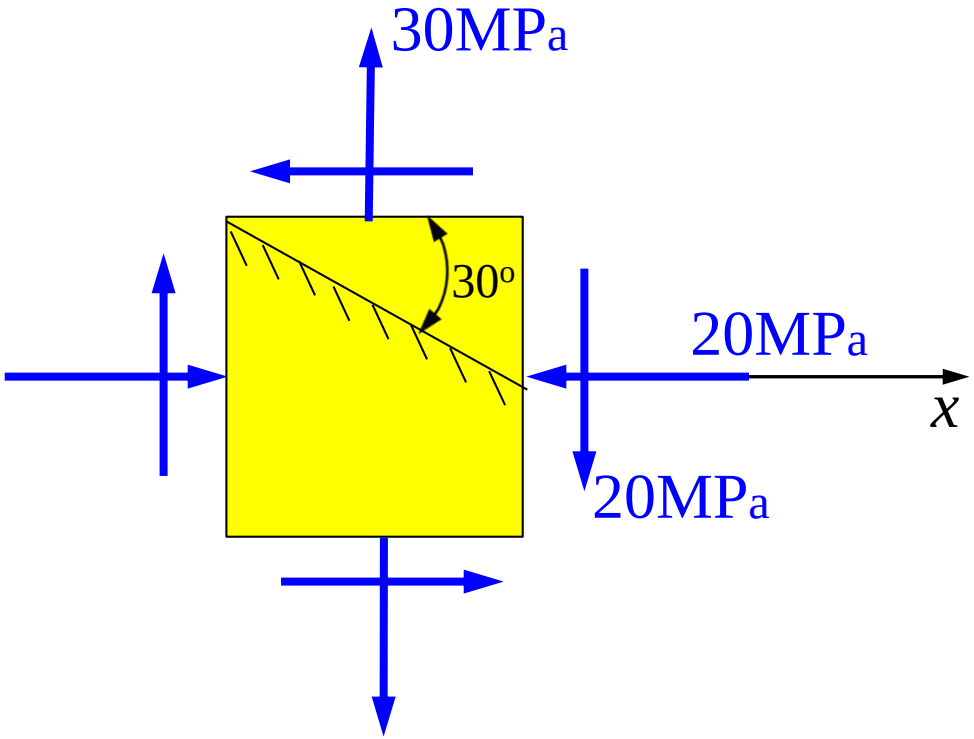
（2）思考问题的工具，分析和解决难度较大的应力状态问题。

应力圆做法

- 建 $\sigma$ - $\tau$ 坐标系，选定比例尺。
- 作出点D($\sigma_x$, $\tau_{xy}$)和 D ′($\sigma_y$, $\tau_{yx}$）。
- $DD'$连线与$\sigma$ 轴交点$C$ 。
- 以$C$为圆心， $CD$ 为半径作圆。

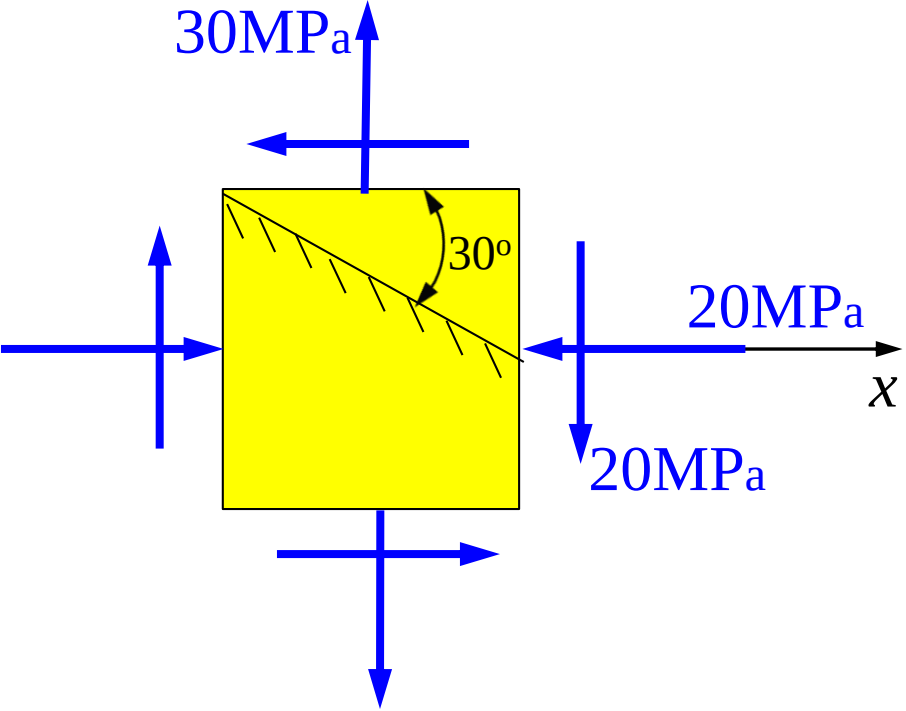
**例题7-2** 单元体的应力状态如图。

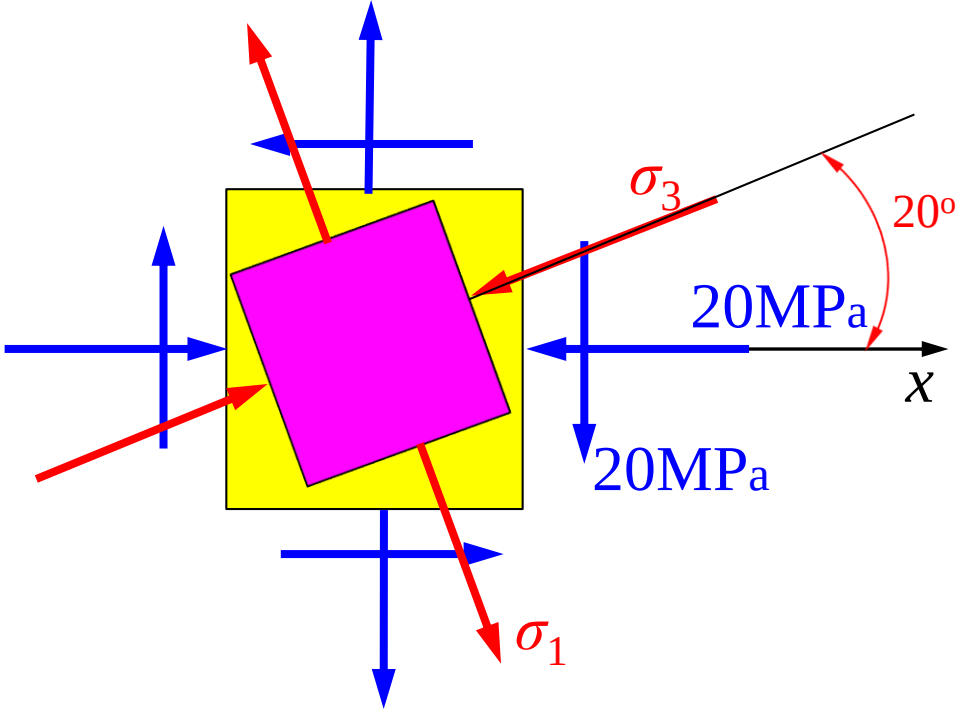
求（1）斜截面上的应力；（2）主应力大小，主平面位置；在单元体上绘出主平面位置及主应力方向；（3）图示平面内最大切应力。

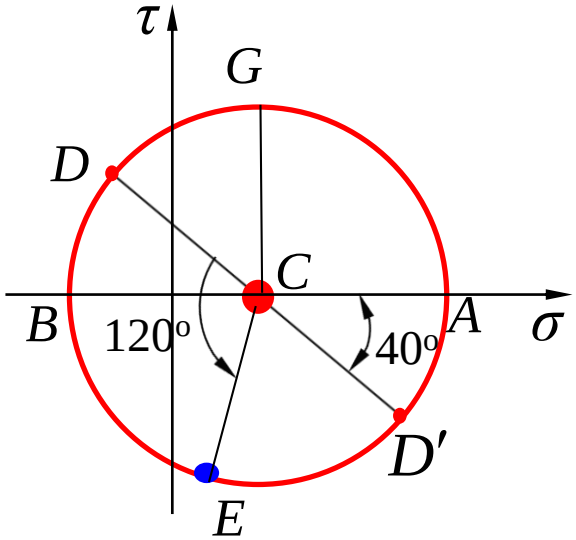



$$\sigma_x = -20\text{MPa} \qquad \sigma_y = 30\text{MPa}$$

$$\tau_{xy} = 20\text{MPa} \qquad \alpha = 60^\circ$$




$D(-20,20)\quad D'(30,-20)$

$\sigma_{60}=0.2\text{MPa}$

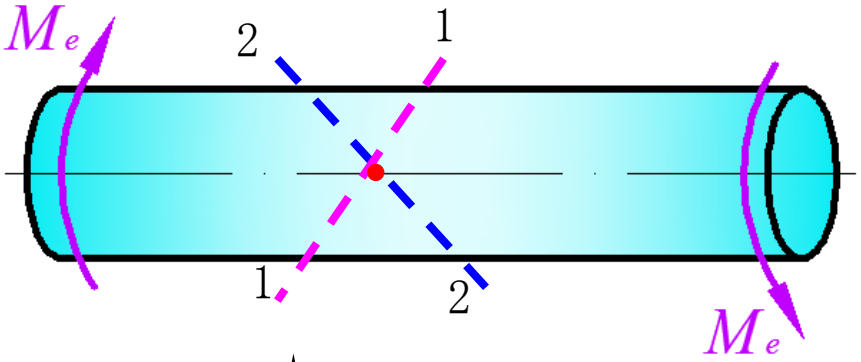
$\tau_{60}=-31.7\text{MPa}$

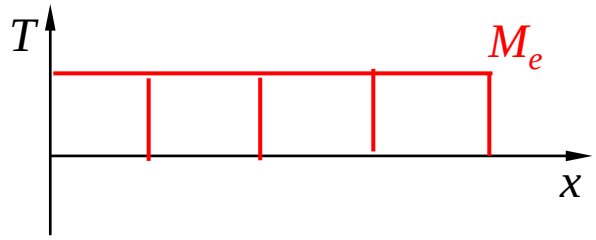
$\sigma_1=37\text{MPa},\quad \sigma_2=0$

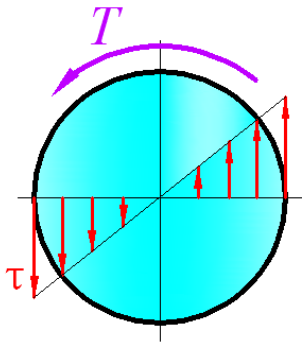
$\sigma_3=-28\text{MPa}$

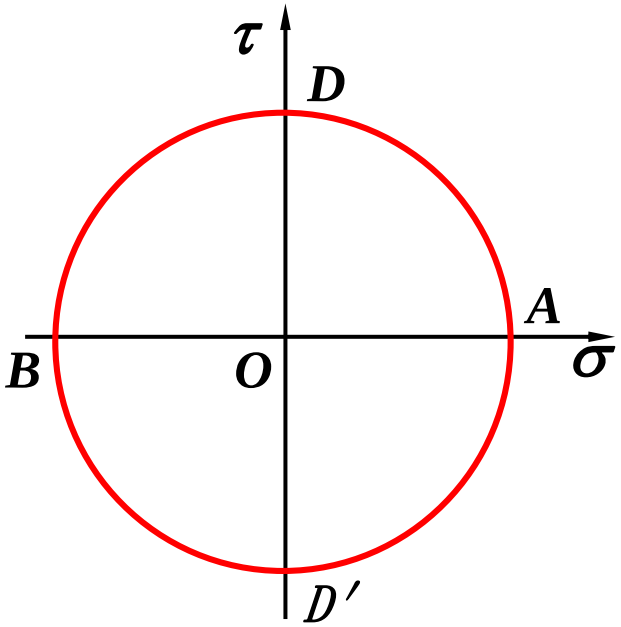
$\alpha_0=20^0,\quad \tau_{\max}=32.5\text{MPa}$

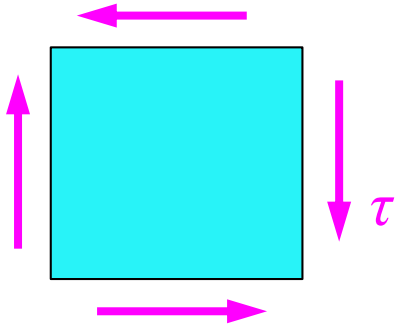
**例题7-3** 为什么铸铁扭转破坏时在45°的斜截面？

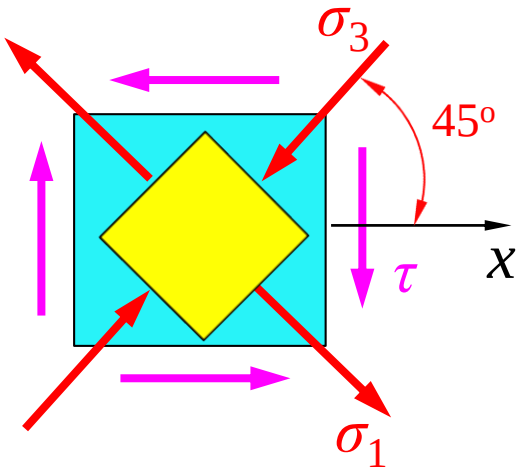


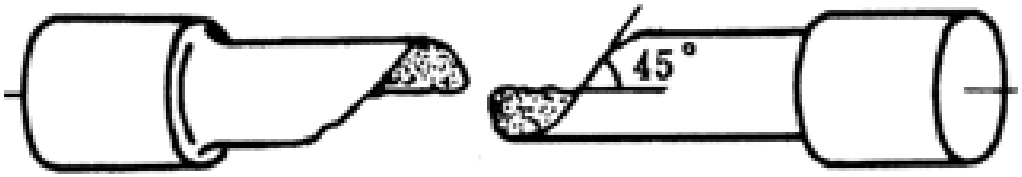



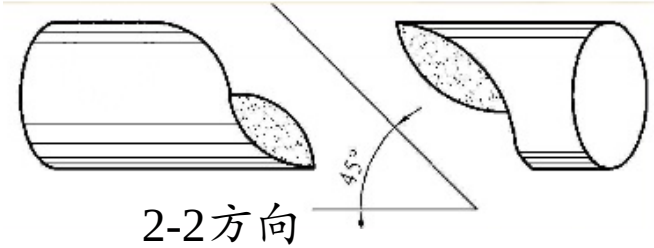




$\sigma_1 = \tau \quad \sigma_2 = 0 \quad \sigma_3 = -\tau$



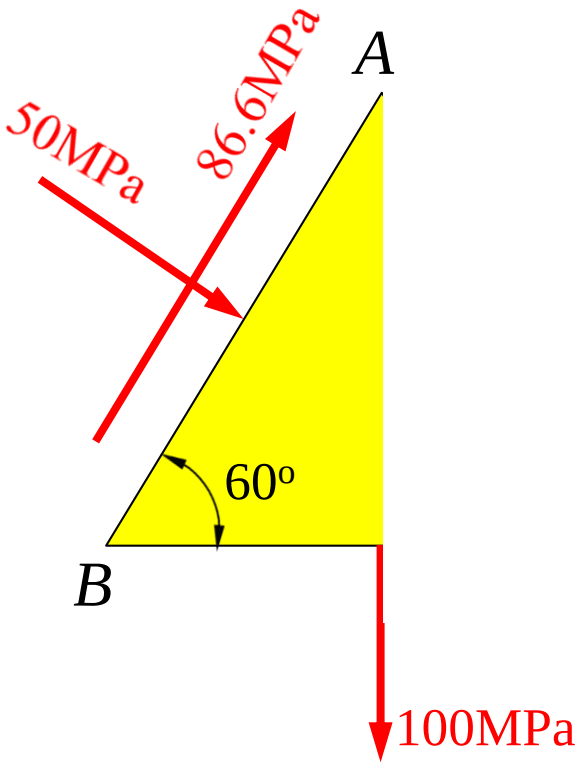

1-1方向

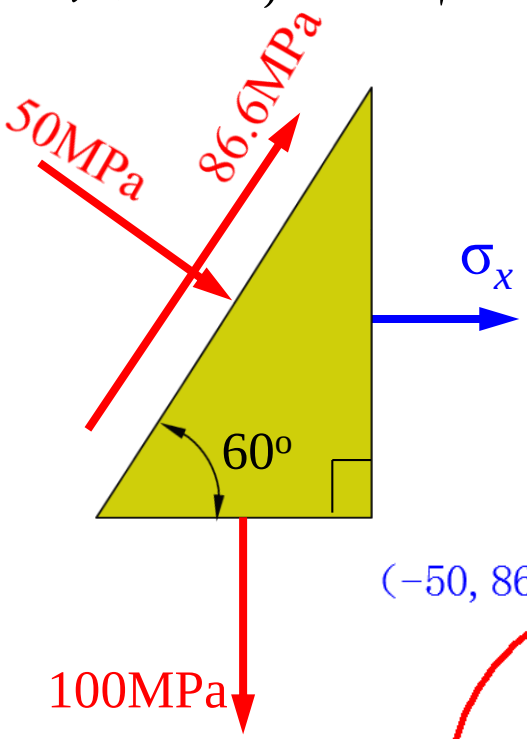


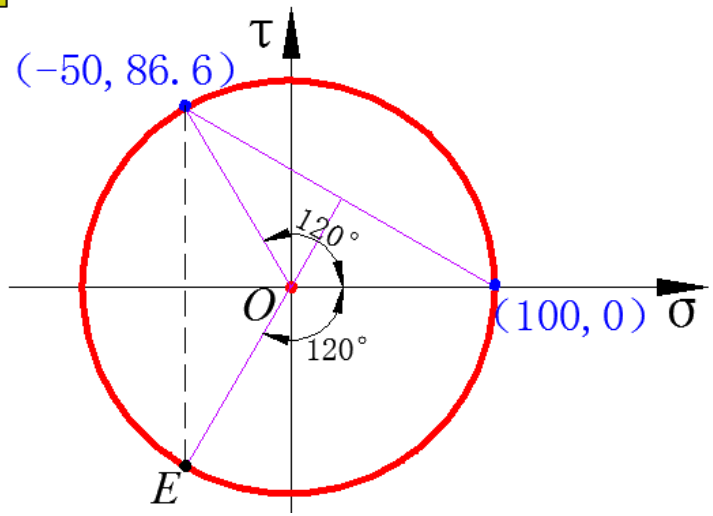

2-2方向

**例题7-4** 一处于平面应力状态的受力物体中，用一正三角形取出一单元体，应力状态如图。

求（1）σ和τ；（2）主应力大小，主平面位置。



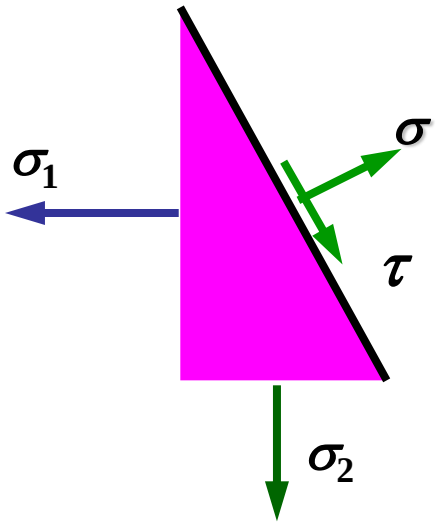



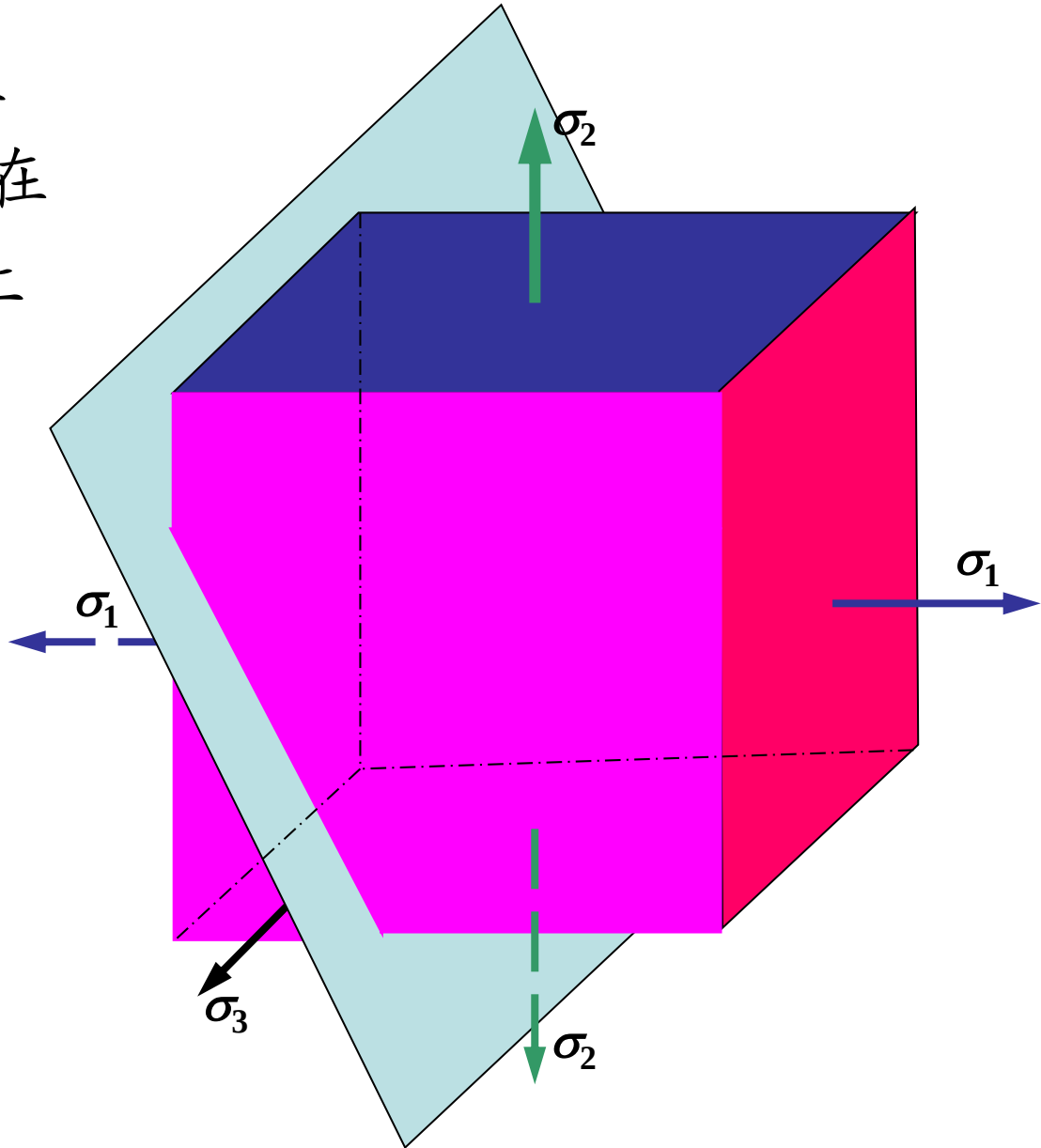

$\sigma$=-50MPa　　$\tau$=86.6MPa　　$\sigma_1$=100MPa　　$\sigma_2$=0　　$\sigma_3$=-100MPa

# §7-4 三向应力状态分析

（analysis of three-dimensional stress-state）

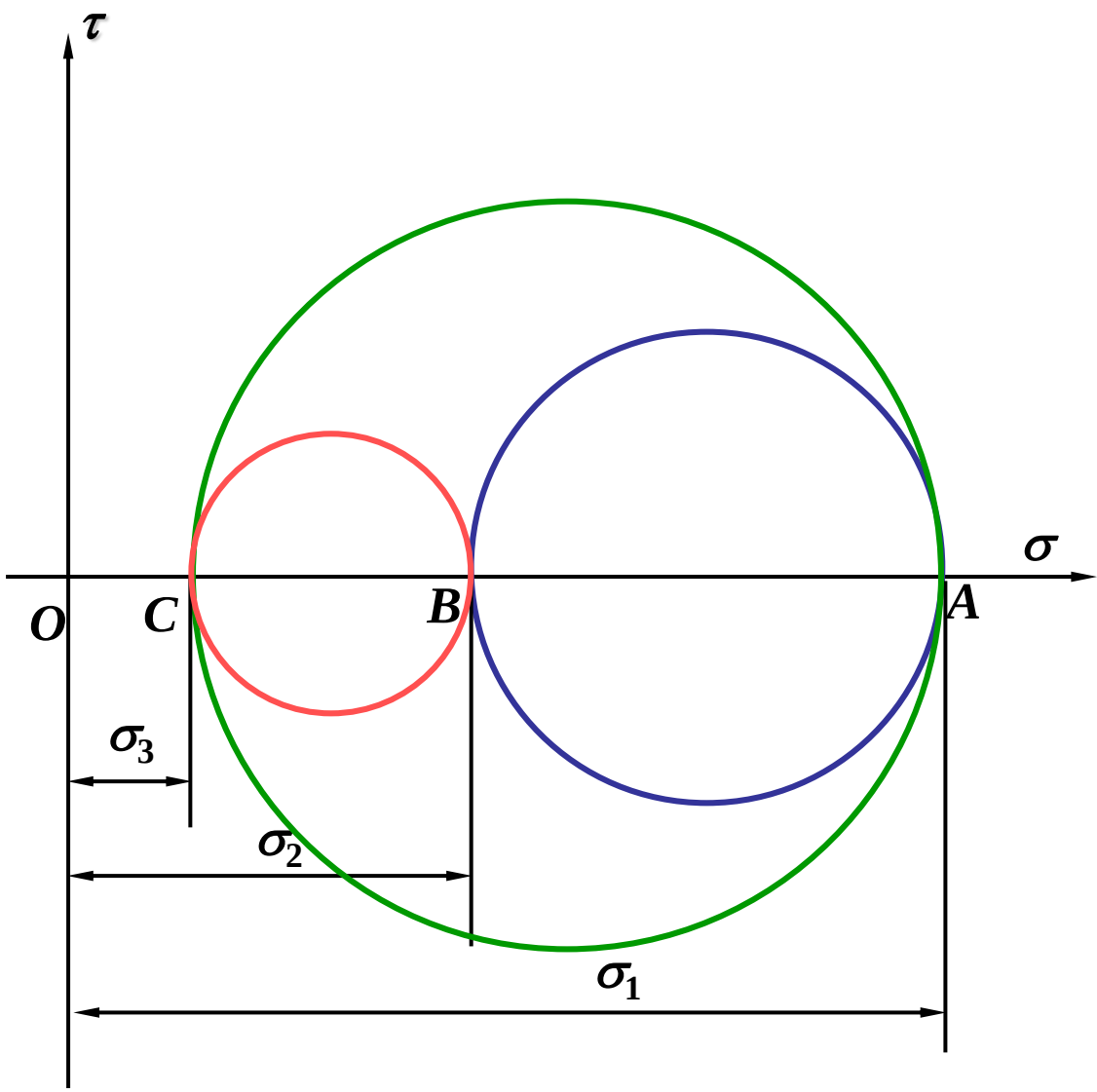
首先研究与其中一个主平面（例如主应力$\sigma_3$所在的平面）垂直的斜截面上的应力，用截面法。

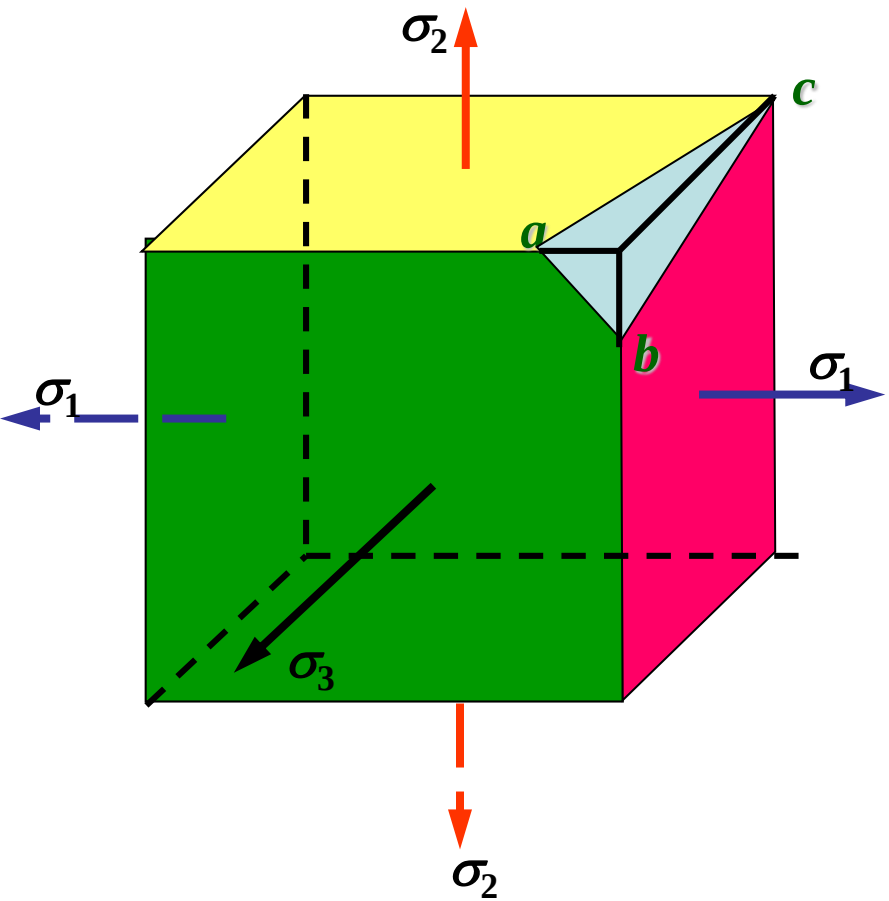


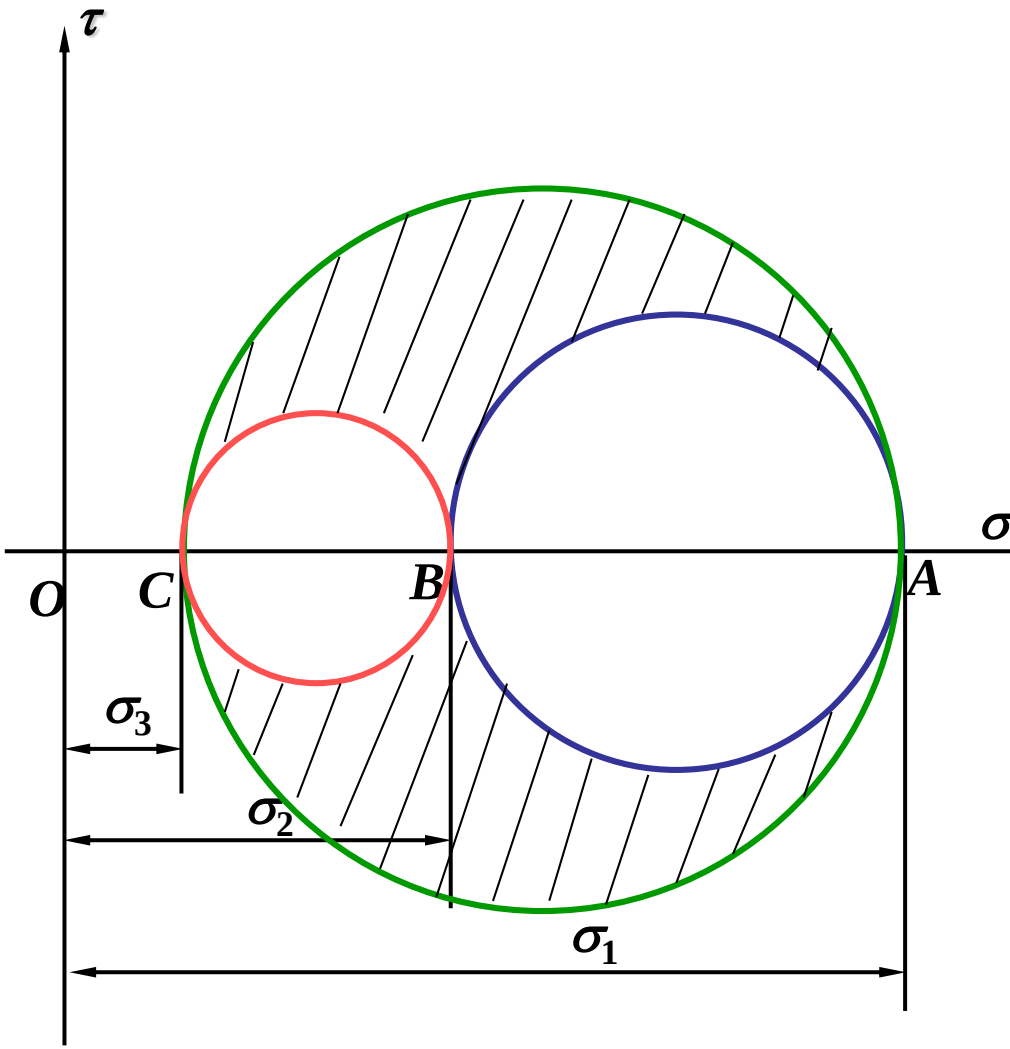



与主应力 $\sigma_3$ 所在主平面垂直的斜截面上的应力σ、τ 可用由$\sigma_1$ ,$\sigma_2$ 作出的应力圆上的点来表示。

结论：与某一主应力所在主平面垂直的斜截面上的应力σ, τ 可用由剩下两个主应力作出的应力圆上的点来表示。





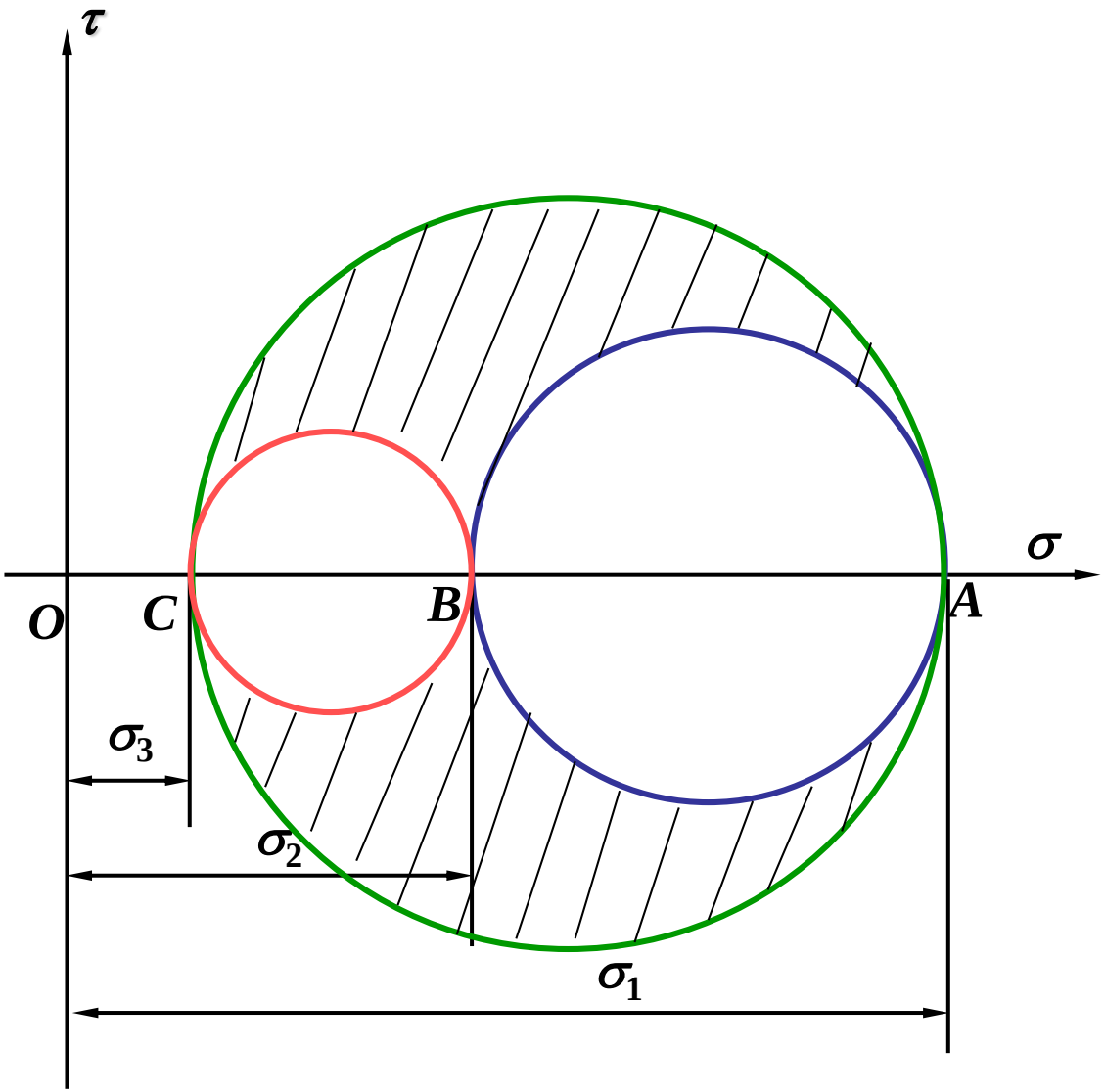




abc 截面表示与三个主平面斜交的任意斜截面。该截面上应力σ和τ对应的点必位于三个应力圆所围成的阴影内。

$$\sigma_{\max} = \sigma_1$$

$$\sigma_{\min} = \sigma_3$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}$$

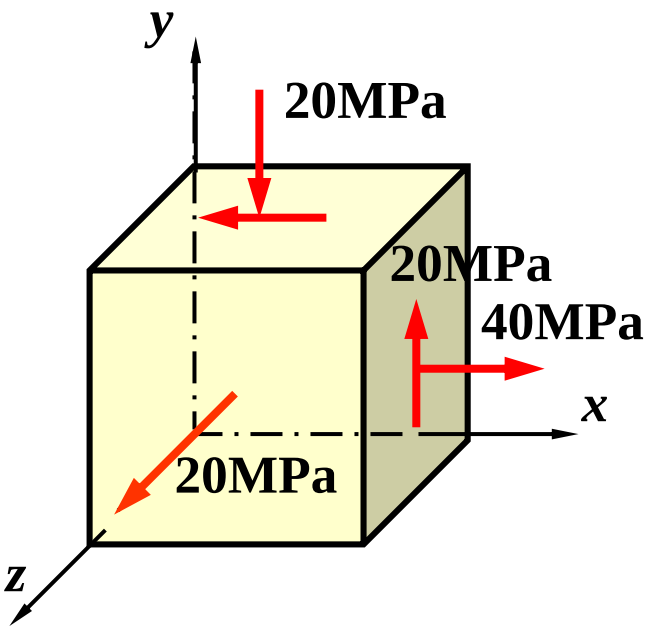


**例题7-5** 单元体的应力如图所示，求出主应力和最大切应力值，作应力圆。

解：该单元体有一个已知主应力

$$\sigma_z=20\text{MPa}$$

因此与该主平面正交的各截面上的应力与主应力$\sigma_z$无关，依据$x$截面和$y$截面上的应力画出应力圆，求另外两个主应力。



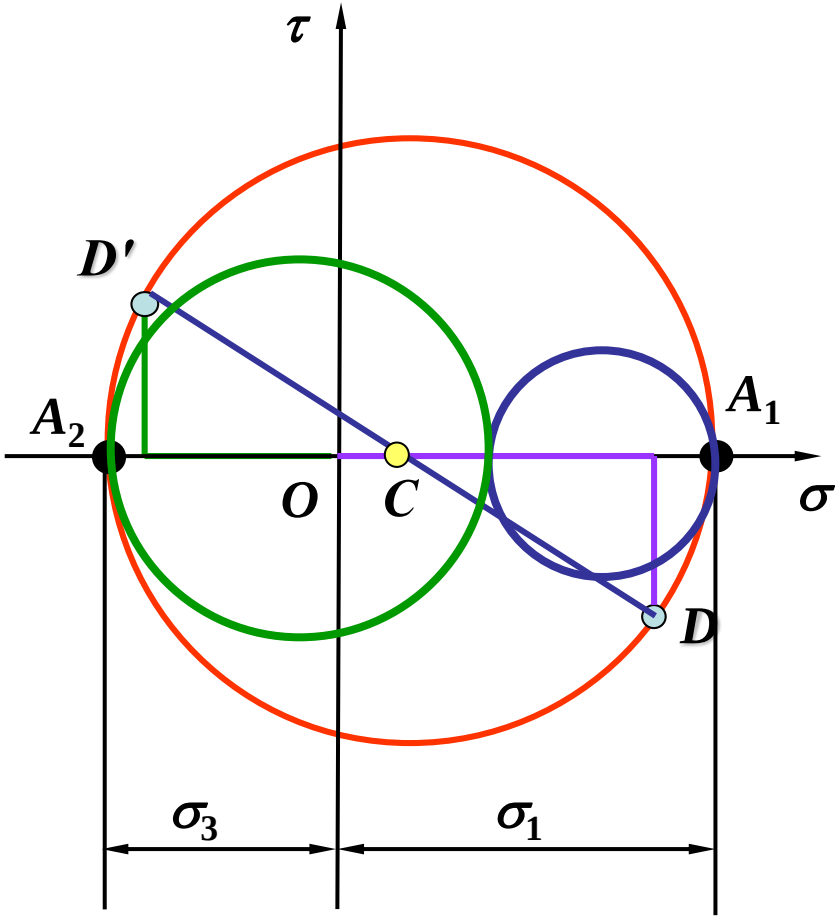
$$D(40,-20) \quad D'(-20,20)$$

$\sigma_1 = 46\text{MPa}$

$\sigma_2 = 20\text{MPa}$

$\sigma_3 = -26\text{MPa}$

$\tau_{max} = 36\text{MPa}$



# 作　业

6版：

7.2

7.4　（a）　（d）

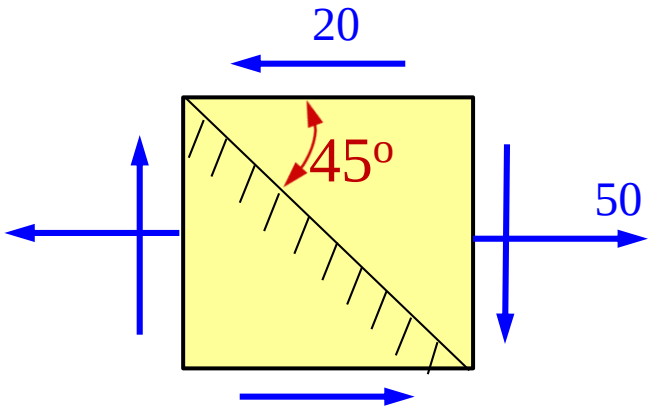
求：（4）斜截面上的应力。

要求：用两种解析法和图解法两种方法。

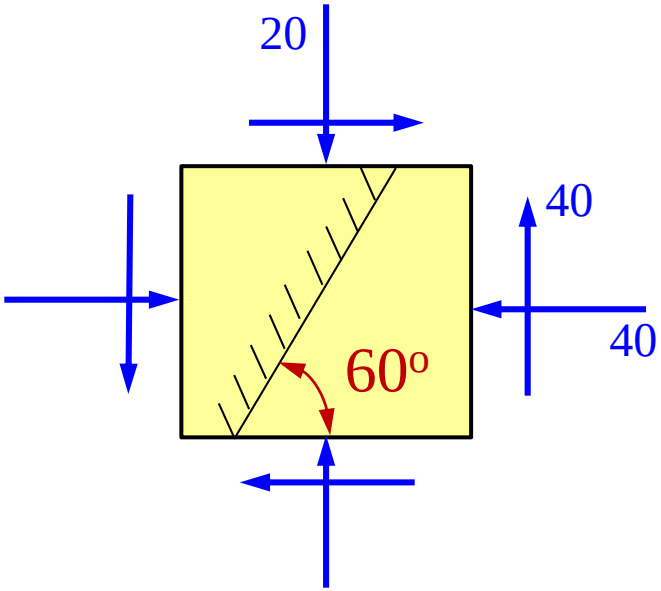
5版

7.1

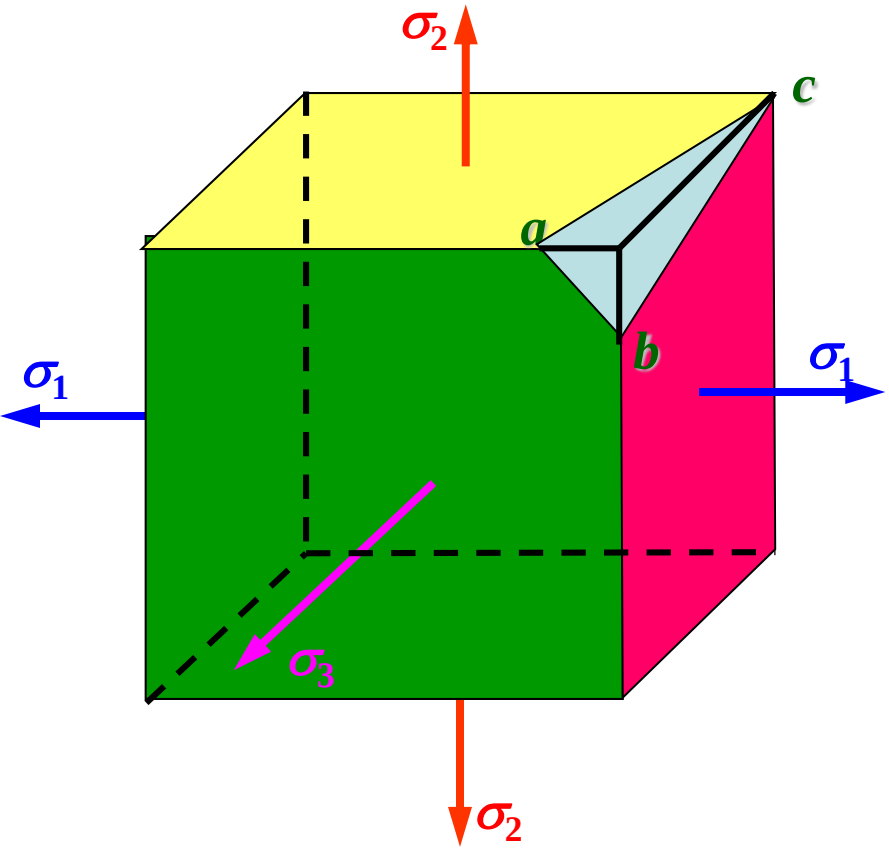
7.3　（a）　（d）

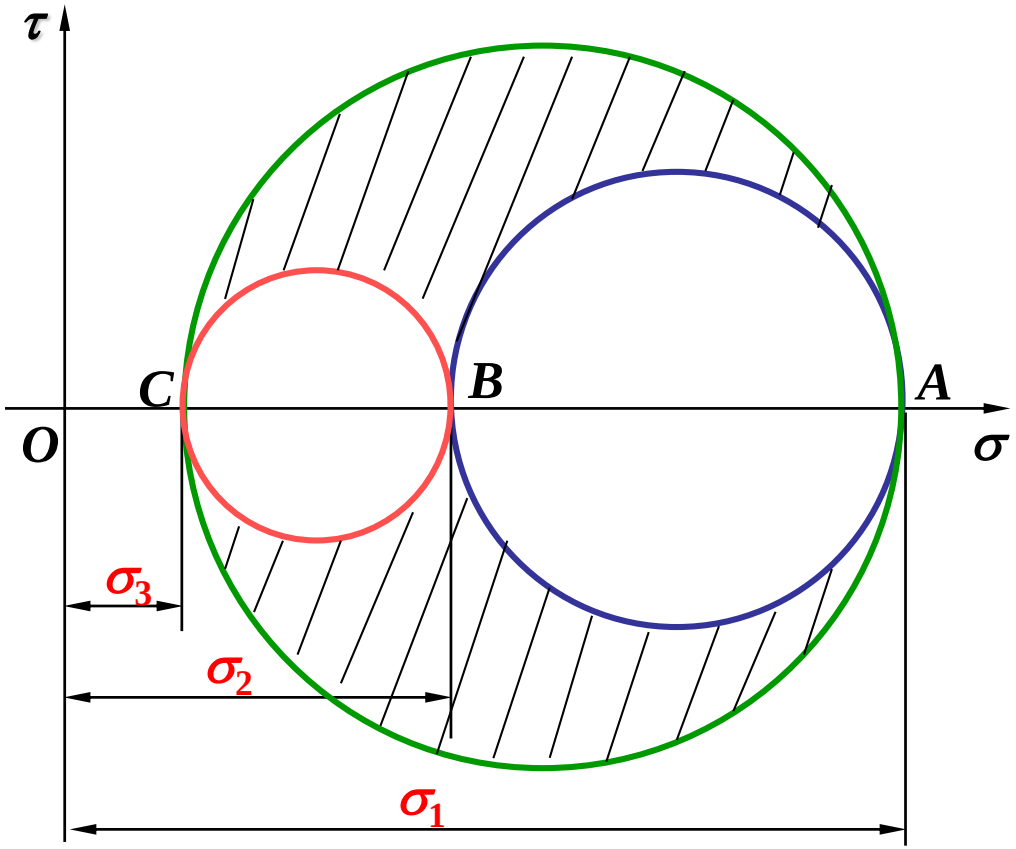



（a）




（d）




与某一主应力所在主平面垂直的斜截面上的应力可用由剩下两个主应力作出的应力圆上的点来表示。与三个主平面斜交的任意斜截面（如*abc* 截面）上应力对应的点必位于三个应力圆所围成的阴影内。

$$\sigma_{\max} = \sigma_1$$

$$\sigma_{\min} = \sigma_3$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}$$

# §7-6 广义胡克定律

Generalized Hooke's law

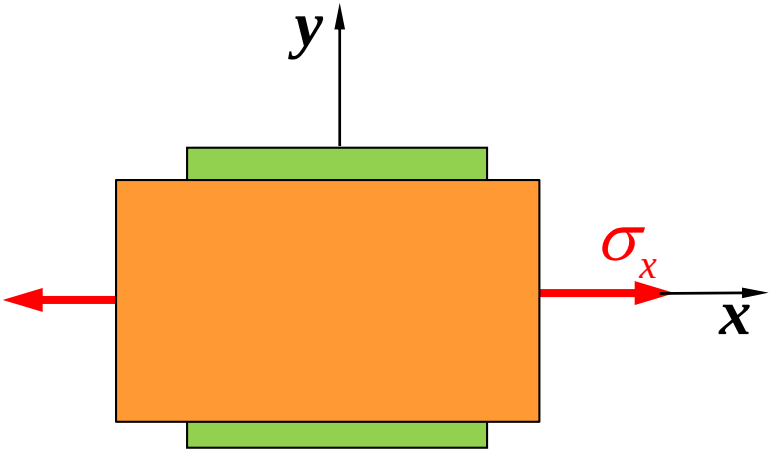
## 一、基本变形时的胡克定律

1、轴向拉压胡克定律

$$\varepsilon_x = \frac{\sigma_x}{E}$$

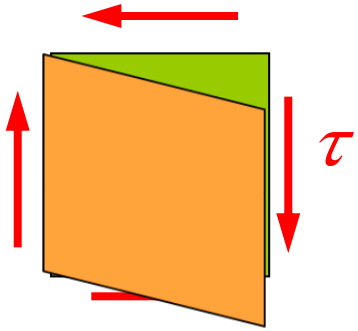
横向变形

$$\varepsilon_y = -\mu\varepsilon_x = -\mu\frac{\sigma_x}{E}$$

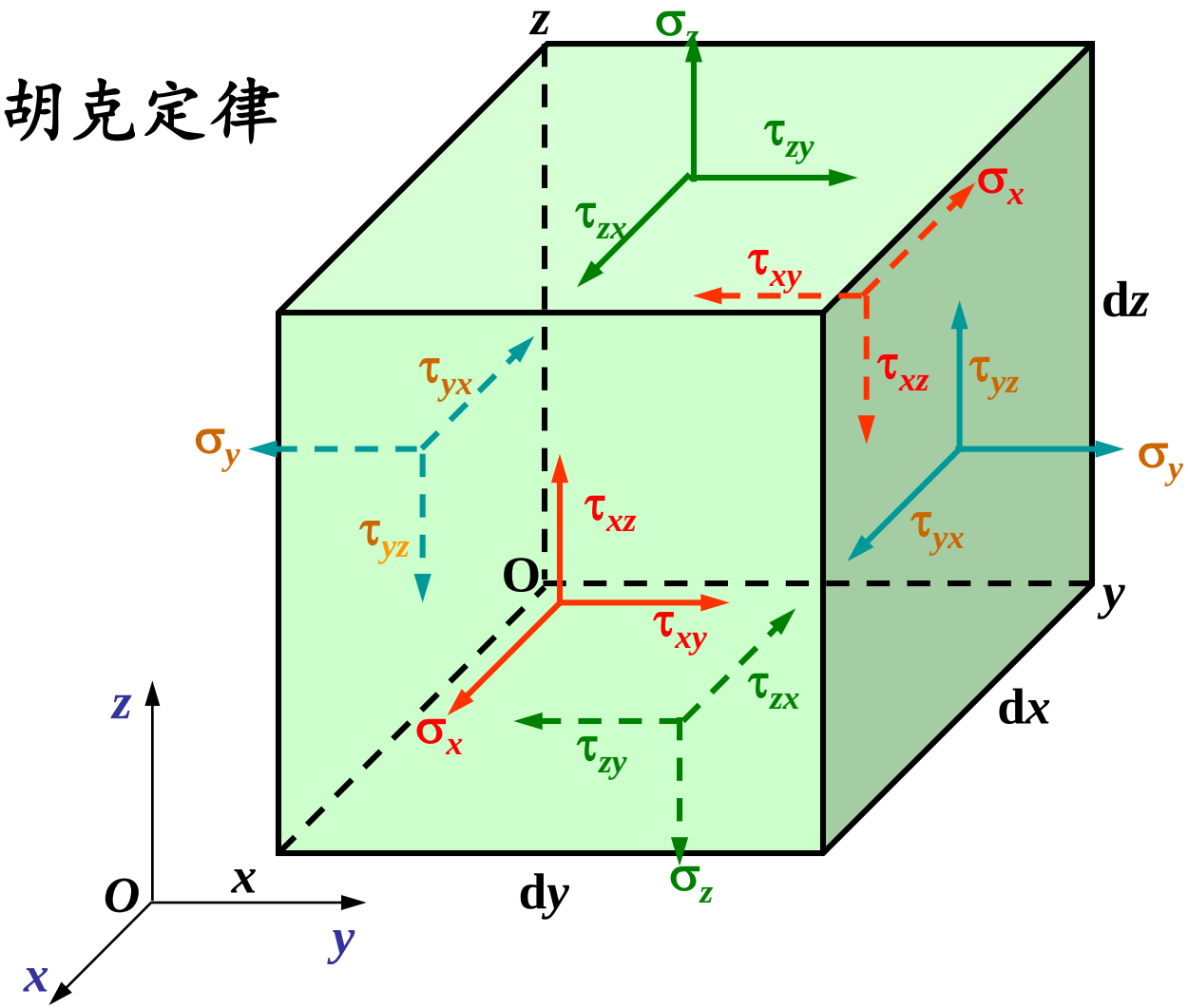



2、纯剪切胡克定律

$$\gamma = \frac{\tau}{G}$$



# 二、广义胡克定律

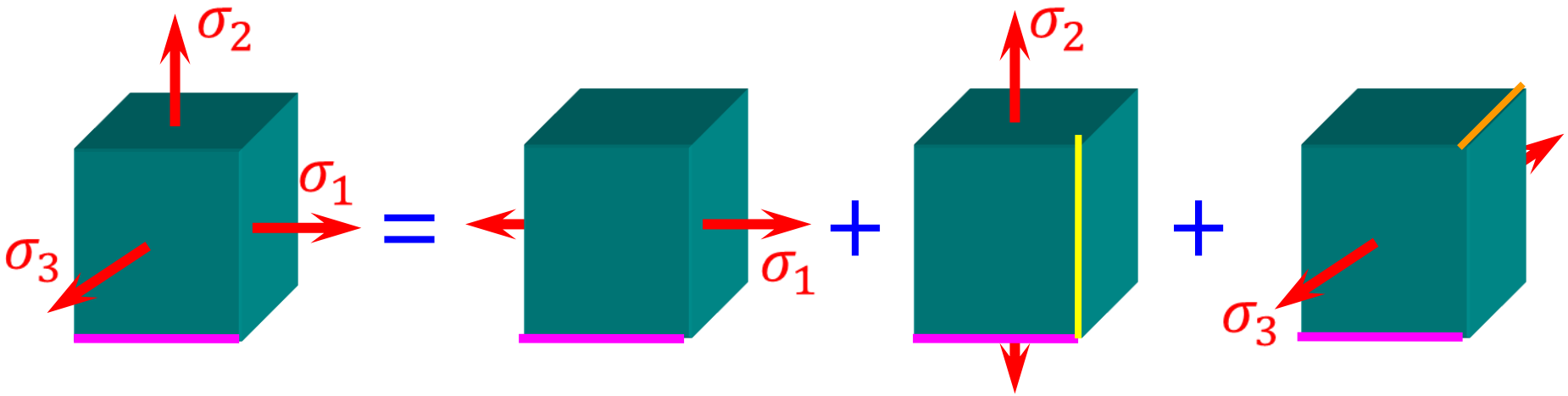



各向同性材料，小变形，弹性范围内时，

正应力只引起线应变，切应力只引起角应变。

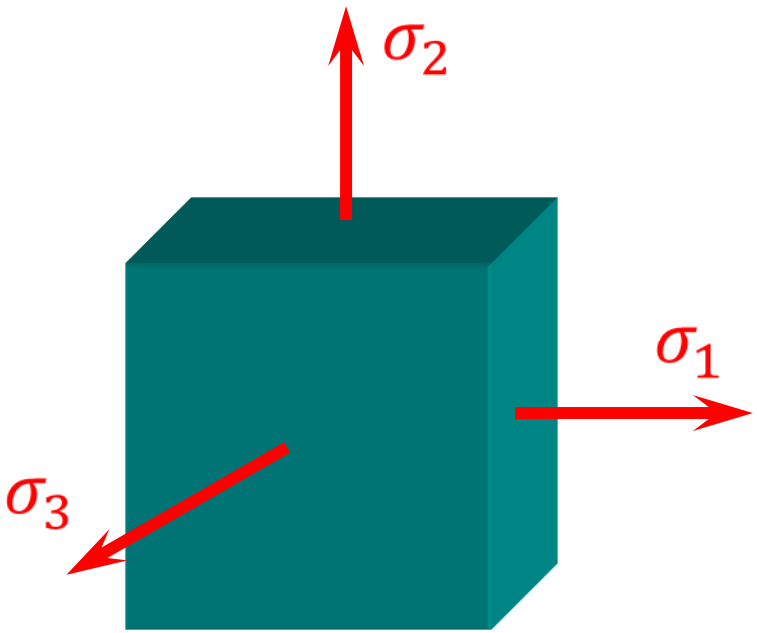
## 1、三向应力状态的广义胡克定律

用叠加原理，分别计算出正应力分别单独存在时，各方向的线应变，然后代数相加。




$$\frac{\sigma_1}{E} \qquad -\mu\frac{\sigma_2}{E} \qquad -\mu\frac{\sigma_3}{E}$$

$$\varepsilon_1 = \frac{1}{E}\left[\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right)\right]$$




$$\varepsilon_1 = \frac{1}{E}\left[\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right)\right]$$

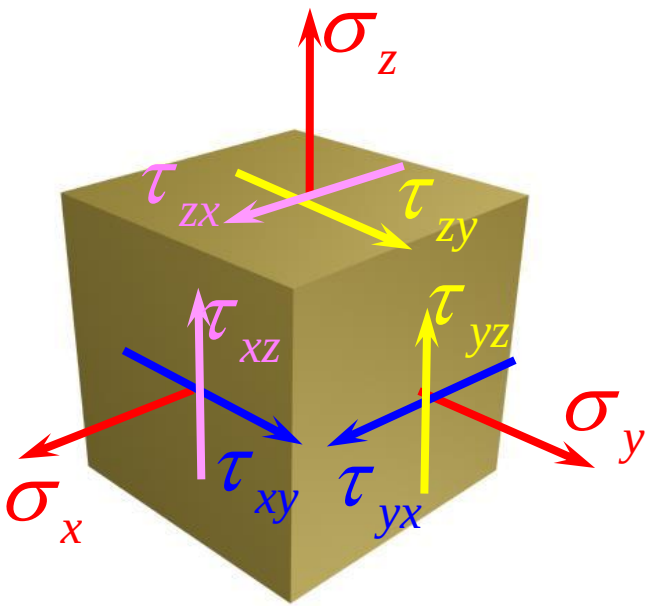
$$\varepsilon_2 = \frac{1}{E}\left[\sigma_2 - \mu\left(\sigma_1 + \sigma_3\right)\right]$$

$$\varepsilon_3 = \frac{1}{E}\left[\sigma_3 - \mu\left(\sigma_1 + \sigma_2\right)\right]$$

$$\gamma_{xy} = 0 \quad \gamma_{xz} = 0 \quad \gamma_{yz} = 0$$

主应变，方向沿主应力方向。

## 2、广义胡克定律的一般形式




$$\varepsilon_x = \frac{1}{E}\left[\sigma_x - \mu\left(\sigma_y + \sigma_z\right)\right]$$

$$\varepsilon_y = \frac{1}{E}\left[\sigma_y - \mu\left(\sigma_x + \sigma_z\right)\right]$$

$$\varepsilon_z = \frac{1}{E}\left[\sigma_z - \mu\left(\sigma_x + \sigma_y\right)\right]$$

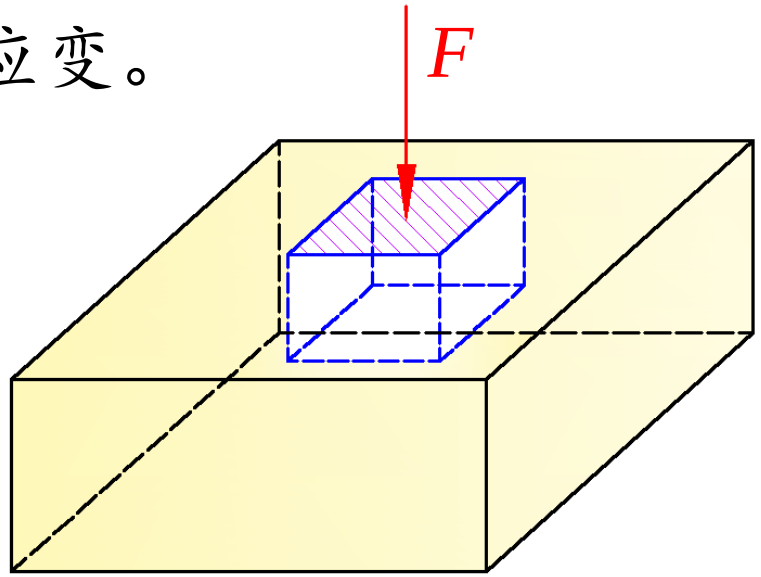
$$\gamma_{xy} = \frac{\tau_{xy}}{G} \quad \gamma_{xz} = \frac{\tau_{xz}}{G} \quad \gamma_{yz} = \frac{\tau_{yz}}{G}$$

- 应变无量纲。
- 线应变只与正应力有关，切应变只与切应力有关。

各应力怎么求？

$\varepsilon_{max}$=? $\varepsilon_{min}$=?

**例题7-6** 边长$a$= 0.1m的铜立方块，无间隙地放入体积较大，变形可略去不计的钢凹槽中，如图所示。已知铜的弹性模量$E$=100GPa，泊松比$\mu$=0.34，当受到$F$=300kN的均布压力作用时，求该铜块的主应力和主应变。



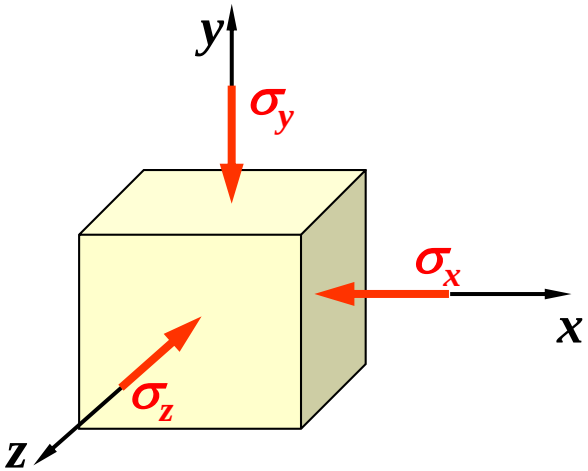

解： 已知条件：

铜块横截面上的压应力

$$\sigma_y = \frac{F}{A} = 30\,\mathbf{MPa}$$

$$\varepsilon_x = 0$$

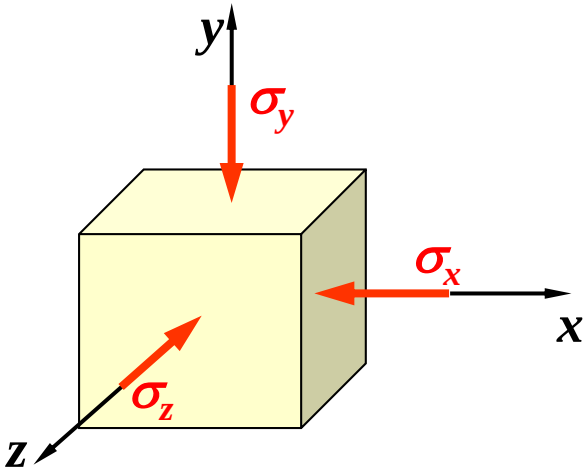
$$\varepsilon_z = 0$$



$$\varepsilon_x = \frac{1}{E}\left[-\sigma_x - \mu\left(-\sigma_y - \sigma_z\right)\right] = 0$$

$$\varepsilon_z = \frac{1}{E}\left[-\sigma_z - \mu\left(-\sigma_y - \sigma_x\right)\right] = 0$$

$$\varepsilon_y = \frac{1}{E}\left[-\sigma_y - \mu\left(-\sigma_x - \sigma_z\right)\right]$$




$$\sigma_x = \sigma_z = \frac{\mu}{1-\mu}\sigma_y = 15.5\mathbf{MPa}$$

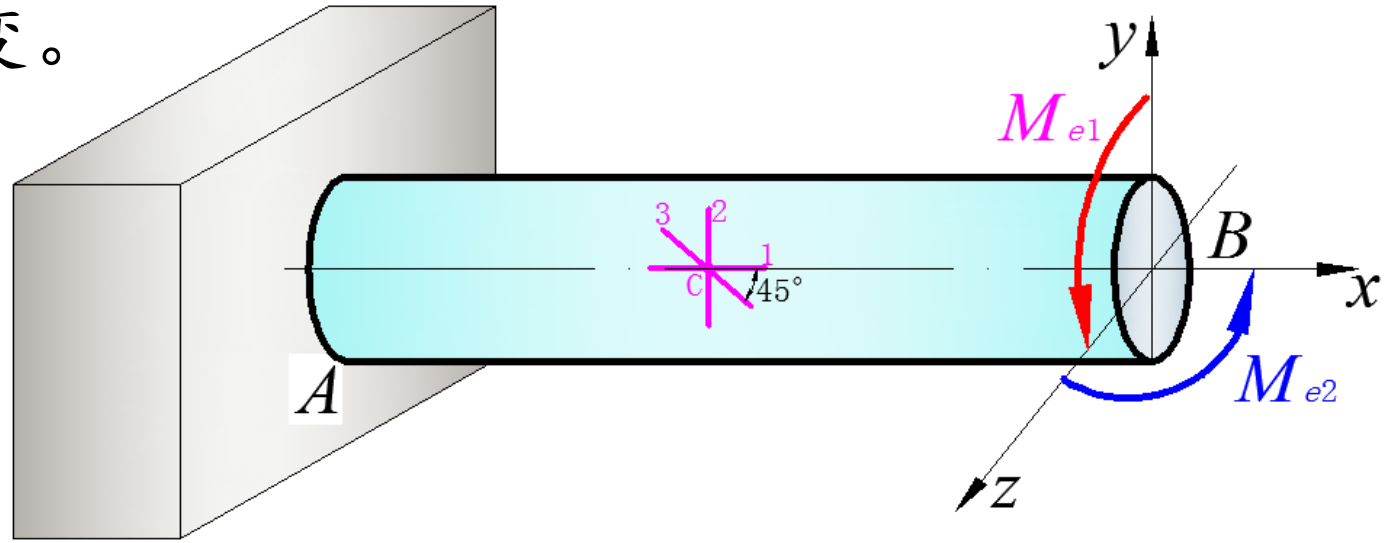
$\varepsilon_y$=-0.000195

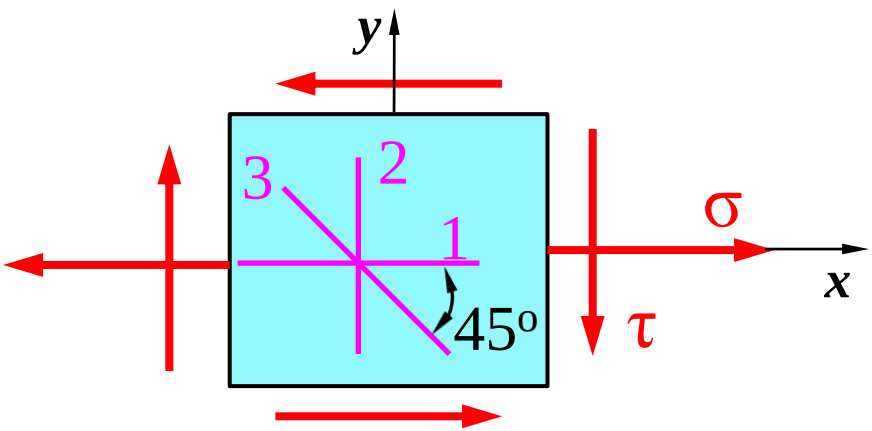
铜块的主应力主应变为

$\sigma_1$= =-15.5MPa　　$\sigma_2$=-15.5MPa　　$\sigma_3$=-30MPa

$\varepsilon_1$=0　　$\varepsilon_2$=0　　$\varepsilon_3$=-0.000195

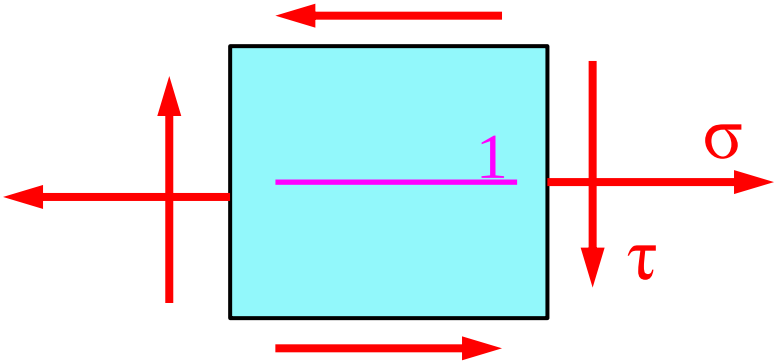
**例题7-7** 如图所示$M_{e1}$=150N m，$M_{e2}$=120N m，轴$d$=0.03m，弹性模量$E$=200GPa，泊松比$\mu$=0.3，求C点1、2、3个方向的应变。

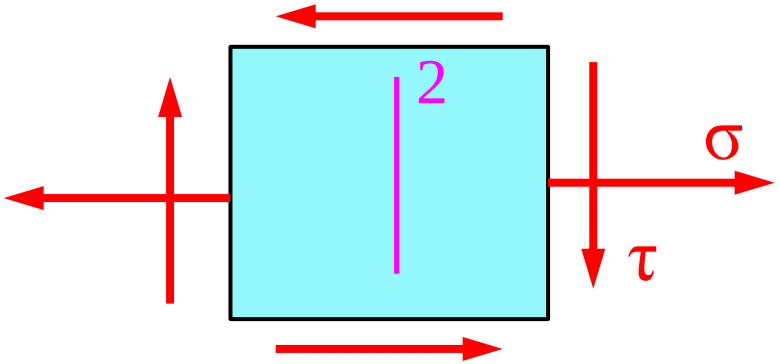


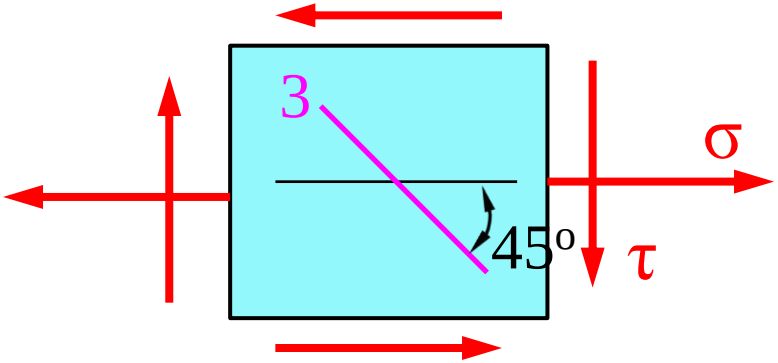



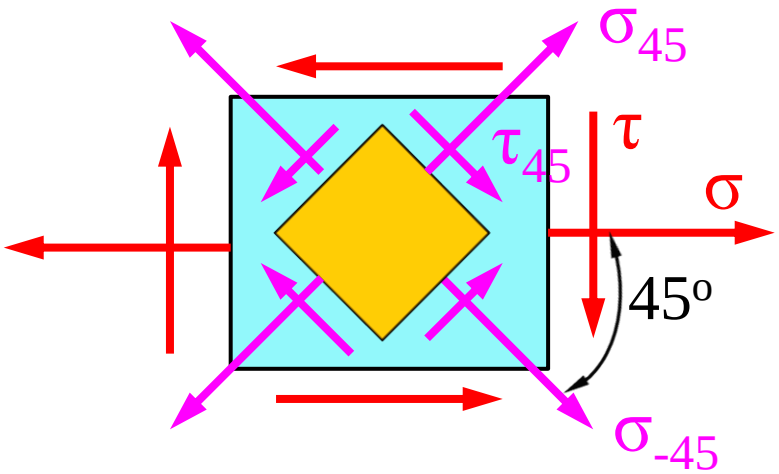

$$\tau = \frac{16M_{e1}}{\pi d^3} = 28.29\,\mathbf{MPa}$$

$$\sigma = \frac{32M_{e2}}{\pi d^3} = 45.27\,\mathbf{MPa}$$



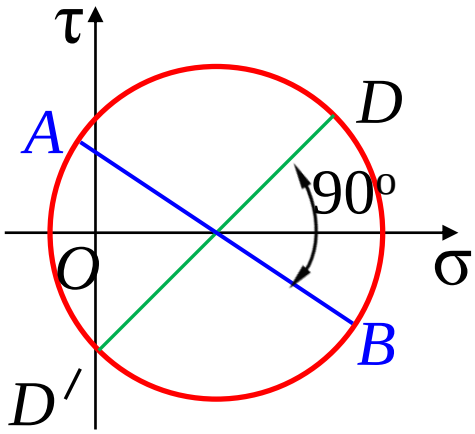

$\varepsilon^1=0.000226$

$\varepsilon^2=-0.0000679$

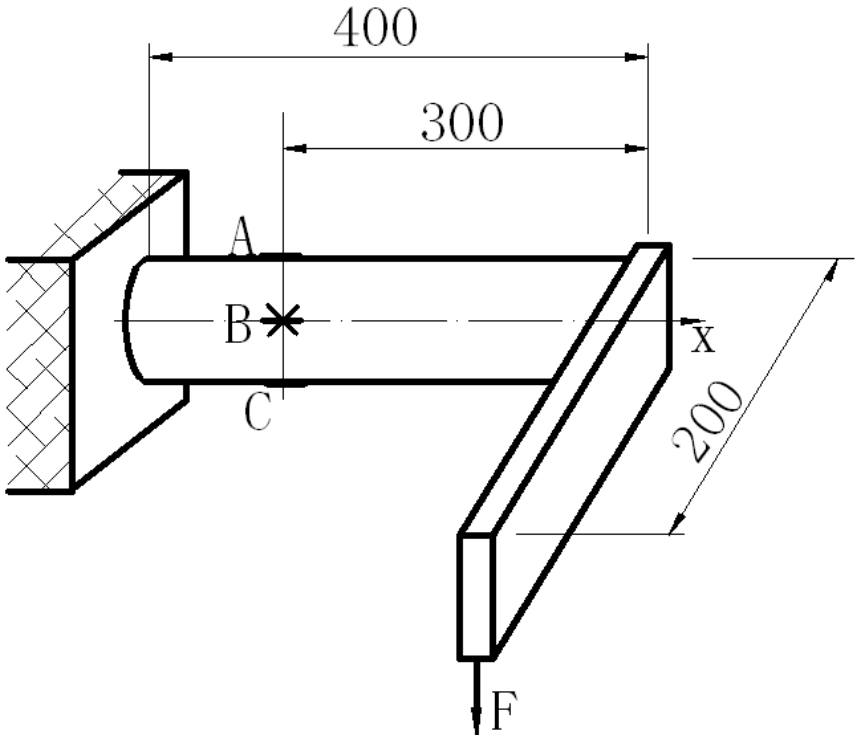
$\sigma_{45}=-5.655\text{MPa}$

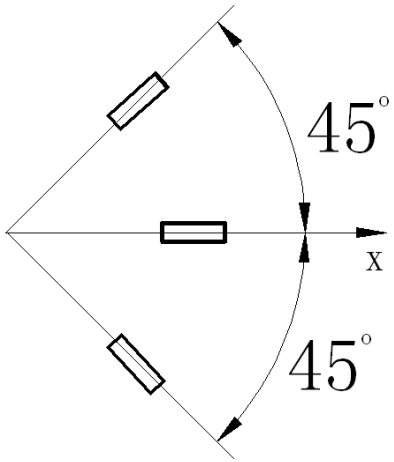
$\sigma_{-45}=50.925\text{MPa}$

$\varepsilon^3=0.000263$

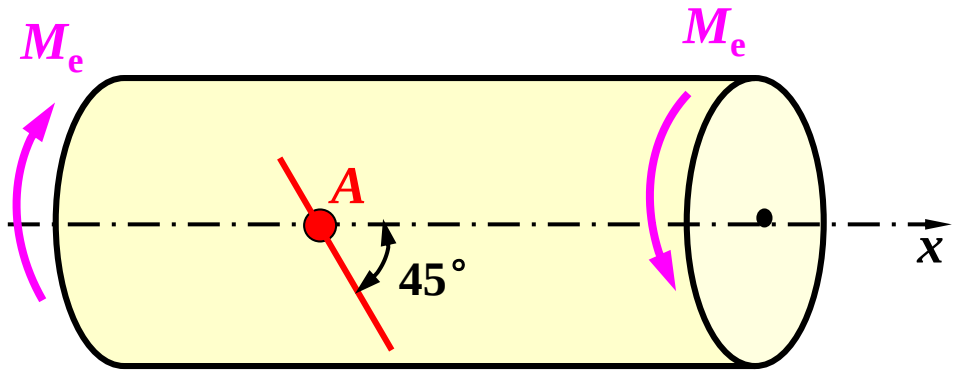


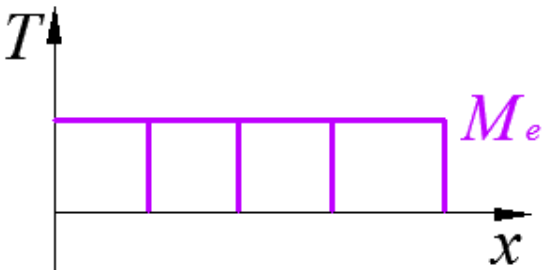
# 弯扭组合作用下薄壁圆筒应变的测定

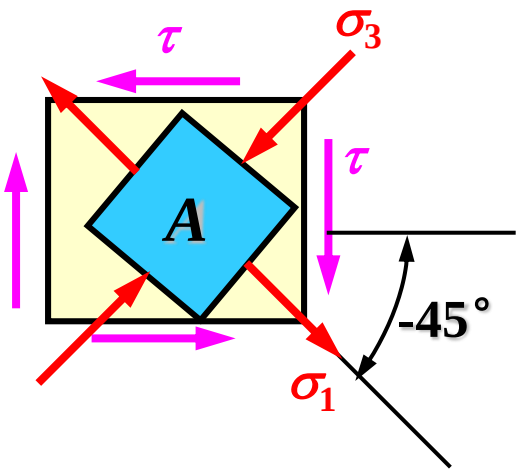


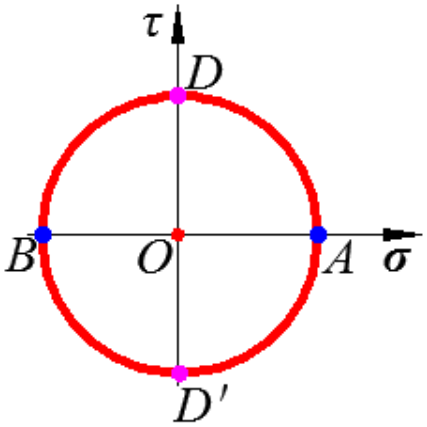



**例题7-8** 一直径$d$=20mm的实心圆轴，在轴的的两端加力偶$M_e$，在轴的表面上某一点$A$处用变形仪测出与轴线成-45方向的应变 $\varepsilon=5.0\times10^{-4}$。弹性模量$E$=200GPa，泊松比$\mu$=0.3。求：（1）作出该轴扭矩图；（2）用单元体表示$A$点的应力状态；（3）$A$点处的主应力；（4）$M_e$的大小。




$\sigma_1=\tau=76.92\text{MPa}$

$\sigma_2=0$

$\sigma_3=-\tau=-76.92\text{MPa}$

$$M_e=\frac{E\varepsilon\pi d^3}{16(1+\mu)}=120.8\,\mathbf{N\cdot m}$$

# §7-7 复杂应力状态的应变能密度

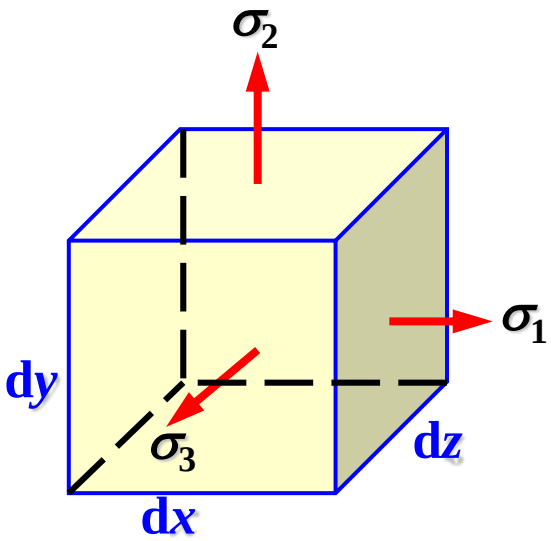
Strain-energy density in general stress-state

## 一、各向同性材料的体积应变

The volumetric strain for isotropic materials

构件每单位体积的体积变化，称为体积应变用$\theta$表示。

各向同性材料在三向应力状态下的体应变如图所示的单元体，三个边长为 d$x$，d$y$，d$z$。




变形后的边长分别为

$$dx(1+\varepsilon_1),\ dy(1+\varepsilon_2),\ dz(1+\varepsilon_3)$$

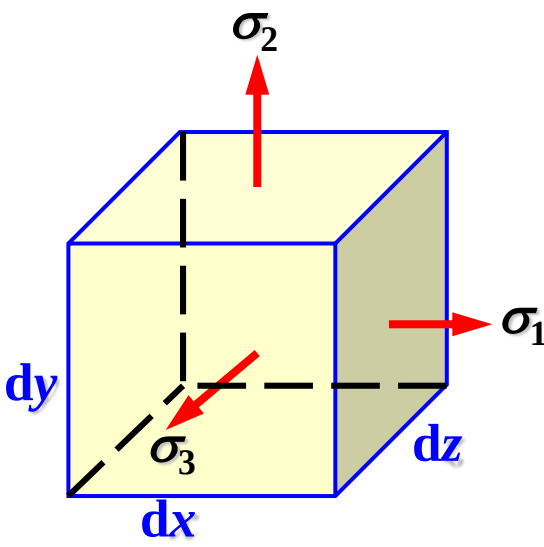
变形后单元体的体积为

$$V_1=dx(1+\varepsilon_1)\cdot dy(1+\varepsilon_2)\cdot dz(1+\varepsilon_3)$$

# 体积应变

$$\theta = \frac{V_1 - V}{V}$$

$$= \frac{\mathrm{d}x(1+\varepsilon_1)\cdot \mathrm{d}y(1+\varepsilon_2)\cdot \mathrm{d}z(1+\varepsilon_3) - \mathrm{d}x\cdot \mathrm{d}y\cdot \mathrm{d}z}{\mathrm{d}x\cdot \mathrm{d}y\cdot \mathrm{d}z}$$




$$= (1+\varepsilon_1)\cdot(1+\varepsilon_2)\cdot(1+\varepsilon_3) - 1$$

$$= 1 + \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_1\varepsilon_2 + \varepsilon_1\varepsilon_3 + \varepsilon_3\varepsilon_2 + \varepsilon_1\varepsilon_2\varepsilon_3 - 1$$

$$\approx \varepsilon_1 + \varepsilon_2 + \varepsilon_3$$

$$= \frac{1-2\mu}{E}(\sigma_1 + \sigma_2 + \sigma_3)$$

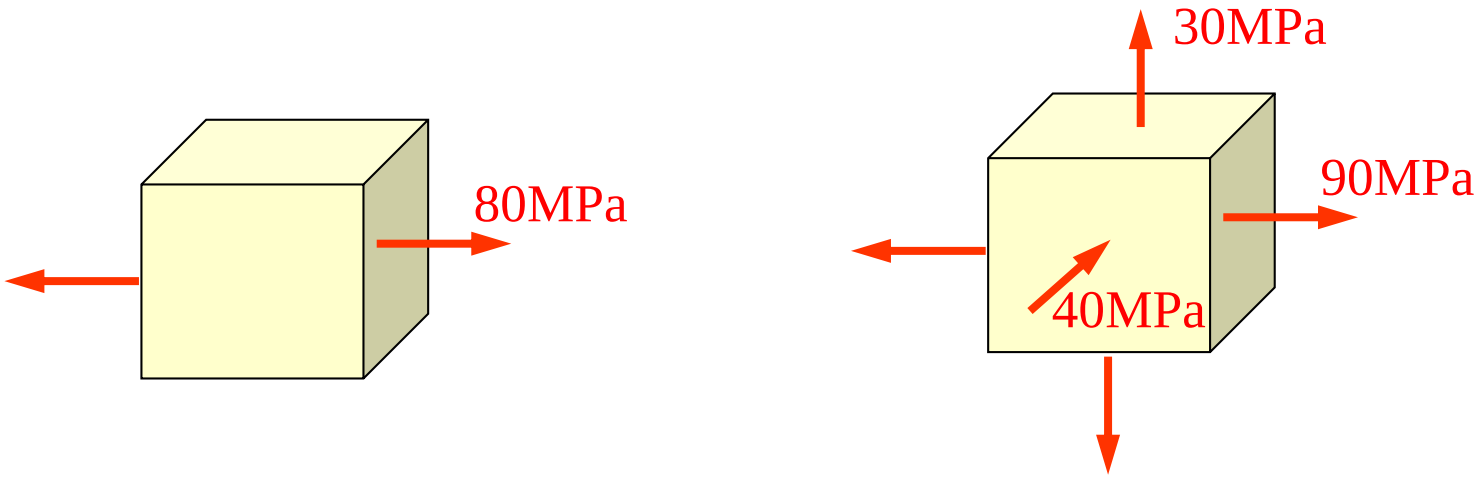
$$\varepsilon_1 = \frac{1}{E}\left[\sigma_1 - \mu(\sigma_2 + \sigma_3)\right]$$

$$\varepsilon_2 = \frac{1}{E}\left[\sigma_2 - \mu(\sigma_1 + \sigma_3)\right]$$

$$\varepsilon_3 = \frac{1}{E}\left[\sigma_3 - \mu(\sigma_1 + \sigma_2)\right]$$

$$\left.\begin{aligned}\theta &= \frac{1-2\mu}{E}\left(\sigma_1+\sigma_2+\sigma_3\right)\\ \sigma_m &= \frac{\sigma_1+\sigma_2+\sigma_3}{3}\end{aligned}\right\}\quad \theta = \frac{1-2\mu}{E}\cdot 3\sigma_m$$

体积应变与三个主应力的数值之比无关，取决于三个主应力的代数和。




二者的体积应变是否相同？

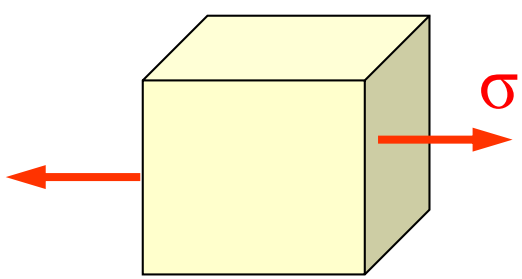
## 二、应变能密度的定义 (Definition of Strain-energy density )

物体在单位体积内所积蓄的应变能。

## 三、应变能密度的计算公式

(Calculation formula for Strain-energy density)

1、单向应力状态下，物体内所积蓄的应变能密度为




$$v_\varepsilon = \frac{V_\varepsilon}{V} = \frac{\frac{1}{2}F \cdot \Delta l}{Al} = \frac{1}{2}\sigma\varepsilon$$

$$v_\varepsilon = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E} = \frac{E}{2}\varepsilon^2$$

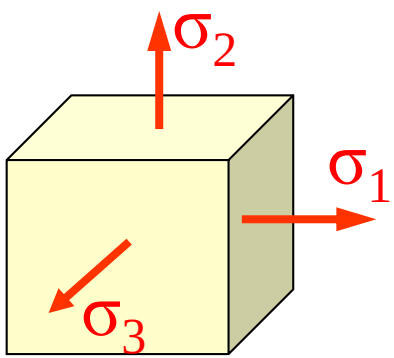
2、三个主应力同时存在时，单元体的应变能密度为

应变能只与外力的最终值有关，而与加载过程和加载次序无关。

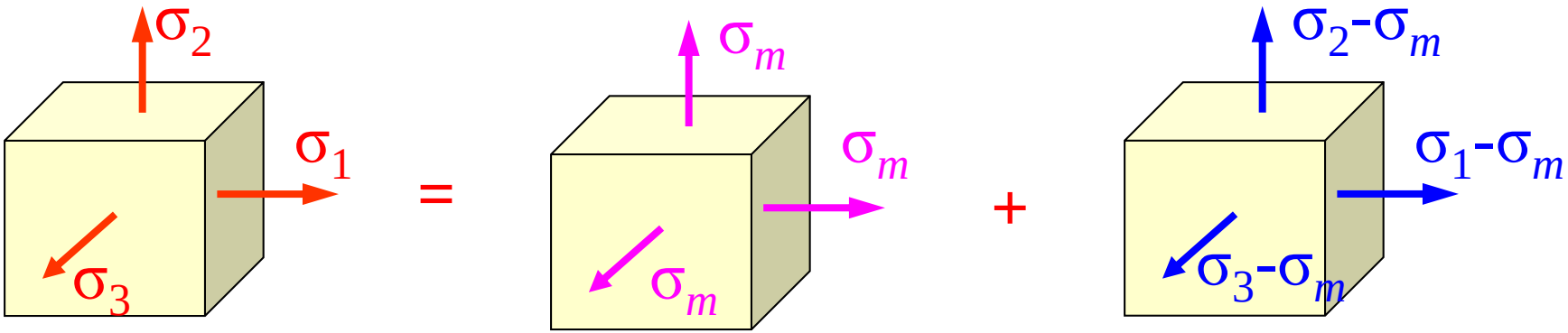
$$v_\varepsilon = \frac{1}{2}\left(\sigma_1\varepsilon_1 + \sigma_2\varepsilon_2 + \sigma_3\varepsilon_3\right)$$

将广义胡克定律代入上式，经整理得

$$v_\varepsilon = \frac{1}{2E}\left[\sigma_1 + \sigma_2 + \sigma_3 - 2\mu\left(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3\right)\right]$$




$$\boxed{\begin{aligned}\varepsilon_1 &= \frac{1}{E}\left[\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right)\right]\\ \varepsilon_2 &= \frac{1}{E}\left[\sigma_2 - \mu\left(\sigma_1 + \sigma_3\right)\right]\\ \varepsilon_3 &= \frac{1}{E}\left[\sigma_3 - \mu\left(\sigma_1 + \sigma_2\right)\right]\end{aligned}}$$

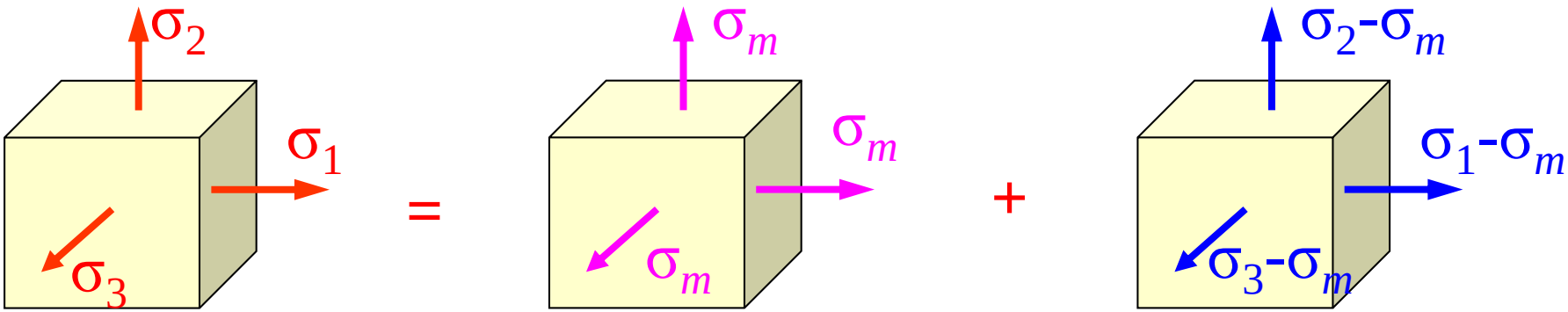



（a）图　　（b）图　　（c）图

图（a）所示单元体的三个主应力不相等，因而，变形后既发生体积改变也发生形状改变。

$$\sigma_m = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

图（b）所示单元体的三个主应力相等，因而，变形后的形状与原来的形状相似，即只发生体积改变而无形状改变。




（a）图　　（b）图　　（c）图

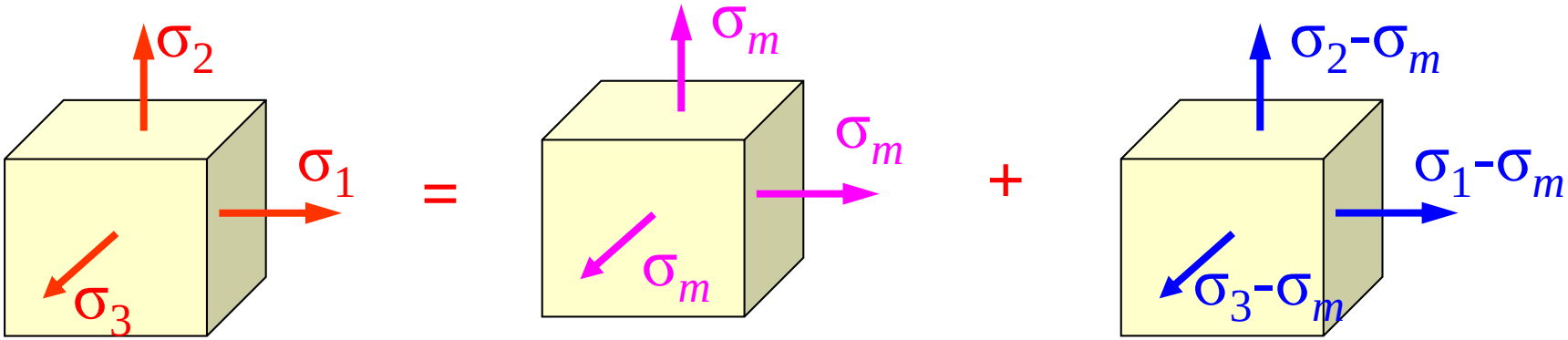
（b）图　　体积应变 $\theta = \dfrac{1-2\mu}{E}\cdot 3\sigma_m$

用 $v_V$ 表示单元体体积改变能密度

$$v_\varepsilon = \frac{1}{2E}\left[\sigma_1+\sigma_2+\sigma_3-2\mu\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)\right]$$

$$\boldsymbol{v_V = \frac{1}{2E}[(\sigma_m^2+\sigma_m^2+\sigma_m^2-2\mu(\sigma_m^2+\sigma_m^2+\sigma_m^2)]}$$

$$\boldsymbol{= \frac{3(1-2\mu)}{2E}\sigma_m^2 = \frac{1-2\mu}{6E}(\sigma_1+\sigma_2+\sigma_3)^2}$$




（a）图　　（b）图　　（c）图

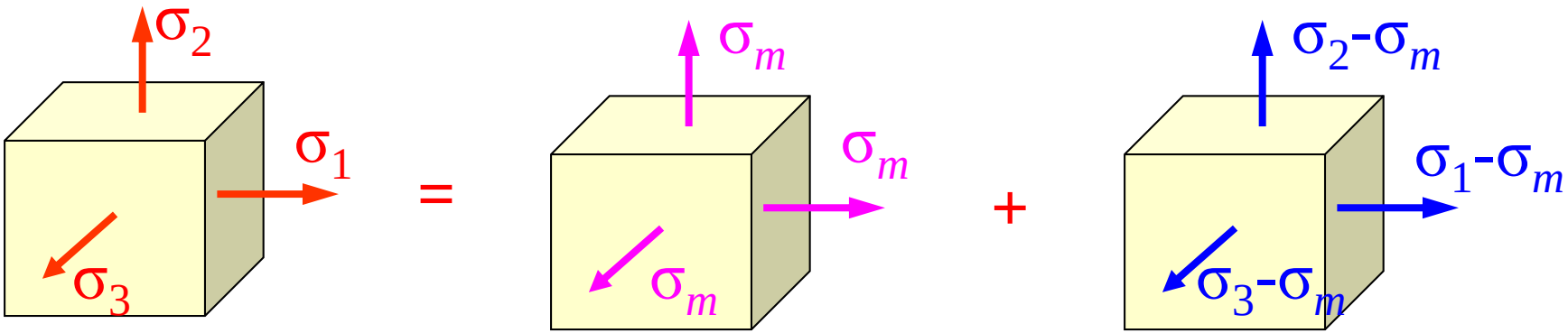
（c）图　体积应变　$\sigma_m = \dfrac{\sigma_1+\sigma_2+\sigma_3}{3}$

$$\theta = \frac{1-2\mu}{E}\left(\sigma_1+\sigma_2+\sigma_3\right) = \frac{1-2\mu}{E}\left(\sigma_1-\sigma_m+\sigma_2-\sigma_m+\sigma_3-\sigma_m\right) = 0$$

用$v_d$表示与单元体形状改变相应的那部分应变能密度，称为**畸变能密度。**

$$v_\varepsilon = \frac{1}{2E}\left[\sigma_1+\sigma_2+\sigma_3-2\mu\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)\right]$$

$$v_d = \frac{1+\mu}{6E}\left[\left(\sigma_1-\sigma_2\right)^2+\left(\sigma_1-\sigma_3\right)^2+\left(\sigma_3-\sigma_2\right)^2\right]$$




（a）图　　（b）图　　（c）图

应变能密度 $$v_\varepsilon = \frac{1}{2E}\left[\sigma_1+\sigma_2+\sigma_3-2\mu\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)\right]$$

$$v_\varepsilon = v_V + v_d$$

体积改变能密度 $$v_V = \frac{1-2\mu}{6E}(\sigma_1+\sigma_2+\sigma_3)^2$$

畸变能密度 $$v_d = \frac{1+\mu}{6E}\left[(\sigma_1-\sigma_2)^2+(\sigma_1-\sigma_3)^2+(\sigma_3-\sigma_2)^2\right]$$

# §7-8 强度理论 (The failure criteria)

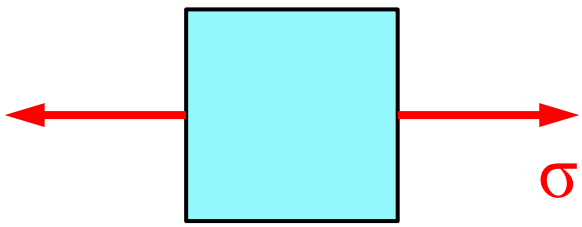
## 一、强度理论的概念（Concepts of failure criteria)

### 1、引言 (introduction)

正应力强度条件

拉压 $$\sigma = \frac{F_{N\max}}{A} \le [\sigma]$$

弯曲 $$\sigma_{\max} = \frac{M_{\max}}{W_z} \le [\sigma]$$

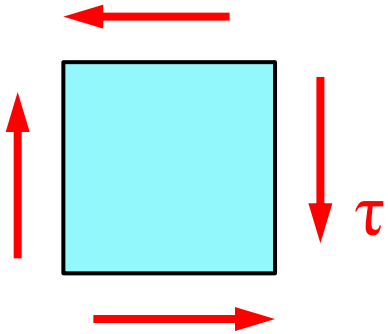



切应力强度条件

剪切 $$\tau = \frac{F_S}{A} \le [\tau]$$

扭转 $$\tau_{\max} = \frac{T_{\max}}{W_\rho} \le [\tau]$$

弯曲 $$\tau_{\max} = \frac{F_{S\max} S^*_{z\max}}{b I_z} \le [\tau]$$



上述强度条件具有如下特点

（1）危险点处于单向应力状态或纯剪切应力状态；

（2）材料的许用应力，是通过拉（压）试验或纯剪试验测定试件在破坏时其横截面上的极限应力，以此极限应力作为强度指标，除以适当的安全因数而得，即根据相应的试验结果建立的强度条件。

2、强度理论的概念(Concepts for failure criteria)

是关于“**构件发生强度失效起因**”的假说。

根据材料在复杂应力状态下破坏时的一些现象与形式，进行分析，提出破坏原因的假说。在这些假说的基础上，可利用材料在**单向应力状态**时的试验结果，来建立材料在复杂应力状态下的强度条件。

基本观点：

构件受外力作用而发生破坏时，不论破坏的表面现象如何复杂，其破坏形式总不外乎几种类型，而**同一类型的破坏则可能是某一个共同因素所引起的**。

## 3、材料破坏的两种类型（常温、静载荷）

(Two failure types for materials in normal temperature and static loads)

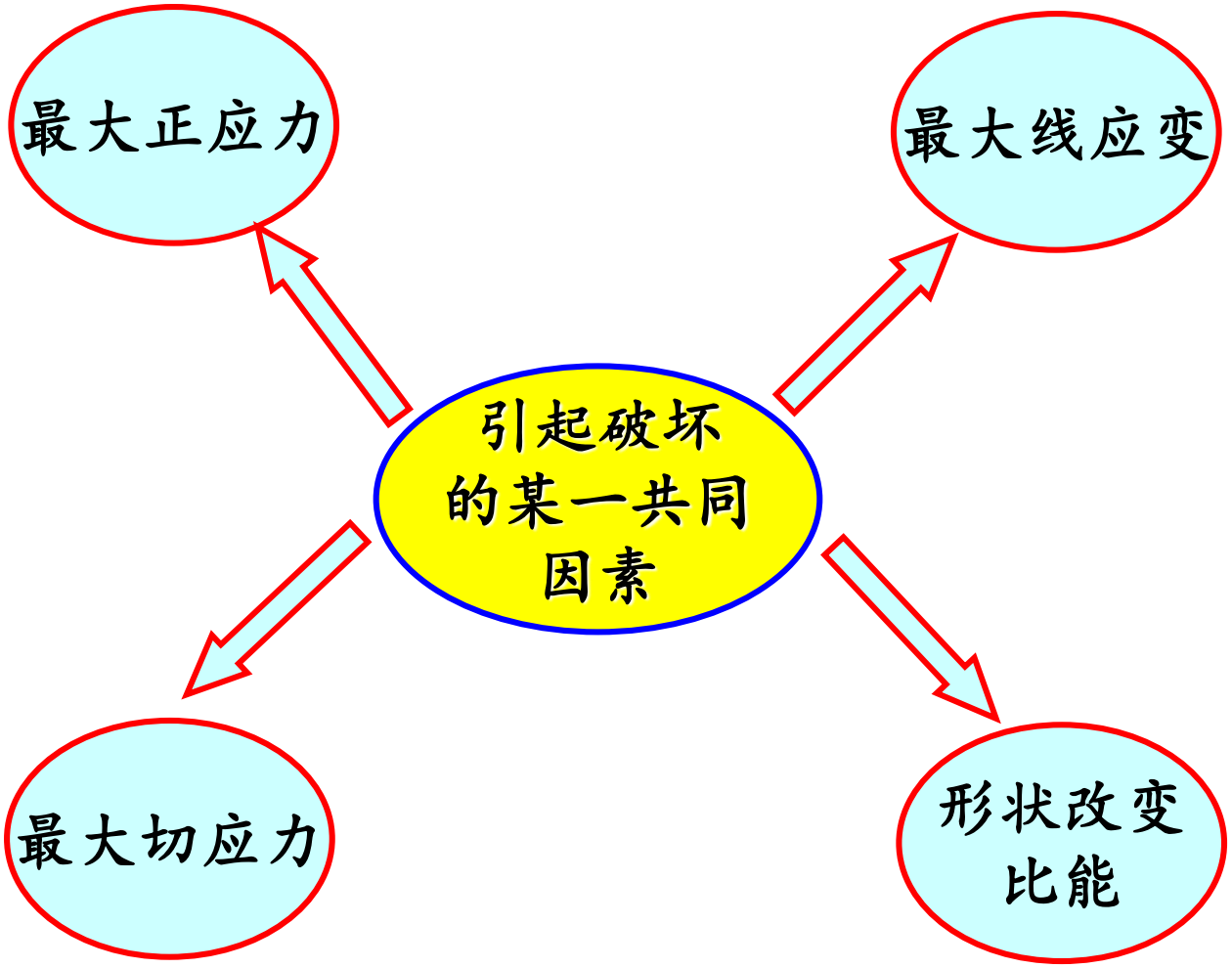
（1）**屈服**失效（Yielding failure）

材料出现显著的塑性变形而丧失其正常的工作能力。

（2） **断裂**失效（Fracture failure）

（a）脆性断裂：无明显的变形下突然断裂。

（b）韧性断裂：产生大量塑性变形后断裂。


最大正应力
最大线应变
引起破坏的某一共同因素
最大切应力
形状改变比能

## 二、四个强度理论（Four failure criteria）

### 1、概述

（1）伽利略播下了第一强度理论的种子；

（2）马里奥特关于变形过大引起破坏的论述，是第二强度理论的萌芽；

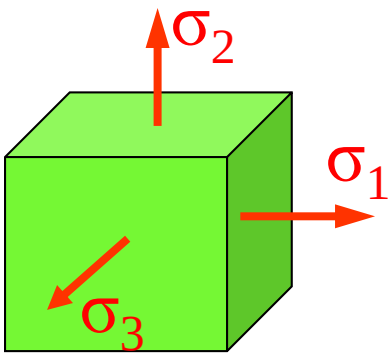
（3）杜奎特（C.Duguet）提出了最大切应力理论；

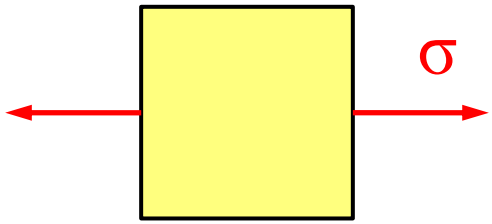
（4）麦克斯威尔最早提出了最大畸变能理论，这是后来人们在他的书信出版后才知道的。

（5） 第一类强度理论——以**脆断**作为破坏的标志。

包括：**最大拉应力理论**和**最大伸长线应变理论**。

（6）第二类强度理论——以出现**屈服**现象作为破坏的标志。包括：**最大切应力理论**和**形状改变比能理论**。

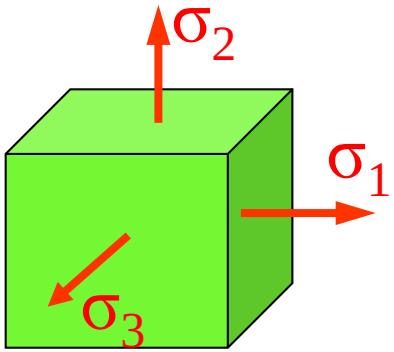



$\sigma_1$， $\sigma_2$， $\sigma_3$　　$\sigma_1 = \sigma_b$（$\sigma_s$），$\sigma_2 = \sigma_3 = 0$。

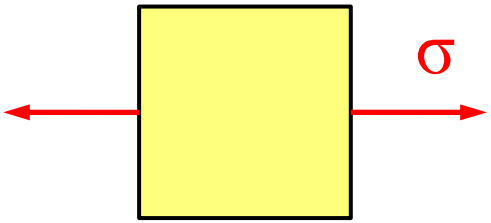
## 2、最大拉应力理论（第一强度理论）

（Maximum-normal-stress criterion）




根据：当作用在构件上的外力过大时，其危险点处的材料就会沿**最大拉应力**所在截面发生脆断破坏。

基本假说：最大拉应力$\sigma_1$是引起材料脆断破坏的因素。




脆断破坏的条件：　$\sigma_1 = \sigma_b$

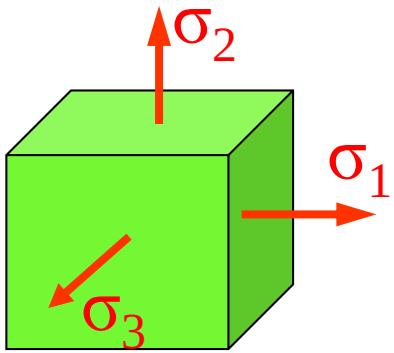
强度条件：　$\boxed{\sigma_1 \le [\sigma]}$

## 3、最大伸长线应变理论（第二强度理论）

（Maximum-normal-strain criterion）

根据：当作用在构件上的外力过大时，其危险点处的材料就会沿垂直于**最大伸长线应变**方向的平面发生破坏。

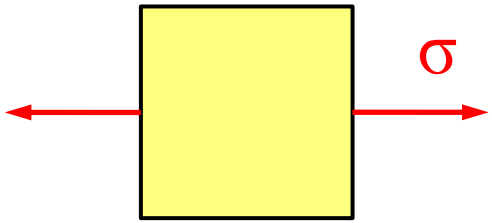
基本假说：最大伸长线应变$\varepsilon_1$是引起材料脆断破坏的因素。




脆断破坏的条件：$\varepsilon_1 = \dfrac{\sigma_b}{E}$

最大伸长线应变：$\varepsilon_1 = \dfrac{1}{E}\left[\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right)\right] = \dfrac{\sigma_b}{E}$

强度条件：$\boxed{\sigma_1 - \mu\left(\sigma_2 + \sigma_3\right) \le \left[\sigma\right]}$

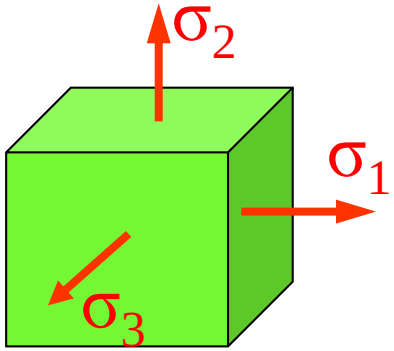


## 4、最大切应力理论 （第三强度理论）

（Maximum-shear-stress criterion）

根据：当作用在构件上的外力过大时，其危险点处的材料就会沿**最大切应力**所在截面滑移而发生屈服失效。

基本假说： 最大切应力$\tau_{max}$是引起材料屈服的因素。

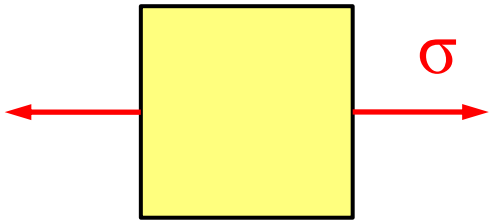



屈服条件： $\tau_{\max} = \dfrac{\sigma_s}{2}$

在复杂应力状态下一点处的最大切应力为

$$\tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{\sigma_s}{2}$$

$$\sigma_1 - \sigma_3 = \sigma_s$$

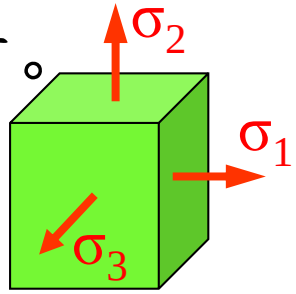



强度条件： $\boxed{\sigma_1 - \sigma_3 \le [\sigma]}$

## 5、畸变能密度理论（第四强度理论）

（Maximum-distortion-energy criterion）

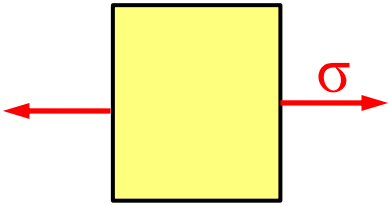
基本假说：**畸变能密度**$v_d$是引起材料屈服的因素。




$$V_d=\frac{1+\mu}{6E}\left[(\sigma_1-\sigma_m)^2+(\sigma_2-\sigma_m)^2+(\sigma_3-\sigma_m)^2\right]$$

单向拉伸下，$\sigma_1=\sigma_s$，$\sigma_2=\sigma_3=0$，材料的极限值

$$V_d=\frac{1+\mu}{6E}\cdot 2\sigma_s^2$$




屈服准则：$V_d=\frac{1+\mu}{6E}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]$

$$=\frac{1+\mu}{6E}\cdot 2\sigma_s^2$$

强度条件：$\boxed{\sqrt{\frac{1}{2}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]}\le[\sigma]}$

## 6、相当应力（Equivalent stress）

把各种强度理论的强度条件写成统一形式

$$\sigma_{\mathrm{r}} \leq [\sigma]$$

$\sigma_{\mathrm{r}}$ 称为复杂应力状态的相当应力。

$$\sigma_{r1} = \sigma_1$$

$$\sigma_{r2} = \sigma_1 - \mu(\sigma_2 + \sigma_3)$$

$$\sigma_{r3} = \sigma_1 - \sigma_3$$

$$\sigma_{r4} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]}$$

## 三、 各种强度理论的适用范围及其应用

（The appliance range and application for all failure criteria）

1、适用范围（The appliance range）

（1）一般脆性材料选用第一或第二强度理论；

（2）塑性材料选用第三或第四强度理论；

（3）在二向和三向等拉应力时，无论是塑性还是脆性都发生脆性破坏，故选用第一或第二强度理论；

（4）在二向和三向等压应力状态时，无论是塑性还是脆性材料都发生塑性破坏，故选用第三或第四强度理论。

## 2、强度计算的步骤（Steps of strength calculation）

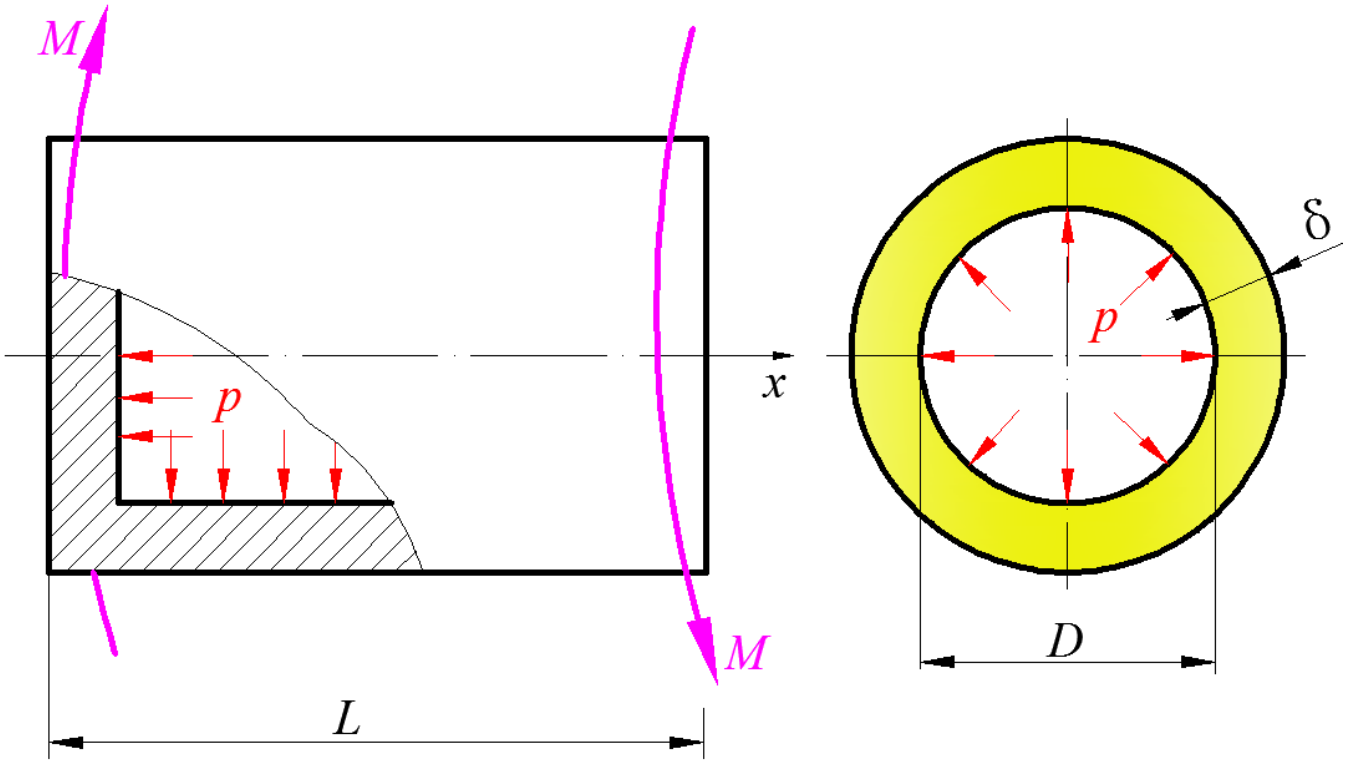
（1）外力分析：确定所需的外力值；

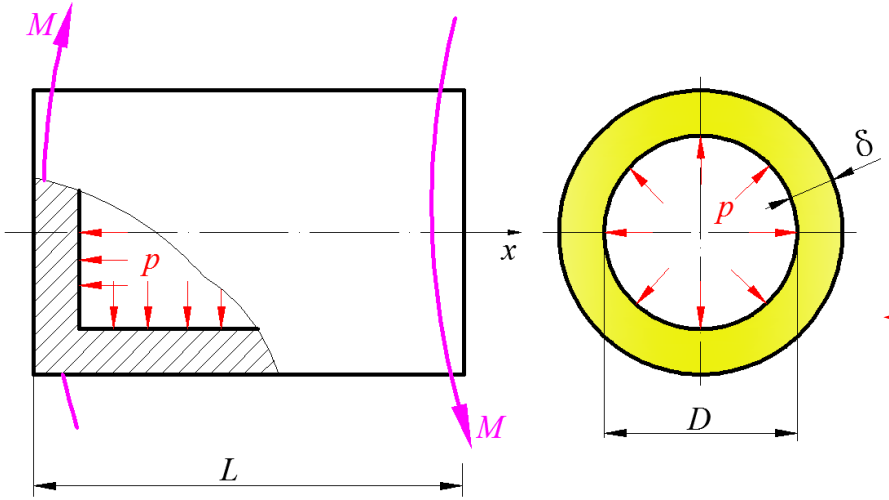
（2）内力分析：画内力图，确定可能的危险面；

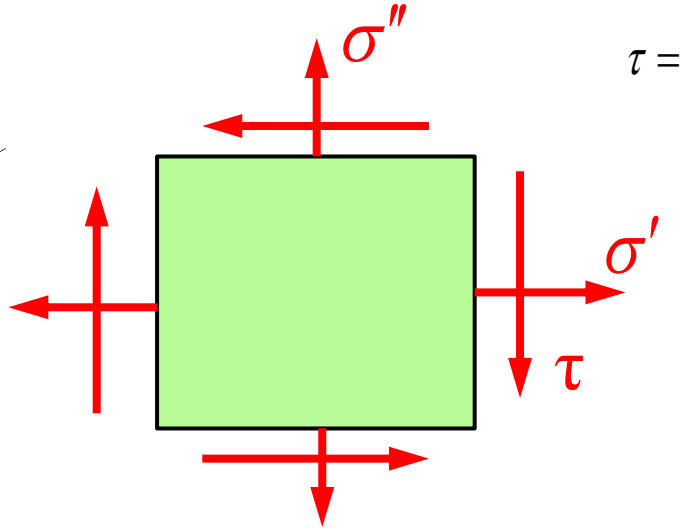
（3）应力分析：画危面应力分布图,确定危险点并画出单元体，求主应力；

（4）强度分析：选择适当的强度理论，计算相当应力，然后进行强度计算。

**例题7-9** 薄壁圆筒同时受内压$p$和力偶$M$的作用。圆筒部分的内径为$D$，筒长为$L$。弹性模量为$E$，泊松比为$\mu$许用应力为[σ]。$M=\dfrac{\pi D^3}{4}p$ 。（1）试用第三、四强度理论设计筒厚。（2）如果筒厚为δ，计算筒体的轴向变形。






$$\tau = \frac{M}{2\pi\left(\frac{D}{2}\right)^2\delta} = \frac{pD}{2\delta}$$

$$\sigma' = \frac{pD}{4\delta}$$

$$\sigma'' = \frac{pD}{2\delta}$$

$$\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix} = \frac{\sigma'+\sigma''}{2} \pm \sqrt{\left(\frac{\sigma'+\sigma''}{2}\right)^2 + \tau^2} = \frac{3pD}{8\delta} \pm \frac{\sqrt{17}pD}{8\delta}$$

$$\sigma_1 = \frac{\left(3+\sqrt{17}\right)pD}{8\delta} \quad \sigma_2 = 0 \quad \sigma_3 = \frac{\left(3-\sqrt{17}\right)pD}{8\delta}$$

$$\sigma_{r3} = \sigma_1 - \sigma_3 = \frac{\sqrt{17}pD}{4\delta}$$

$$\sigma_{r4} = \sqrt{\frac{1}{2}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right]}$$

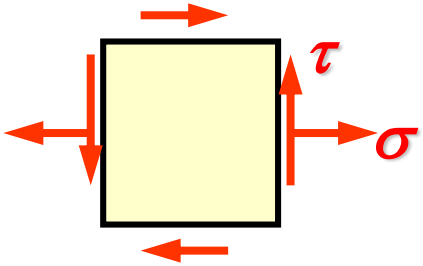
$$= \frac{\sqrt{15}pD}{4\delta}$$

第三强度理论 $\delta \geq \frac{\sqrt{17}Dp}{4[\sigma]}$

第四强度理论 $\delta \geq \frac{\sqrt{15}Dp}{4[\sigma]}$

$$\Delta L_x = \varepsilon_x \cdot L = \frac{DpL}{4E\delta}(1-2\mu)$$

$$\begin{matrix}\sigma_{\max}\\ \sigma_{\min}\end{matrix}=\frac{\sigma}{2}\pm\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$$




$$\sigma_1=\frac{\sigma}{2}+\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}\quad \sigma_2=0\quad \sigma_3=\frac{\sigma}{2}-\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$$

$$\sigma_{r1}=\sigma_1=\frac{\sigma}{2}+\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$$

$$\sigma_{r2}=\sigma_1-\mu(\sigma_2+\sigma_3)=\frac{(1-\mu)\sigma}{2}+(1+\mu)\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}$$

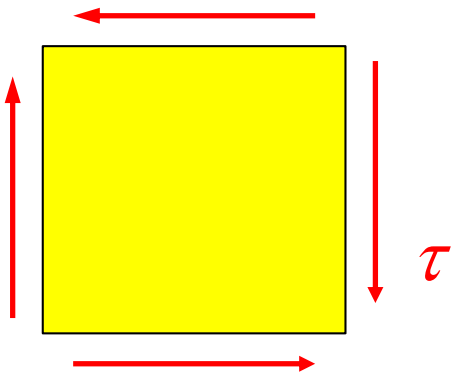
$$\sigma_{r3}=\sigma_1-\sigma_3=2\sqrt{\left(\frac{\sigma}{2}\right)^2+\tau^2}=\boxed{\sqrt{\sigma^2+4\tau^2}}$$

$$\sigma_{r4}=\sqrt{\frac{1}{2}\left[(\sigma_1\text{-}\sigma_2)^2+(\sigma_2\text{-}\sigma_3)^2+(\sigma_3\text{-}\sigma_1)^2\right]}=\boxed{\sqrt{\sigma^2+3\tau^2}}$$

第三强度理论和第四强度理论哪个更安全？

$\sigma_{r3}$=71.7MPa　　$\sigma_{r4}$=62.2MPa

**例题7-10** 根据强度理论，可以从材料在单轴拉伸时的$[\sigma]$可推知低碳钢类塑性材料在纯剪切应力状态下的$[\tau]$。




解:纯剪切应力状态下

$$\sigma_1 = \tau \ , \ \sigma_2 = 0 \ , \ \sigma_3 = -\tau$$

按第三强度理论得强度条件为：

$$\sigma_{r3} \le [\sigma] \qquad \sigma_1 - \sigma_3 = \tau - (-\tau) = 2\tau \le [\sigma] \qquad \tau \le \frac{1}{2}[\sigma]$$

另一方面，剪切的强度条件是: $\tau \le [\tau]$

材料在纯剪切应力状态下的许用切应力为$[\tau]$。

所以 $[\tau] = 0.5[\sigma]$

按第四强度理论强度条件为： $\sigma_{r4} \leq [\sigma]$

$$\sigma_{r4} = \sqrt{\frac{1}{2}\left[\left(\sigma_1 - \sigma_2\right)^2 + \left(\sigma_1 - \sigma_3\right)^2 + \left(\sigma_2 - \sigma_3\right)^2\right]}$$

$$= \sqrt{\frac{1}{2}\left[\left(\tau - 0\right)^2 + \left(\tau + \tau\right)^2 + \left(0 + \tau\right)^2\right]}$$

$$= \sqrt{3}\tau$$

$$\tau \leq \frac{1}{\sqrt{3}}\left[\sigma\right]$$

$$\left[\tau\right] = \frac{1}{\sqrt{3}}\left[\sigma\right] = 0.577\left[\sigma\right] \approx 0.6\left[\sigma\right]$$

按第三强度理论得到：$[\tau] = 0.5\,[\sigma]$

按第四强度理论得到：$[\tau] \approx 0.6\,[\sigma]$

# 第7章结束